

STOR 155 - Lecture 1

Case Study & Introduction

Panagiotis Andreou

UNC Chapel Hill

January 8, 2025

Treating Chronic Fatigue Syndrome

- ▶ **Objective:** Evaluate the effectiveness of cognitive-behavior therapy for chronic fatigue syndrome.
- ▶ **Participant pool:** 142 patients who were recruited from referrals by primary care physicians and consultants to a hospital clinic specializing in chronic fatigue syndrome.
- ▶ **Actual participants:** Only 60 of the 142 referred patients entered the study. Some were excluded because they didn't meet the diagnostic criteria, some had other health issues, and some refused to be a part of the study.

Study design

- ▶ Patients randomly assigned to treatment and control groups, 30 patients in each group:
 - ▶ **Treatment group:** Cognitive behavior therapy – collaborative, educative, and with a behavioral emphasis. Patients were shown on how activity could be increased steadily and safely without exacerbating symptoms.
 - ▶ **Control group:** Relaxation – No advice was given about how activity could be increased. Instead progressive muscle relaxation, visualization, and rapid relaxation skills were taught.
- ▶ 7 patients dropped out of the study later: 3 from the treatment and 4 from the control group.

Results

The table below shows the distribution of patients with good outcomes at 6-month follow-up. Note that 7 patients dropped out of the study: 3 from the treatment and 4 from the control group.

		<i>Good outcome</i>		Total
		Yes	No	
<i>Group</i>	Treatment	19	8	27
	Control	5	21	26
	Total	24	29	53

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- Proportion with good outcomes in control group:

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Understanding the results

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- ▶ Suppose you flip a coin 100 times. While the chance a coin lands heads in any given coin flip is 50%, we probably won't observe exactly 50 heads. This type of fluctuation is part of almost any type of data generating process.
- ▶ The observed difference between the two groups ($70 - 19 = 51\%$) may be real, or may be due to natural variation.
- ▶ Since the difference is quite large, it is more believable that the difference is real.
- ▶ We need statistical tools to determine if the difference is so large that we should reject the notion that it was due to chance.

Generalizing the results

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Are the results of this study **generalizable** to all patients with chronic fatigue syndrome?

- ▶ These patients had specific characteristics and volunteered to be a part of this study.
- ▶ Therefore, maybe not representative of all patients with chronic fatigue syndrome.
- ▶ While we cannot immediately generalize the results to all patients, this first study is encouraging.
- ▶ The method works for patients with some narrow set of characteristics, and that gives hope that it will work, at least to some degree, with other patients.

Key concepts

Some key concepts we'll see:

- ▶ Randomization
- ▶ Confounding Variables
- ▶ Statistical Significance

Randomization

- ▶ Randomization is a cornerstone of experimental design.
- ▶ It helps eliminate bias in treatment assignment.
- ▶ Ensures any differences observed are due to treatment, not confounding variables.
- ▶ *Example:* Assigning patients to treatment/control groups using a random number generator.

Confounding Variables

- ▶ A confounding variable is one that influences both the independent variable (treatment) and the dependent variable (outcome).
- ▶ Can distort the perceived effect of the treatment.
- ▶ *Example:* Age or pre-existing symptoms in a medical study.
- ▶ Randomization helps mitigate confounding effects (e.g., age groups get distributed more evenly across groups).

Statistical Significance

- ▶ Statistical significance quantifies whether an observed effect is likely due to chance.
- ▶ A result is statistically significant if it is unlikely to occur under the null hypothesis.
- ▶ *Example:* Comparing the proportions of good outcomes between treatment and control groups.

Coin Flip Simulation

- ▶ Imagine flipping a fair coin 10 times.
- ▶ True probability of heads: 50%.
- ▶ Will we always observe exactly 5 heads?
- ▶ How about **simulating** this experiment?

References

- ▶ Section 1.1 of the textbook
- ▶ Coin flip simulator

STOR 155 - Lecture 2

Data Basics

Panagiotis Andreou

UNC Chapel Hill

January 10, 2025

Goals of this lecture

- ▶ Classification of variables
- ▶ Scatterplots
- ▶ Observational studies vs. Experiments
- ▶ Correlation does not imply causation

Classroom survey

A survey was conducted on students in an introductory statistics course. Below are a few of the questions on the survey, and the corresponding variables the data from the responses were stored in:

- ▶ `gender`: What is your gender?
- ▶ `intro_extra`: Do you consider yourself introverted or extraverted?
- ▶ `sleep`: How many hours do you sleep at night, on average?
- ▶ `bedtime`: What time do you usually go to bed?
- ▶ `countries`: How many countries have you visited?
- ▶ `dread`: On a scale of 1-5, how much do you dread being here?

Data matrix

Data collected on students in a statistics class on a variety of variables:

variable
↓

Stu.	gender	intro_extra	...	dread
1	male	extravert	...	3
2	female	extravert	...	2
3	female	introvert	...	4
4	female	extravert	...	2
⋮	⋮	⋮	⋮	⋮
86	male	extravert	...	3

←
observation

Types of variables

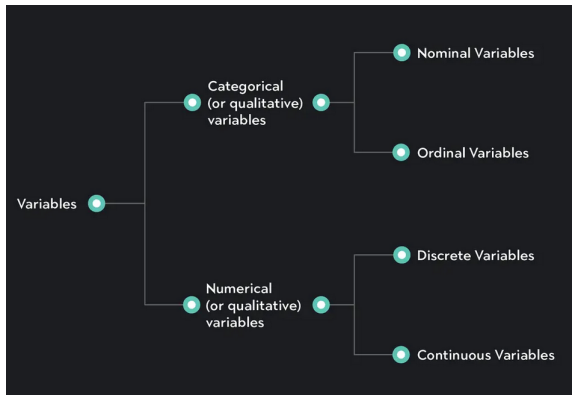


FIGURE: Types of variables in statistics. Source: [Understanding Variables in Statistics](#)

Types of variables (cont.)

	gender	sleep	bedtime	countries	dread
1	male	5	12-2	13	3
2	female	7	10-12	7	2
3	female	5.5	12-2	1	4
4	female	7	12-2		2
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- ▶ sleep: *numerical, continuous*
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- ▶ countries: *numerical, discrete*
- ▶ dread: *categorical, ordinal - could also be used as numerical*

Practice

What type of variable is a telephone area code?

- (a) numerical, continuous
- (b) numerical, discrete
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What type of variable is the count of apples in a basket?

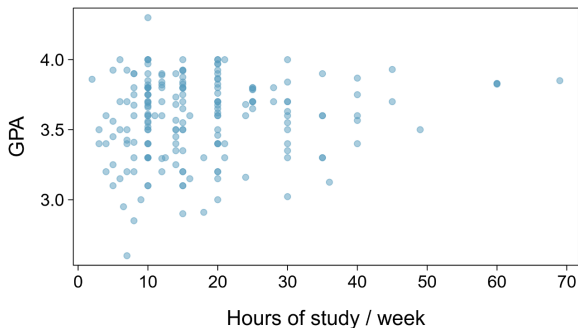
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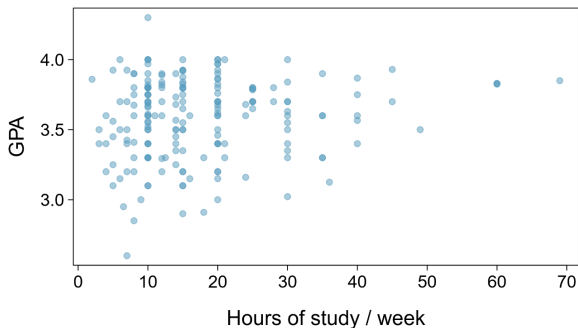
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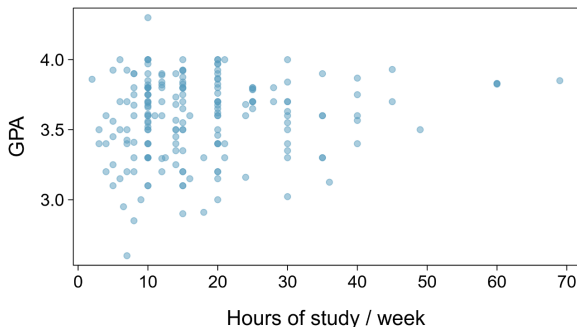


- Can you spot anything unusual about any of the data points?



Scatterplot

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- Can you spot anything unusual about any of the data points?
- *There is one student with $GPA > 4.0$, which is not possible. This looks like a data error.*

Explanatory and Response Variables

- ▶ **Explanatory Variables (Predictors):** These are the variables that are used to predict or explain variations in the response variable. Often referred to as independent variables.
- ▶ **Response Variable (Outcome):** This is the variable that we want to predict or explain. It is also known as the dependent variable.
- ▶ **Relationship:** The relationship between explanatory and response variables may be causal or correlative and is often expressed in a mathematical or statistical model.

Explanatory and response variables

- ▶ To identify the explanatory variable in a pair of variables, identify which of the two is suspected of affecting the other:

explanatory variable $\xrightarrow{\text{might affect}}$ response variable

- ▶ Labeling variables as explanatory and response does not guarantee the relationship between the two is actually causal, even if there is an association identified between the two variables.
- ▶ We use these labels only to keep track of which variable we suspect affects the other.
- ▶ **Correlation DOES NOT imply causation!**

Two primary types of data collection

- ▶ **Observational studies:** Collect data in a way that does not directly interfere with how the data arise (e.g., surveys).
 - ▶ Can provide evidence of a naturally occurring association between variables, but they cannot by themselves show a causal connection.

Two primary types of data collection

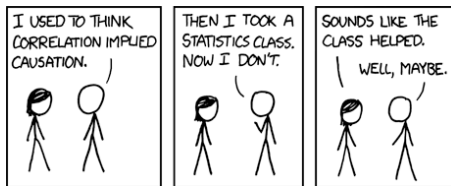
- ▶ **Observational studies:** Collect data in a way that does not directly interfere with how the data arise (e.g., surveys).
 - ▶ Can provide evidence of a naturally occurring association between variables, but they cannot by themselves show a causal connection.
- ▶ **Experiment:** Researchers randomly assign subjects to various treatments in order to establish causal connections between the explanatory and response variables.
 - ▶ In general, association does not imply causation, and causation can only be inferred from a randomized experiment.

Correlation vs. Causation

- ▶ When two variables show some connection with one another, they are called **correlated/associated** variables.
 - ▶ Associated variables can also be called **dependent** variables and vice-versa.
- ▶ If two variables are not associated, i.e., there is no evident connection between the two, then they are said to be **independent**.
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- ▶ A statistical measure that describes the extent to which two variables change together.
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- ▶ *Example:* Imagine you're examining the relationship between the amount of time students spend studying for a test and their respective test scores. You might find that, generally, as the number of hours spent studying increases, test scores also increase. This would be a positive correlation.

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 1. There's a correlation between the two variables.
 2. The cause happens before the effect.
 3. There aren't any other factors that could explain the relationship.
- ▶ *Example:* Taking a painkiller (like ibuprofen) to reduce a headache. If headaches consistently diminish or disappear after taking the pill, and no other factors are at play (e.g., you didn't also rest in a dark room), then we can infer a causal relationship between the painkiller and the reduction of headache pain.

Correlation DOES NOT imply causation

*Examples:*¹

1. Ice Cream Sales & Shark Attacks
2. Pool Drownings vs. Nuclear Energy Production
3. High School Graduates vs. Pizza Consumption

¹see References for details

Simple Example: *Ice Cream Sales and Drowning Incidents*

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- ▶ Does this mean buying more ice cream causes more drownings?
- ▶ Of course not! Potential reason?
- ▶ The **confounding variable** of temperature.
- ▶ During the summer, when it's hot, people are more likely to buy ice cream and also more likely to go swimming, which increases the chances of drowning.

The good stuff: *Science funding and Suicides*

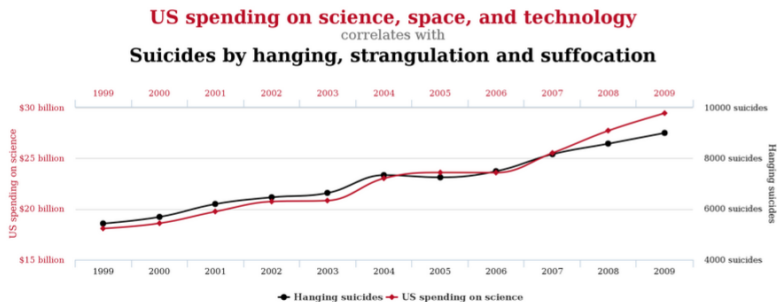


FIGURE: Example of “correlation does not imply causation”.

References

- ▶ Section 1.2 of the textbook
- ▶ Understanding Variables in Statistics
- ▶ Correlation Does Not Imply Causation: 5 Real-World Examples

STOR 155 - Lecture 3

Sampling Principles and Strategies

Panagiotis Andreou

UNC Chapel Hill

January 13, 2025

Anecdotal evidence and early smoking research

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- ▶ It was concluded that “smoking is a complex human behavior, by its nature difficult to study, confounded by human variability.”
- ▶ In time, researchers were able to examine larger samples of cases (smokers), and trends showing that smoking has negative health impacts became much clearer.

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 2. Populations rarely stand still. Even if you could take a census, the population changes constantly, so it's never possible to get a perfect measure.
 3. Taking a census may be more complex than sampling.

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- ▶ When you taste a spoonful of soup and decide the spoonful you tasted isn't salty enough, that's **exploratory analysis**.
- ▶ If you generalize and conclude that your entire soup needs salt, that's an **inference**.
- ▶ For your inference to be valid, the spoonful you tasted (the sample) needs to be **representative** of the entire pot (the population).
 - ▶ If your spoonful comes only from the surface and the salt is collected at the bottom of the pot, what you tasted is probably not representative of the whole pot.
 - ▶ If you first stir the soup thoroughly before you taste, your spoonful will more likely be representative of the whole pot.

Sampling bias

- ▶ **Non-response:** If only a small fraction of the randomly sampled people choose to respond to a survey, the sample may no longer be representative of the population.
 - ▶ *Example:* A sample of students in the class are given a survey form but only one or two students fill it up.

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 - ▶ *Example:* A sample of students in the class are given a survey form but only one or two students fill it up.
- ▶ **Voluntary response:** Occurs when the sample consists of people who volunteer to respond because they have strong opinions on the issue. Such a sample will also not be representative of the population.
 - ▶ *Example:* A survey asks whether the students are enjoying STOR 155. Those who don't like it are more eager to complain, and fill up the survey, than those who like the course.

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 - ▶ *Example:* A survey asks whether the students are enjoying STOR 155. Those who don't like it are more eager to complain, and fill up the survey, than those who like the course.
- ▶ **Convenience sample:** Individuals who are easily accessible are more likely to be included in the sample.
 - ▶ *Example:* A survey has a question: "Do you like answering surveys?"

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- ▶ Results of an observational study can generally be used to establish an **association** between the explanatory and response variables.
- ▶ They **cannot** be used to establish **causation**.

Obtaining good samples

- ▶ Almost all statistical methods are based on the notion of implied **randomness**.

Obtaining good samples

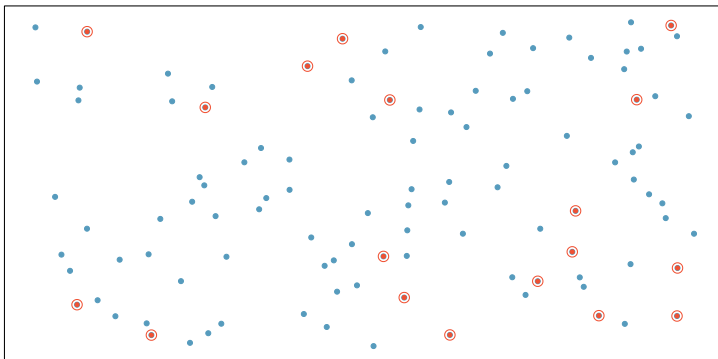
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- ▶ Almost all statistical methods are based on the notion of implied **randomness**.
- ▶ If observational data are not collected in a random framework from a population, these statistical methods – the estimates and errors associated with the estimates – are not reliable.
- ▶ Most commonly used random sampling techniques are **simple**, **stratified**, **cluster**, and **multistage** sampling.

Simple random sample

Randomly select cases from the population, where there is no implied connection between the points that are selected.



Example

- ▶ Objective: study the salaries of MLB players.

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- ▶ Each player is a member of one of the league's 30 teams.

Example

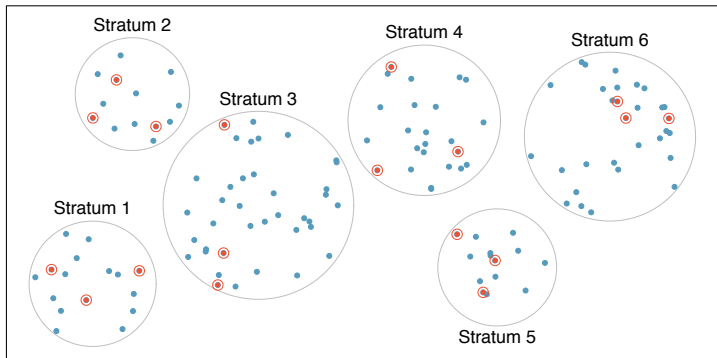
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- ▶ Suppose we want to take a random sample of 120 players.

Example

- ▶ Objective: study the salaries of MLB players.
- ▶ Each player is a member of one of the league's 30 teams.
- ▶ Suppose we want to take a random sample of 120 players.
- ▶ Execution: write all the players' names on slips of paper, mix them inside a bucket, and select 120 of them.

Stratified sample

Strata are made up of similar observations. We take a simple random sample from each stratum.



Example

- ▶ Back to the MLB players example.
- ▶ Goal: sample 120 players.

Example

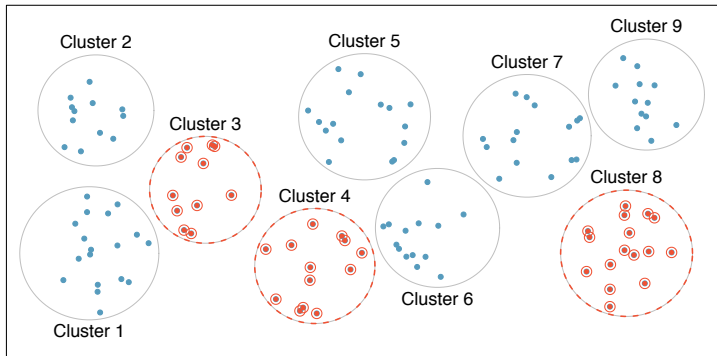
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Example

- ▶ Back to the MLB players example.
- ▶ Goal: sample 120 players.
- ▶ Insight: teams might differ a lot in terms of compensation, so it makes sense to use different teams as different strata.
- ▶ Stratified sampling: split the players population naturally in 30 teams, and then sample 4 players from each team.

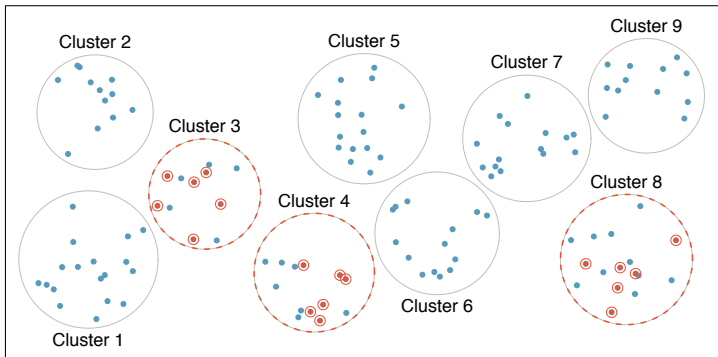
Cluster sample

Clusters are usually not made up of homogeneous observations. We take a simple random sample of clusters, and then sample all observations in that cluster. Usually preferred for economical reasons.



Multistage sample

Clusters are usually not made up of homogeneous observations. We take a simple random sample of clusters, and then take a simple random sample of observations from the sampled clusters.



References

- ▶ Section 1.3 of the textbook
- ▶ Sampling methods in Clinical Research; an Educational Review (1,000+ citations)

STOR 155 - Lecture 4

Experiments

Panagiotis Andreou

UNC Chapel Hill

January 15, 2025

Experiments

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- ▶ Studies where the researchers assign treatments to cases are called **experiments**.
- ▶ When this assignment includes randomization, e.g., using a coin flip to decide which treatment a patient receives, it is called a **randomized experiment**.
- ▶ Randomized experiments are fundamentally important when trying to show a causal connection between two variables.

Principles of experimental design

Randomized experiments are generally built on four principles:

1. **Control:** Control for the (potential) effect of variables other than the ones directly being studied.

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Randomized experiments are generally built on four principles:

1. **Control:** Control for the (potential) effect of variables other than the ones directly being studied.
2. **Randomize:** Randomly assign subjects to treatments, and randomly sample from the population whenever possible.
3. **Replicate:** Within a study, replicate by collecting a sufficiently large sample. Or replicate the entire study.
4. **Block:** If there are variables that are known or suspected to affect the response variable, first group subjects into **blocks** based on these variables, and then randomize cases within each block to treatment groups.

Controlling

- ▶ Researchers assign treatments to cases, and they do their best to control any other differences in the groups.

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- ▶ Researchers assign treatments to cases, and they do their best to control any other differences in the groups.
- ▶ *Example:* When patients take a drug in pill form, some patients take the pill with only a sip of water while others may have it with an entire glass of water. To control for the effect of water consumption, a doctor may ask all patients to drink a 12 ounce glass of water with the pill.

Randomization

- ▶ Researchers randomize patients into treatment groups to account for variables that cannot be controlled.

Randomization

- ▶ Researchers randomize patients into treatment groups to account for variables that cannot be controlled.
- ▶ *Example:* Some patients may be more susceptible to a disease than others due to their dietary habits. Randomizing patients into the treatment or control group helps even out such differences, and it also prevents accidental bias from entering the study.

Replication

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- ▶ In a single study, we replicate by collecting a sufficiently large sample.
- ▶ Additionally, a group of scientists may replicate an entire study to verify an earlier finding.

Blocking

- ▶ Sometimes, variables other than the explanatory ones might affect the response. These are called **blocking variables**.

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- ▶ Sometimes, variables other than the explanatory ones might affect the response. These are called **blocking variables**.
- ▶ To account for them, we might first group the subjects into **blocks** and then randomize cases within each block to the treatment groups. This process is called **blocking**.
- ▶ *Example:* If we are looking at the effect of a drug on heart attacks, we might first split patients in the study into low-risk and high-risk blocks, then randomly assign half the patients from each block to the control group and the other half to the treatment group.

Blocking - Example

- ▶ We would like to design an experiment to investigate if energy gels makes you run faster:

Blocking - Example

- ▶ We would like to design an experiment to investigate if energy gels makes you run faster:
 - ▶ Treatment: energy gel
 - ▶ Control: no energy gel

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Blocking - Example

- ▶ We would like to design an experiment to investigate if energy gels makes you run faster:
 - ▶ Treatment: energy gel
 - ▶ Control: no energy gel
- ▶ It is suspected that energy gels might affect pro and amateur athletes differently, therefore we block for pro status:
 - ▶ Divide the sample to pro and amateur
 - ▶ Randomly assign pro athletes to treatment and control groups
 - ▶ Randomly assign amateur athletes to treatment and control groups
 - ▶ Pro/amateur status is equally represented in the resulting treatment and control groups

Difference between blocking and explanatory variables

- ▶ **Factors** are conditions we can impose on the experimental units.

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Difference between blocking and explanatory variables

- ▶ **Factors** are conditions we can impose on the experimental units.
- ▶ **Blocking variables** are characteristics that the experimental units come with, that we would like to control for.
- ▶ Blocking is like stratifying, except used in experimental settings when **randomly assigning**, as opposed to when **randomly sampling**.

Experimental Design terminology

- ▶ **Placebo:** fake treatment, often used as the control group for medical studies.

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- ▶ **Placebo:** fake treatment, often used as the control group for medical studies.
- ▶ **Placebo effect:** experimental units showing improvement simply because they believe they are receiving a special treatment.
- ▶ **Blinding:** when experimental units do not know whether they are in the control or treatment group.
- ▶ **Double-blind:** when both the experimental units and the researchers who interact with the patients do not know who is in the control and who is in the treatment group.

Random Assignment vs. Random Sampling

<i>ideal experiment</i>	Random assignment	No random assignment	<i>most observational studies</i>
Random sampling	Causal conclusion, generalized to the whole population.	No causal conclusion, correlation statement generalized to the whole population.	Generalizability
No random sampling	Causal conclusion, only for the sample.	No causal conclusion, correlation statement only for the sample.	No generalizability
<i>most experiments</i>	Causation	Correlation	<i>bad observational studies</i>

References

- ▶ Section 1.4 of the textbook
- ▶ Experimental Design

STOR 155 - Lecture 5

Numerical Data: centrality and dispersion measures

Panagiotis Andreou

UNC Chapel Hill

January 17, 2025

Centrality and Dispersion Measures

- ▶ **Centrality measures** are important statistical tools to analyze numerical data. They provide insight into the “central” or typical values of datasets.
- ▶ Common centrality measures: **mean, median, mode**
- ▶ **Measures of dispersion** are non-negative real numbers that help to gauge the spread of data about a central value. These measures help to determine how stretched or squeezed the given data is.
- ▶ Common measures of dispersion: **range, variance, standard deviation**

Mean

- ▶ The **mean**, often referred to as the **average**, is calculated by summing all values in the dataset and dividing by the total number of observations.
- ▶ It is highly sensitive to outliers.
- ▶ The mean is an appropriate measure of central tendency when the data is roughly **symmetrically** distributed.

- ▶ The **sample mean**, denoted as $\bar{x}$, can be calculated as

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

where $x_1, x_2, \cdots, x_n$ represent the n observed values.

- ▶ The **population mean** is also computed the same way but is denoted as μ . It is often not possible to calculate μ since population data are rarely available.
- ▶ The sample mean is a **sample statistic**, and serves as a **point estimate** of the population mean. This estimate may not be perfect, but if the sample is good (representative of the population), it is usually a pretty good estimate.

Median

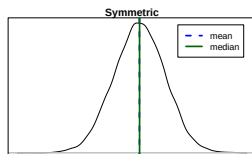
- ▶ The **median** is the value that separates the higher half from the lower half of a dataset.
- ▶ If the dataset has an **odd** number of observations, the median is the middle number. If it's **even**, the median is the average of the two middle numbers.
- ▶ Unlike the mean, the median is not affected by extreme values (outliers), i.e., it is a **robust** statistic.

To *compute* the median:

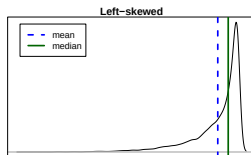
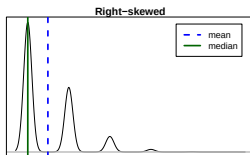
1. Order the numbers in increasing order.
2. If the multitude of numbers is odd, then the median is the number in the middle.
3. If the multitude of numbers is even, then the median is the average of the two middle numbers.
4. *Example:* The median of the 8 numbers 2, 5, 3, 8, 1, 6, 3, 2 can be found by ordering them first. The ordered numbers are 1, 2, 2, 3, 3, 5, 6, 8. There is an even (8) number of numbers, hence the median is the average of the two middle numbers 3 and 3, i.e., $(3 + 3)/2 = 3$.

Mean vs. median

- ▶ If the distribution is **symmetric**, the center is often defined as the mean: $\text{mean} \approx \text{median}$



- ▶ If the distribution is **skewed** or has extreme outliers, center is often defined as the median
 - ▶ Right-skewed: $\text{mean} > \text{median}$
 - ▶ Left-skewed: $\text{mean} < \text{median}$



Mean vs. Median: A Comparison

- ▶ **Sensitivity to Outliers:**

- ▶ Mean: Highly sensitive
- ▶ Median: Not sensitive (Robust)

- ▶ **Applicability:**

- ▶ Mean: Best for symmetric distributions
- ▶ Median: Useful for skewed distributions

- ▶ **Calculation Complexity:**

- ▶ Mean: Requires all data points
- ▶ Median: If the data is ordered, not all points are required

- ▶ **Representation:**

- ▶ Mean: Might not be an actual data point
- ▶ Median: Always an actual data point for odd sample sizes

Q1 and Q3

- ▶ The 25th percentile is also called the first quartile, **Q1**. It is computed as the median of the ordered data to the left of the median.
- ▶ The 50th percentile is also called the median.
- ▶ The 75th percentile is also called the third quartile, **Q3**. It is computed as the median of the ordered data to the right of the median.

Example: Odd sample size

Data: 2, 4, 5, 7, 10

- ▶ Sample size = 5 (odd)
- ▶ Median (Q_2) = 5
- ▶ Lower half: 2, 4
- ▶ Upper half: 7, 10
- ▶ $Q_1 = \text{Median of Lower half} = \frac{2+4}{2} = 3$
- ▶ $Q_3 = \text{Median of Upper half} = \frac{7+10}{2} = 8.5$

Example: Even sample size

Data: 1, 2, 4, 5, 7, 10

- ▶ Sample size = 6 (even)
- ▶ Median (Q_2) = 4.5
- ▶ Lower half: 1, 2, 4
- ▶ Upper half: 5, 7, 10
- ▶ Q_1 = Median of Lower half = 2
- ▶ Q_3 = Median of Upper half = 7

Mode

- ▶ The **mode** is the value that appears most frequently in a dataset.
- ▶ A dataset may have one mode (unimodal), more than one mode (multimodal), or no mode at all.
- ▶ The mode can be used for both numerical and categorical data.
- ▶ *Example:* The mode of $\{4, 2, 2, 3, 5, 4, 6, 2\}$ is 2, because it occurs three times, which is more than any other number in the dataset.

Range

- ▶ The range is the difference between the largest and smallest values in the dataset. It provides a measure of total spread.
- ▶ Mathematically, the range of n numbers $x_1, \dots, x_n$ is defined to be the statistic

$$R = \max\{x_1, \dots, x_n\} - \min\{x_1, \dots, x_n\}.$$

- ▶ *Example:* The range of the 8 numbers 2, 5, 3, 8, 1, 6, 3, 2 is $8 - 1 = 7$.
- ▶ Between Q1 and Q3 is the middle 50% of the data. The range these data span is called the **interquartile range**, or the **IQR**.

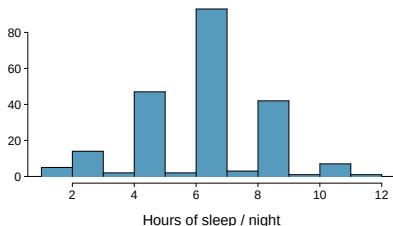
$$IQR = Q3 - Q1$$

Variance

Variance is roughly the average squared deviation from the mean.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- ▶ The sample mean is $\bar{x} = 6.71$, and the sample size is $n = 217$.
- ▶ The variance of amount of sleep students get per night can be calculated as:



$$s^2 = \frac{(5 - 6.71)^2 + (9 - 6.71)^2 + \dots + (7 - 6.71)^2}{217 - 1} = 4.11 \text{ hours}^2$$

Different Idea

- ▶ Why use squares and not just take the average of the differences from the mean?
- ▶ In that case, we would always get

$$\begin{aligned}\frac{(x_1 - \bar{x}) + \cdots + (x_n - \bar{x})}{n} &= \frac{(x_1 + \cdots + x_n) - n\bar{x}}{n} \\ &= \frac{x_1 + \cdots + x_n}{n} - \bar{x} = 0,\end{aligned}$$

which is not informative.

Standard deviation

The **standard deviation** is the square root of the variance, and has the same units as the data:

$$s = \sqrt{s^2}$$

- ▶ The standard deviation of the amount of sleep students get per night can be calculated as:

$$s = \sqrt{4.11} = 2.03 \text{ hours}$$

- ▶ We can see that all of the data are within 3 standard deviations of the mean.
- ▶ Same unit of measurement as that of the initial data.

References

- ▶ Section 2.1 of the textbook
- ▶ See pdf *NumericalData_problems* on Canvas.

STOR 155 - Lecture 6

Data Visualization

Panagiotis Andreou

UNC Chapel Hill

January 22, 2025

Numerical vs. Categorical Data: What's the Difference?

▶ Numerical Data

- ▶ *Quantities*: Things you can **measure** or count.
- ▶ Examples: Age, height, salary, temperature.
- ▶ Operations: You can add, subtract, find averages, etc.

▶ Categorical Data

- ▶ *Qualities*: Things you can **categorize** but not measure.
- ▶ Examples: Colors, gender, types of cuisine, yes/no answers.
- ▶ Operations: You can sort, group, but can't perform arithmetic.

Overall:

- ▶ *Numerical* is about *numbers you can calculate with*
- ▶ *Categorical* is about *categories you can sort into*.

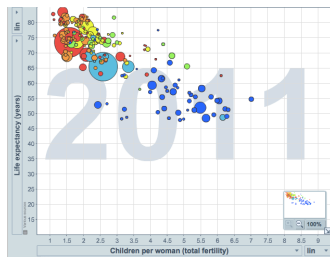
Visualization of Numerical Data

Four main visualization objects:

- ▶ Scatterplot
- ▶ Dot plot
- ▶ Histogram
- ▶ Box-plot

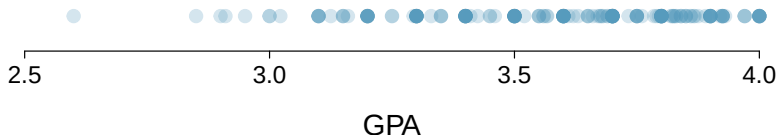
Scatterplot

- ▶ Useful for visualizing the relationship between **two** numerical variables.
- ▶ They provide information on positive/negative association.
- ▶ *Example:* Fertility and life expectancy appear to be *linearly* and *negatively associated*: as fertility increases, life expectancy decreases.



Dot plots

- ▶ Useful for visualizing **one** numerical variable. Darker colors represent areas where there are more observations.
- ▶ Darker colors represent areas where there are more observations.
- ▶ *Example:* Below is a dot plot of the GPA of the students in a class. We can see that most of the students have GPA higher than 3.5, quite some between 3 and 3.5, very few below 3, no one below 2.5, etc.

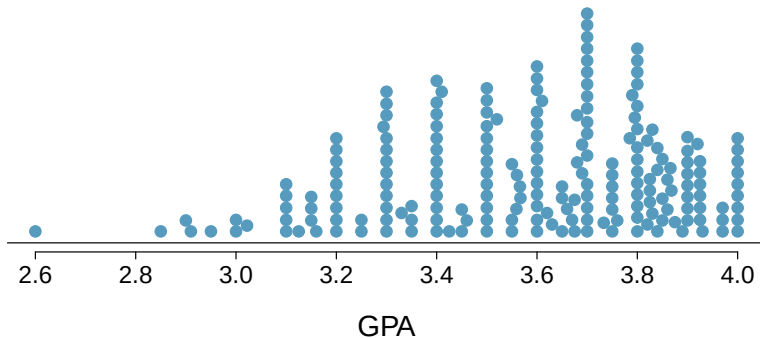


- ▶ Sometimes we indicate the mean of the data in the dot plot to give an idea about the center of the distribution.
- ▶ *Example:* The previous GPA dot plot with the mean (around 3.6).



Stacked Dot Plot

- ▶ Higher bars represent areas where there are more observations.
- ▶ Makes it a little easier to judge the center and the shape of the distribution.

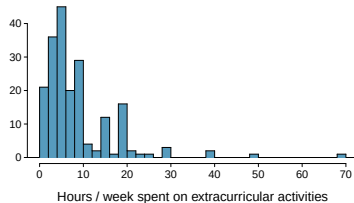
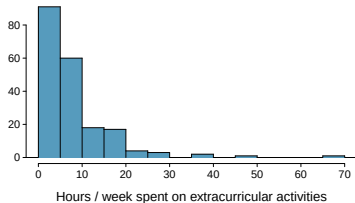
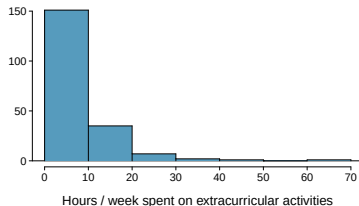
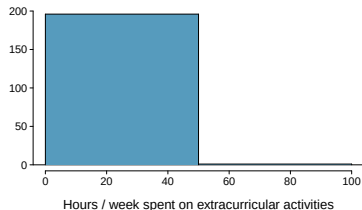


Histograms

- ▶ Useful for visualizing **one** numerical variable.
- ▶ A histogram is a plot with rectangles whose area represents the **frequency** of data and width represents consecutive, non-overlapping intervals of a variable.
- ▶ The x-axis of a histogram represents the possible values that each data point can take. The y-axis represents the frequency of each of these values.
- ▶ The horizontal range of the histogram is divided into regular sub-ranges called **bins**. The number of data points that fall into each bin is represented by the height of the corresponding rectangle.
- ▶ Histograms provide a visual summary of large amounts of data, and are particularly useful for understanding the **skewness** and the **kurtosis** of the data distribution.

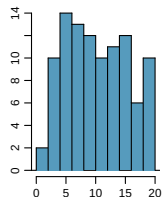
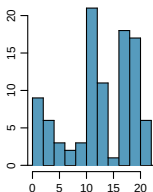
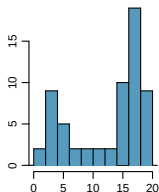
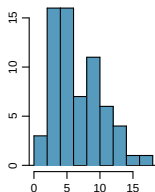
Bin width

- ▶ The same dataset can lead to very different histograms based on the chosen bin width.
- ▶ Preferred: neither too many nor too few bins.



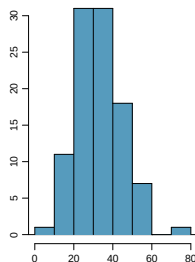
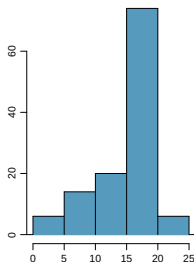
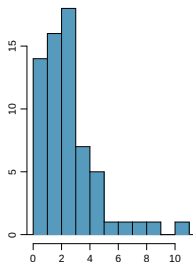
Histograms and Modality

- ▶ The number of peaks in the histogram gives insight for the modality of the underlying data.
- ▶ *Example:* Below, from left to right, we have unimodal, bimodal, multimodal, uniform.



Histograms and Skewness

- ▶ Skewness is a measure of **asymmetry**.
- ▶ A histogram can visually represent skewness.
- ▶ If the distribution of data is skewed to the left, the left tail is long and the mass of the distribution is concentrated on the right; if it's skewed to the right, the right tail is long and the mass of the distribution is concentrated on the left.



Commonly observed shapes of distributions

► modality

unimodal



bimodal



multimodal



uniform



► skewness

right skew



left skew

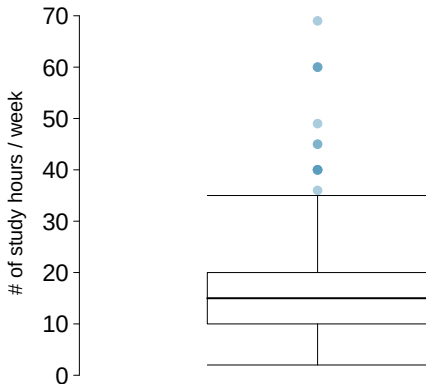


symmetric

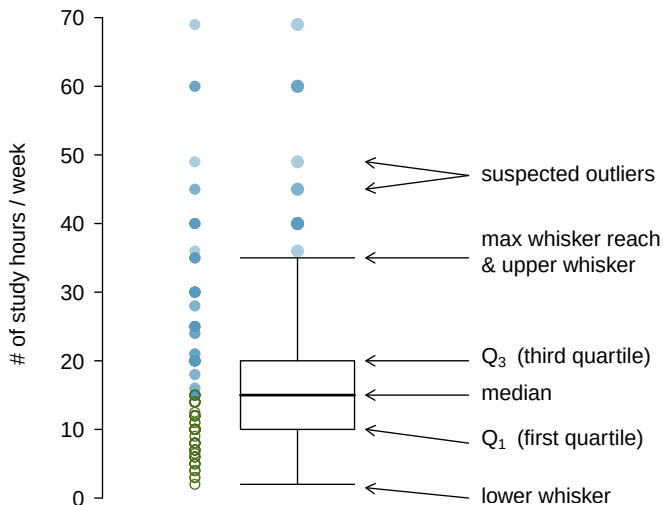


Box plot

The box in a **box plot** represents the middle 50% of the data, and the thick line in the box is the median.



Anatomy of a box plot



Whiskers and outliers

- ▶ Define the **interquartile range** to be $IQR = Q_3 - Q_1$.
- ▶ **Whiskers** of a box plot can extend up to $1.5 \times IQR$ away from the quartiles.

$$\text{max upper whisker reach} = Q_3 + 1.5 \times IQR$$

$$\text{max lower whisker reach} = Q_1 - 1.5 \times IQR$$

$$IQR : 20 - 10 = 10$$

$$\text{max upper whisker reach} = 20 + 1.5 \times 10 = 35$$

$$\text{max lower whisker reach} = 10 - 1.5 \times 10 = -5$$

- ▶ A potential **outlier** is defined as an observation beyond the maximum reach of the whiskers. It is an observation that appears extreme relative to the rest of the data.

Robust statistics

Median and IQR are more **robust** to skewness and outliers than mean and SD. Therefore,

- ▶ for skewed distributions it is often more helpful to use median and IQR to describe the center and spread
- ▶ for symmetric distributions it is often more helpful to use the mean and SD to describe the center and spread

If you would like to estimate the typical household income for a student, would you be more interested in the mean or median income?

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Median and IQR are more **robust** to skewness and outliers than mean and SD. Therefore,

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Median

Categorical Data

- ▶ **Definition:** Categorical data represent *characteristics* such as a person's gender, marital status, hometown, or the types of movies they like.
- ▶ **Types:** Two types exist - **nominal** (no order) and **ordinal** (ordered).
- ▶ **Ubiquity and Usefulness:** Categorical data is common and valuable in many fields, from market research to health science. Its ability to classify individuals or items into different groups makes it ideal for comparison, identification of patterns, and decision-making.

Visualization

5 main ways of visualizing categorical data:

- ▶ Contingency tables
- ▶ Bar plots
- ▶ Mosaic plots
- ▶ Pie charts
- ▶ Side-by-side box plots

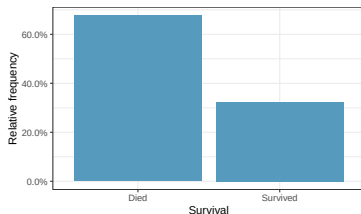
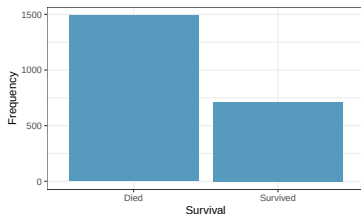
Contingency tables

- ▶ A table that summarizes data for two categorical variables is called a **contingency table**.
- ▶ The contingency table below shows the distribution of survival and ages of passengers on the Titanic.

		Survival		Total
		Died	Survived	
Age	Adult	1438	654	2092
	Child	52	57	109
	Total	1490	711	2201

Bar plots

- ▶ A **bar plot** is a common way to display a single categorical variable.
- ▶ A bar plot where proportions instead of frequencies are shown is called a **relative frequency bar plot**.

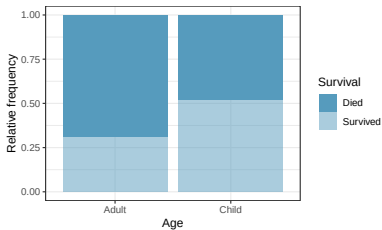
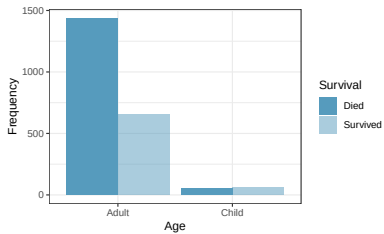
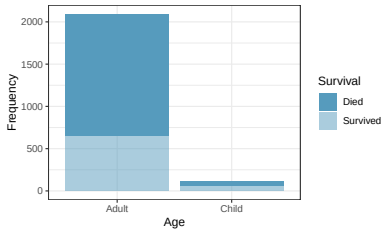


Bar plots vs. Histograms

- ▶ Bar plots are used for displaying distributions of categorical variables.
- ▶ Histograms are used for numerical variables.
- ▶ The x-axis in a histogram is a number line, hence the order of the bars cannot be changed.
- ▶ In a bar plot, the categories can be listed in any order (though some orderings make more sense than others, especially for ordinal variables.)

Bar plots with two variables

- ▶ **Stacked bar plot:** Graphical display of contingency table information, for counts.
- ▶ **Side-by-side bar plot:** Displays the same information by placing bars next to, instead of on top of, each other.
- ▶ **Standardized stacked bar plot:** Graphical display of contingency table information, for proportions.



Mosaic Plots

- ▶ **Definition:** Mosaic plots are a graphical representation of the cell frequencies of a contingency table. They divide the area into rectangular regions corresponding to categories of variables, with the area of each rectangle proportional to the cell frequencies.
- ▶ **Visualizing Relationships:** Mosaic plots are useful for visualizing the relationships between categorical variables in a dataset. The distribution of categories in one variable can be compared across levels of other variables.
- ▶ **Coloring:** Colors or shading can be used in mosaic plots to show additional dimensions, such as the outcome of a statistical test, or to highlight particular aspects of the data.

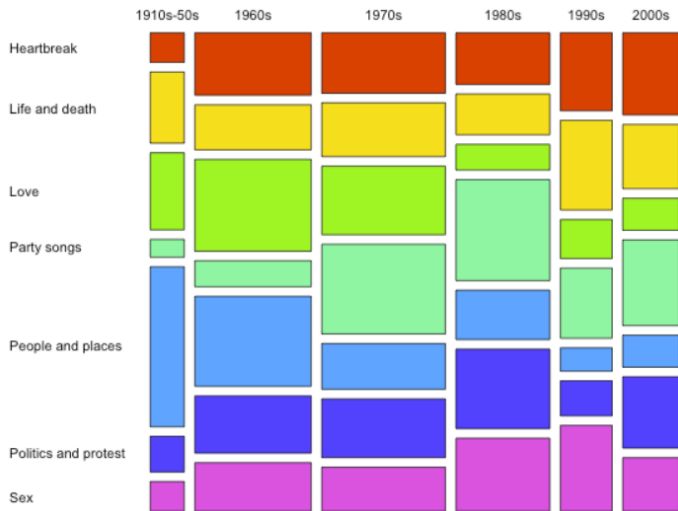


FIGURE: Source: Wikipedia

References

- ▶ Sections 2.1 & 2.2 of the textbook
- ▶ Data & Information Visualization

STOR 155 - Lecture 7

Basic Probability

Panagiotis Andreou

UNC Chapel Hill

January 24, 2025

Random Processes (informally)

- ▶ A **random process** is like a game of chance that keeps going over time. It's like rolling a dice, not just once, but repeatedly at specific intervals.
- ▶ The idea is that what happens at any given roll (or time point) doesn't necessarily depend on what happened before, and you can't predict exactly what will happen next. You might have some idea about the possibilities (like knowing the dice can land on any number from 1 to 6), but you can't say for sure.
- ▶ It's everywhere in real life, like checking the weather every day, watching the stock market, or even studying the habits of a bunch of ants!

Random Processes (more formally)

- ▶ A **random process** (also known as stochastic process) is a collection of random variables indexed by time or space. Each random variable in the collection represents an event or outcome at a particular time or location.
- ▶ The randomness implies that the outcomes of the process (events or values it produces) are not fully predictable, although they may be governed by certain probabilistic laws.
- ▶ Depending on the properties of the process, different points in time may exhibit varying degrees of dependency or independence.
- ▶ Random processes are fundamental in the study of systems that evolve over time or space under uncertainty, such as queueing systems, stock prices, signal processing, and many natural phenomena.

Examples

- ▶ **Coin Tossing:** Each toss of a fair coin is a simple random process. The outcome of each toss (head or tail) is random and does not depend on the outcome of any previous toss.
- ▶ **Temperature Measurements:** If you measure the temperature at a specific location every day at noon, the result is a random process. Each day's temperature is influenced by a variety of factors (weather, seasons, etc.), making the sequence of temperatures a random process.
- ▶ **Stock Prices:** The daily closing price of a stock is a random process. Various factors like market conditions, company performance, and global events contribute to the randomness of stock prices.

Outcome and Sample Space

- ▶ An **outcome** is a possible result of a random process or experiment.
- ▶ The **sample space** is the set of all possible outcomes of an experiment.
- ▶ *Examples:*
 - ▶ A coin toss has sample space $S = \{H, T\}$.
 - ▶ A sequence of two coin tosses has sample space $S = \{HH, HT, TH, TT\}$.
 - ▶ When rolling a die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$, each number is an outcome.

Event

- ▶ An **event** is a set of outcomes of an experiment (a subset of the sample space).
- ▶ *Examples:*
 - ▶ An event when rolling a die could be “the result is an even number”. This event includes is written $\{2, 4, 6\}$.
 - ▶ In a coin toss, “tails” is an event and we denote it as $\{T\}$.
 - ▶ In a sequence of two coin tosses, the event of getting “at least one head” is $\{HH, HT, TH\}$.
- ▶ The sample space itself is an event.

Event Operations

- ▶ The **union** of events A and B is the event that either A or B (or both) occur. It's represented as $A \cup B$.
- ▶ The **intersection** of events A and B is the event that both A and B occur. It's represented as $A \cap B$.
- ▶ The **complement** of an event A is the event that A does not occur, denoted as A^c .
- ▶ The **set difference** of two events A and B consists of all elements of A which are not in B . It is denoted by $A \setminus B$.

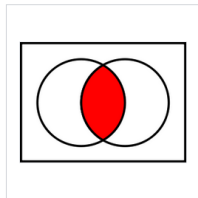
Disjoint Events

- ▶ Events A and B are **disjoint** (or mutually exclusive) if they cannot both occur at the same time. In other words, if A occurs, B cannot, and vice versa.
- ▶ For disjoint events, the intersection of A and B is the empty set, denoted as $A \cap B = \emptyset$.
- ▶ *Examples:*
 - ▶ Suppose a dice is rolled. Then, $S = \{1, 2, 3, 4, 5, 6\}$. The events $A = \{1\}$ and $B = \{3\}$ are disjoint, since we cannot get a 1 and a 3 at the same dice roll.
 - ▶ For two coin tosses, consider the events $A = \{HH, HT\}$, $B = \{HT, TH\}$, $C = \{TT\}$. Then, A and C are disjoint, B and C are disjoint, A and B are NOT disjoint.

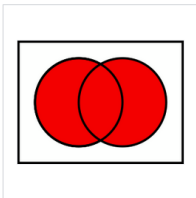
Venn Diagrams

- ▶ A **Venn diagram** is a diagrammatic representation of events using circles or other shapes.
- ▶ Each circle represents an event, and the space inside the circle represents the elements of the event.
- ▶ Overlapping regions of circles represent common elements between events.
- ▶ The sample space is usually represented by a rectangle.
- ▶ Venn diagrams are very useful for visualizing operations between events.

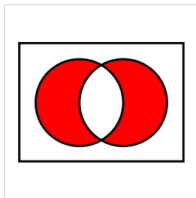
Venn Diagrams



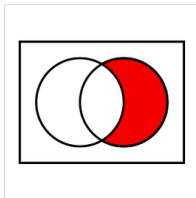
Intersection of two sets $A \cap B$



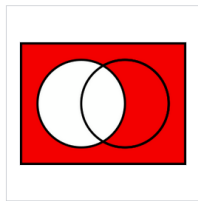
Union of two sets $A \cup B$



Symmetric difference of two sets
 $A \triangle B$



Relative complement of A (left) in B
(right) $A^c \cap B = B \setminus A$



Absolute complement of A in U
 $A^c = U \setminus A$

FIGURE: Source: Wikipedia

A cool Venn diagram



FIGURE: The uppercase letters shared by the Latin, Russian and Greek alphabets (source: Wikipedia)

Example: Dice Rolling

Problem: Two six-sided dice are rolled.

- ▶ Event D : The sum of the numbers shown is 7.
- ▶ Event E : The first die shows a 4.

Questions:

1. What is the sample space S for this experiment?
2. What is $D \cap E$? Interpret its meaning.
3. If Event F is that the second die shows an even number, what is $D \cap F$?
4. What is $E \setminus D$? Interpret its meaning.

Solution

1. $S = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}.$
2. $D \cap E = \{(4, 3)\}.$ The first die shows a 4 and the sum is 7.
3. $D \cap F = \{(1, 6), (3, 4), (5, 2)\}.$ The sum is 7 and the second die shows an even number.
4. $E \setminus D = \{(4, 1), (4, 2), (4, 4), (4, 5), (4, 6)\}.$ The first die shows a 4 but the sum is not 7.

Example: Selecting Students

Problem: A class of 30 students is composed of 18 males and 12 females. Out of these, 10 males and 7 females have brown eyes.

- ▶ Event G : A randomly selected student is male.
- ▶ Event H : A randomly selected student has brown eyes.
- ▶ Event I : A randomly selected student is female.

Questions:

1. What does the event $G \cap H$ represent? How many elements does it have?
2. What does the event G^c represent? How many elements does it have?
3. What does the event $H \setminus I$ represent, and how many elements does it have?

Solution

Solutions:

1. $G \cap H$ represents males with brown eyes, which are 10.
2. G^c represents all females, which are 12.
3. $H \setminus I$ represents all male students with brown eyes, which are $17 - 12 = 5$.

Example: Music Preferences

Problem: A survey of 50 people asked about their preference for Jazz and Rock music.

- ▶ 20 people like Jazz
- ▶ 25 people like Rock
- ▶ 15 people like both Jazz and Rock
- ▶ Event J : A person likes Jazz.
- ▶ Event K : A person likes Rock.

Questions:

1. How many people do not like Jazz or Rock?
2. What is $J \cup K$? How many elements does it have?
3. How many people only like Rock music?
4. If Event L is that a person likes neither Jazz nor Rock, what is $J^c \cap K^c$ and how many elements does it have?

Solution

1. Total people liking Jazz or Rock or both = $20 + 25 - 15 = 30$. So, $50 - 30 = 20$ people do not like either.
2. $|J \cup K| = 30$ people like either Jazz, Rock, or both.
3. $|J \setminus K| = |J| - |J \cap K| = 25 - 15 = 10$ (in words, we are saying "Only Rock = Total liking Rock - Both liking Jazz and Rock = $25 - 15 = 10$ ".)
4. $|J^c \cap K^c| = 20$ people like neither Jazz nor Rock.

References

- ▶ Section 3.1 of the textbook
- ▶ Venn Diagram

STOR 155 - Lecture 8

Basic Probability

Panagiotis Andreou

UNC Chapel Hill

January 27, 2025

Probability (informally)

- ▶ Informally, the probability of an event is the chance of occurrence of the event.
- ▶ **Frequentist interpretation:** The probability of an outcome is the proportion of times the outcome would occur if we observed the random process an infinite number of times.
- ▶ **Bayesian interpretation:** A Bayesian interprets probability as a subjective degree of belief: for the same event, two separate people could have different viewpoints and so assign different probabilities.

Probability (formally)

- ▶ We denote the probability of an event A by $P(A)$.
- ▶ S denotes the sample space
- ▶ **Axioms** of probability:
 1. $0 \leq P(A) \leq 1$
 2. $P(S) = 1$
 3. **Union Law:** If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

Two useful rules

- ▶ General Union Law:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- ▶ Probability of the complement:

$$P(A^c) = 1 - P(A)$$

Independence (informally)

- ▶ Two events A and B are independent if the occurrence of one does not give any information on the occurrence of the other.
- ▶ *Example:* If we toss a coin twice under the same conditions, the outcome of each coin (H or T) says nothing about the outcome of the other coin.

Independence (formally)

- ▶ Two events A and B are independent, and we write $A \perp\!\!\!\perp B$, if

$$P(A \cap B) = P(A)P(B).$$

- ▶ *Example:* We toss a coin twice. Consider the events $A = \{H \text{ in the first toss}\}$ and $B = \{T \text{ in the second toss}\}$. Then,

$$\begin{aligned} P(H \text{ in first}, T \text{ in second}) &= \frac{1}{4} \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= P(H \text{ in first}) \cdot P(T \text{ in second}). \end{aligned}$$

Hence, $A \perp\!\!\!\perp B$.

Independent $\nRightarrow$ Disjoint

- ▶ One might think that the concepts “independent” and “mutually exclusive (disjoint)” are similar, or even equivalent. This is wrong!
- ▶ **Independent:** the occurrence of A does not affect the occurrence of B , and vice versa. Mathematically,

$$P(A \cap B) = P(A)P(B).$$

Example: In tossing a coin twice, the outcomes of the first and the second toss define independent events.

- ▶ **Mutually exclusive:** the events A and B cannot happen simultaneously. Mathematically,

$$P(A \cap B) = 0.$$

Example: In tossing a coin twice, the events that we get tails twice and that we get heads twice are mutually exclusive.

Law of Large Numbers

- ▶ As more observations are collected, the proportion of occurrences with a particular outcome, $\hat{p}_n$, converges to the probability of that outcome, p .
- ▶ *Examples:*
 1. If we keep tossing a coin, the proportion of times we observe Tails approximates $1/2$.
 2. If we keep throwing a die, the proportion of times we see the number 1 approximates $1/6$.

Probability distributions

A **probability distribution** lists all possible events and the probabilities with which they occur.

- ▶ The probability distribution for the gender of one kid:

Event	Male	Female
Probability	0.5	0.5

Probability distributions

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- ▶ Rules for probability distributions:
 1. The events listed must be disjoint
 2. Each probability must be between 0 and 1
 3. The probabilities must total 1

Probability distributions

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- ▶ The probability distribution for the gender of one kid:

Event	Male	Female
Probability	0.5	0.5

- ▶ Rules for probability distributions:

1. The events listed must be disjoint
2. Each probability must be between 0 and 1
3. The probabilities must total 1

- ▶ The probability distribution for the genders of two kids:

Event	MM	FF	MF	FM
Probability	0.25	0.25	0.25	0.25

Example 1: Tossing a Coin

Problem: You toss a fair coin. What is the probability of getting a heads (H)?

Solution:

- ▶ Sample space $S = \{H, T\}$
- ▶ The event $A = \text{Getting a heads} = \{H\}$
- ▶ Since the coin is fair, $P(H) = \frac{1}{2}$

Example 2: Rolling a Die

Problem: You roll a fair six-sided die. What is the probability of getting an even number?

Solution:

- ▶ Sample space $S = \{1, 2, 3, 4, 5, 6\}$
- ▶ The event $B = \text{Getting an even number} = \{2, 4, 6\}$
- ▶ $P(B) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}} = \frac{3}{6} = \frac{1}{2}$

Deck of Cards

- ▶ 52 cards
- ▶ Four suits: Clubs, Diamonds, Hearts, Spades
- ▶ Each suit has cards of 13 ranks: Ace, 2,3,4,5,6,7,8,9,10,Jack, Queen, King.
- ▶ Clubs and Spades are black suits; Diamonds and Hearts are red suits.
- ▶ The above means: there are 13 cards of each suit, 4 cards of each rank, 26 cards of each color.

	Ace	2	3	4	5	6	7	8	9	10	Jack	Queen	King
Clubs													
Diamonds													
Hearts													
Spades													

Example 3: Drawing a Card

Problem: From a standard deck of cards, you draw a card. What is the probability of getting a Queen or a King?

Solution:

- ▶ Sample space $S = 52$ cards
- ▶ Event $C =$ Getting a Queen $= 4$ cards
- ▶ Event $D =$ Getting a King $= 4$ cards
- ▶ $P(C \cup D) = P(C) + P(D)$ [Using the Union Law]
- ▶ $P(C \cup D) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$

Example 4: Independence

Problem: You toss a coin and roll a die. What is the probability of getting a tails (T) and a 5?

Solution:

- ▶ Probability of getting tails $P(T) = \frac{1}{2}$
- ▶ Probability of getting a 5 on the die $P(5) = \frac{1}{6}$
- ▶ Since the two events are independent:

$$P(T \cap 5) = P(T) \cdot P(5) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

Example 5

Problem: You flip a coin 6 times. What is the probability that we get at least one H?

Solution: Call A the event that we get at least one H. The idea is to think of the complement:

$$P(A) = 1 - P(A^c) = 1 - (1/2)^6 = \frac{63}{64}.$$

Example 6: Law of Large Numbers

Problem: Imagine tossing a coin 10 times, 100 times, and 1000 times. What can we expect about the proportion of heads?

Solution:

- ▶ As the number of tosses increases, the proportion of heads should converge to the true probability of getting a head: $\frac{1}{2}$.
- ▶ For 10 tosses, you might see 7 heads or 3 tails, but for 1000 tosses, it's more likely you'll see closer to 500 heads.

References

- ▶ Section 3.1 of the textbook
- ▶ Independence

STOR 155 - Lecture 9

Conditional Probability

Panagiotis Andreou

UNC Chapel Hill

January 29, 2025

Motivation

- ▶ In real-life situations, the probability of an event can often depend on whether another event has occurred. This concept is known as **Conditional Probability**.
- ▶ *Examples:*
 1. Consider a medical test for a disease. The probability that a person has the disease, **given** a positive test result, depends on the overall prevalence of the disease in the population.
 2. Another example is weather prediction. The probability of rain tomorrow may depend on today's weather. If it's cloudy today, the chance of rain tomorrow could be higher.
- ▶ Conditional probability is a powerful tool that allows us to **update** probabilities based on new information.

Example

- ▶ Researchers randomly assigned 72 chronic users of cocaine into three groups: desipramine (antidepressant), lithium (standard treatment for cocaine) and placebo. Results of the study are summarized below.

	relapse	no relapse	total
desipramine	10	14	24
lithium	18	6	24
placebo	20	4	24
total	48	24	72

- ▶ What is the probability that a patient relapsed?
- ▶ What is the probability that a patient took lithium?
- ▶ What is the probability that a patient relapsed given that they took placebo?

- ▶ Last row, 1st column:

$$P(\text{relapse}) = \frac{48}{72}$$

- ▶ 2nd row, last column:

$$P(\text{lithium}) = \frac{24}{72}$$

- ▶ 3rd row, 1st column:

$$P(\text{relapse}|\text{placebo}) = \frac{20}{24}$$

Note that

$$P(\text{relapse}|\text{placebo}) = \frac{P(\text{relapse} \cap \text{placebo})}{P(\text{placebo})}$$

Conditional Probability

- ▶ **Definition:** the conditional probability of an event A given an event B is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- ▶ Essentially, by conditioning on the event B , we **shrink** the sample space.
- ▶ Note: in general, $P(A|B) \neq P(B|A)$.
- ▶ Multiplication Rules:

$$P(A \cap B) = P(A|B)P(B) \quad \text{and} \quad P(A \cap B) = P(B|A)P(A)$$

Understanding Conditional Probability

- ▶ $P(A \cap B)$ means the probability of both A and B occurring.
- ▶ $P(B)$ is the probability of event B .
- ▶ $P(A|B)$ is the proportion of times A occurs when B has already occurred.
- ▶ The conditional probability formula informally reads:
“to find the probability of A given B , check what is the chance of B happening, and for each such occurrence what is the chance of A happening”.

Conditional Probability and Independence

- ▶ Recall: A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

- ▶ *Alternative formulation:* If A and B are independent, then $P(A|B) = P(A)$ and $P(B|A) = P(B)$.
- ▶ *Intuition:* the occurrence of B doesn't affect the probability of A , and vice-versa.
- ▶ If $P(A|B) \neq P(A)$, A and B are not independent.

Motivating Bayes' Theorem: A Medical Example

- ▶ Bayes' Theorem is a fundamental concept in probability that allows us to update our beliefs based on new evidence.
- ▶ Consider a medical test for a disease that affects 1% of the population. The test has a 5% false positive rate and no false negatives.
- ▶ You tested positive. What's the probability you have the disease? It's not 95%!
- ▶ Bayes' Theorem allows us to calculate this probability, incorporating both the test's false positive rate and the prevalence of the disease in the population.

Bayes' Theorem

- ▶ The conditional probability formula we have seen so far is a special case of the Bayes' Theorem, which is applicable even when events have more than just two outcomes.
- ▶ Suppose a variable has k possible outcomes, $A_1, \dots, A_k$. Then, given that an event B has occurred, the conditional probability of A_1 is given by

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + \dots + P(A_k)P(B|A_k)}$$

Example

- ▶ A disease affects 1% of the population (Event D).
- ▶ A test is 99% accurate: it correctly identifies a sick person 99% of the time (true positive), and a healthy person 99% of the time (true negative).
- ▶ However, the test has a 1% false positive rate and a 1% false negative rate.
- ▶ You take the test and get a positive result (Event T). What's the probability you have the disease?

Using Bayes' Theorem

- ▶ We want to find $P(D|T)$, the probability of having the disease given a positive test result.
- ▶ Bayes' Theorem states: $P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T)}$.
- ▶ We know $P(T|D)$ (the probability of a positive test given the disease) is 0.99.
- ▶ $P(D)$ (the probability of the disease) is 0.01.
- ▶ The hard part is usually the computation of the denominator $P(T)$.

Calculating the Answer

- ▶ $P(T)$ (the probability of a positive test) is not just the false positive rate. It's the probability of a true positive or a false positive:

$$\begin{aligned}P(T) &= P(T \cap D) + P(T \cap D^c) \\ &= P(T|D)P(D) + P(T|D^c)P(D^c)\end{aligned}$$

- ▶ This gives us $P(T) = 0.99 \cdot 0.01 + 0.01 \cdot 0.99 = 2 \cdot 0.99 \cdot 0.01$.
- ▶ Plugging into Bayes' Theorem: $P(D|T) = \frac{0.99 \cdot 0.01}{2 \cdot 0.99 \cdot 0.01} = 0.5$.
- ▶ Even with a positive test, there's only a 50% chance you have the disease!

Example

Suppose a student is enrolled in a particularly challenging statistics class. Based on records from previous years:

- ▶ 70% of students who study regularly for the class pass the midterm.
- ▶ 20% of students who do not study regularly for the class pass the midterm.
- ▶ Overall, 50% of students study regularly for the class.

If a randomly selected student passes the midterm, what is the probability that they studied regularly for the class?

Solution

Let:

- ▶ R be the event that a student studies regularly for the class.
- ▶ P be the event that a student passes the midterm.

Given:

- ▶ Probability that a student passes given they study regularly:

$$P(P|R) = 0.7$$

- ▶ Probability that a student passes given they don't study regularly:

$$P(P|R^c) = 0.2$$

- ▶ Probability that a student studies regularly:

$$P(R) = 0.5$$

We want to find $P(R|P)$ (Probability that a student studied regularly given they passed). Bayes' theorem:

$$P(R|P) = \frac{P(P|R) \cdot P(R)}{P(P)},$$

where

$$P(P) = P(R) \cdot P(P|R) + P(R^c) \cdot P(P|R^c).$$

Now, calculating step by step:

- ▶ $P(R^c) = 1 - P(R) = 0.5$ (Probability that a student doesn't study regularly)
- ▶ $P(P) = 0.5 \cdot 0.7 + 0.5 \cdot 0.2 = 0.45$
- ▶ $P(R|P) = \frac{0.7 \cdot 0.5}{0.45} = 0.7778$

Thus, if a student passes the midterm, the probability that they studied regularly is 77.78%.

References

- ▶ Section 3.2 of the textbook
- ▶ Bayes theorem, the geometry of changing beliefs

STOR 155 - Lecture 10

Random Variables

Panagiotis Andreou

UNC Chapel Hill

January 31, 2025

Promising beginning

“The great thing about random variables is that they’re neither random nor variables.”

Introduction to Random Variables

- ▶ We are often interested in numerical outcomes of random processes.
- ▶ For instance: The number of heads in 10 coin tosses.
- ▶ A **random variable** is a function that assigns a real number to each outcome in a sample space.
- ▶ This turns every possible outcome of a random process into a numerical value.
- ▶ We use
 - ▶ a capital letter X to denote a random variable
 - ▶ a lower-case letter x to denote a value of the random variable
 - ▶ we write $P(X = x)$ to denote the probability of a random variable X taking the value x

Examples of Discrete Random Variables

- ▶ A Discrete Random Variable has a countable number of possible values.
- ▶ Examples:
 - ▶ The number of heads in 10 coin tosses.
 - ▶ The number of defective items in a batch.
 - ▶ The number of cars arriving at a toll booth in one hour.
- ▶ These random variables can take on a finite or countably infinite number of values.

Examples of Continuous Random Variables

- ▶ A Continuous Random Variable can take on any value in an interval.
- ▶ Examples:
 - ▶ The amount of rain in a city over a year.
 - ▶ The time taken to run a marathon.
 - ▶ The weight of a randomly chosen adult.
- ▶ These random variables have uncountably many possible values.

Expectation

- ▶ Suppose X is a discrete random variable taking values in the set $\{x_1, \dots, x_k\}$.
- ▶ We are often interested in the average outcome of a random variable.
- ▶ We call this the **expected value** (mean), and it is a weighted average of the possible outcomes

$$\mu = E(X) = \sum_{i=1}^k x_i P(X = x_i)$$

Example

In a game of cards you win \$1 if you draw a heart, \$5 if you draw an ace (including the ace of hearts), \$10 if you draw the king of spades and nothing for any other card you draw. Write the probability model for your winnings, and calculate your expected winning.

Example

In a game of cards you win \$1 if you draw a heart, \$5 if you draw an ace (including the ace of hearts), \$10 if you draw the king of spades and nothing for any other card you draw. Write the probability model for your winnings, and calculate your expected winning.

Event	X	$P(X)$	$X P(X)$
Heart (not ace)	1	$\frac{12}{52}$	$\frac{12}{52}$
Ace	5	$\frac{4}{52}$	$\frac{20}{52}$
King of spades	10	$\frac{1}{52}$	$\frac{10}{52}$
All else	0	$\frac{35}{52}$	0
Total			$E(X) = \frac{42}{52} \approx 0.81$

Variance and Standard Deviation

- ▶ Suppose X is a discrete random variable taking values in the set $\{x_1, \dots, x_k\}$.
- ▶ We are also often interested in the variability in the values of a random variable.
- ▶ **Variance:**

$$\sigma^2 = \text{Var}(X) = \sum_{i=1}^k (x_i - E(X))^2 P(X = x_i)$$

- ▶ **Standard Deviation:**

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$$

Example

X	$P(X)$	$X P(X)$	$(X - E(X))^2$	$P(X) (X - E(X))^2$
1	$\frac{12}{52}$	$1 \cdot \frac{12}{52} = \frac{12}{52}$	$(1 - 0.81)^2 = 0.0361$	$\frac{12}{52} \cdot 0.0361 = 0.0083$
5	$\frac{4}{52}$	$5 \cdot \frac{4}{52} = \frac{20}{52}$	$(5 - 0.81)^2 = 17.5561$	$\frac{4}{52} \cdot 17.5561 = 1.3505$
10	$\frac{1}{52}$	$10 \cdot \frac{1}{52} = \frac{10}{52}$	$(10 - 0.81)^2 = 84.4561$	$\frac{1}{52} \cdot 84.4561 = 1.6242$
0	$\frac{35}{52}$	$0 \cdot \frac{35}{52} = 0$	$(0 - 0.81)^2 = 0.6561$	$\frac{35}{52} \cdot 0.6561 = 0.4416$
		$E(X) = 0.81$		

Example

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Example

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		$E(X) = 0.81$		$V(X) = 3.4246$

Example

X	$P(X)$	$X P(X)$	$(X - E(X))^2$	$P(X) (X - E(X))^2$
1	$\frac{12}{52}$	$1 \cdot \frac{12}{52} = \frac{12}{52}$	$(1 - 0.81)^2 = 0.0361$	$\frac{12}{52} \cdot 0.0361 = 0.0083$
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0	$\frac{35}{52}$	$0 \cdot \frac{35}{52} = 0$	$(0 - 0.81)^2 = 0.6561$	$\frac{35}{52} \cdot 0.6561 = 0.4416$
		$E(X) = 0.81$		$V(X) = 3.4246$ $SD(X) = \sqrt{3.4246} = 1.85$

Linear combinations

- ▶ A **linear combination** of random variables X and Y is given by

$$aX + bY,$$

where a and b are some fixed numbers.

- ▶ The expected value of a linear combination of random variables is given by

$$E(aX + bY) = aE(X) + bE(Y)$$

Example 1

- ▶ On average you take 10 minutes for each statistics homework problem and 15 minutes for each chemistry homework problem. This week you have 5 statistics and 4 chemistry homework problems assigned. What is the total time you expect to spend on statistics and chemistry homework for the week?

Example 1

- ▶ On average you take 10 minutes for each statistics homework problem and 15 minutes for each chemistry homework problem. This week you have 5 statistics and 4 chemistry homework problems assigned. What is the total time you expect to spend on statistics and chemistry homework for the week?
- ▶ We can write

$$\begin{aligned} E(S + S + S + S + S + C + C + C + C) &= 5 \cdot E(S) + 4 \cdot E(C) \\ &= 5 \cdot 10 + 4 \cdot 15 \\ &= 50 + 60 \\ &= 110 \text{ min} \end{aligned}$$

Example 2

- ▶ Let X and Y be two independent random variables, with $E[X] = 2$, $Var(X) = 4$, $E[Y] = 3$ and $Var(Y) = 9$.
- ▶ Consider a new random variable $Z = 3X - 2Y$.
- ▶ Find the expected value and standard deviation of Z .

Solution

- ▶ The expected value of Z is given by
$$E[Z] = 3E[X] - 2E[Y] = 3 \cdot 2 - 2 \cdot 3 = 0.$$
- ▶ To find the standard deviation of Z , we first compute its variance.
- ▶ The variance of Z is given by
$$\text{Var}(Z) = 3^2 \cdot \text{Var}(X) + (-2)^2 \cdot \text{Var}(Y) = 9 \cdot 4 + 4 \cdot 9 = 72.$$
- ▶ The standard deviation of Z is the square root of its variance, which equals $\sqrt{72} \approx 8.49$.

References

- ▶ Section 3.4 of the textbook
- ▶ Random Variables

STOR 155 - Lecture 11

Normal Distribution

Panagiotis Andreou

UNC Chapel Hill

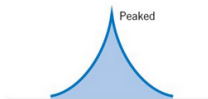
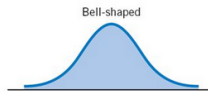
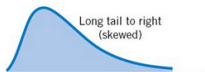
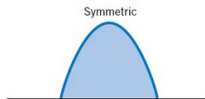
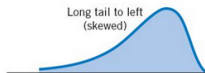
February 3, 2025

Introduction to Continuous Distributions

- ▶ A continuous distribution describes the probabilities of the possible values of a continuous random variable.
- ▶ A **continuous** random variable is a random variable with a set of possible values (known as the range) that is infinite and **uncountable**.
- ▶ Examples include measurements like weight, height, and temperature.

Density Curves

- ▶ A density curve is a graph that shows the probability of a given continuous outcome.
- ▶ The area under a density curve over an interval represents the probability that the variable falls within that interval.
- ▶ Density curves are always above the horizontal axis, and the total area under the curve is equal to 1.



The Normal Distribution

- ▶ The most famous continuous distribution is the **Normal** (or Gaussian) **distribution**.
- ▶ It is characterized by its **bell-shaped** density curve.
- ▶ The curve is **symmetrical** around the mean, reflecting the fact that data near the mean are more frequent in occurrence than data far from the mean.

Parameters of the Normal Distribution

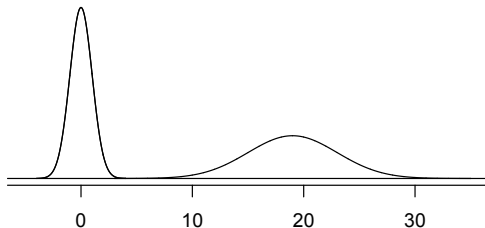
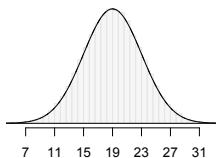
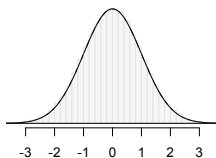
- ▶ The Normal distribution is defined by two parameters: the **mean** (μ) and the **standard deviation** (σ).
- ▶ The mean determines the location of the center of the graph, and the standard deviation determines the height and width of the graph.
- ▶ When $\mu = 0$ and $\sigma = 1$, it is known as the **Standard Normal Distribution**.

Normal distributions with different parameters

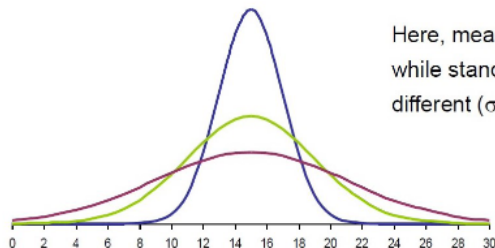
μ : mean, σ : standard deviation

$$N(\mu = 0, \sigma = 1)$$

$$N(\mu = 19, \sigma = 4)$$

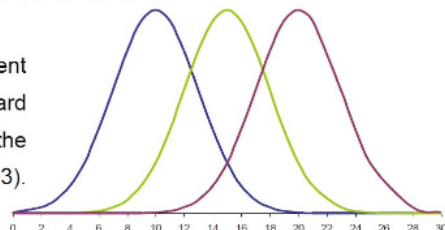


Normal Curves



Here, means are the same ($\mu = 15$) while standard deviations are different ($\sigma = 2, 4$, and 6).

Here, means are different ($\mu = 10, 15$, and 20) while standard deviations are the same ($\sigma = 3$).



Standardizing with Z-scores

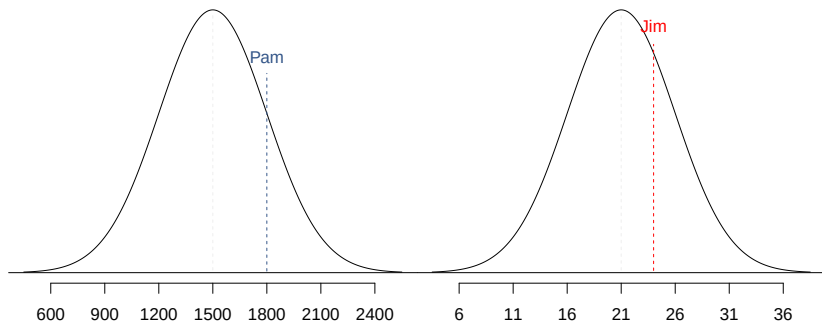
- ▶ **Z-score** of an observation is the number of standard deviations it falls above or below the mean.

$$Z = \frac{\text{observation} - \text{mean}}{SD}$$

- ▶ Z-scores are defined for distributions of any shape, but only when the distribution is normal can we use Z scores to calculate percentiles.
- ▶ Observations that are more than 2 SD away from the mean ($|Z| > 2$) are usually considered unusual.

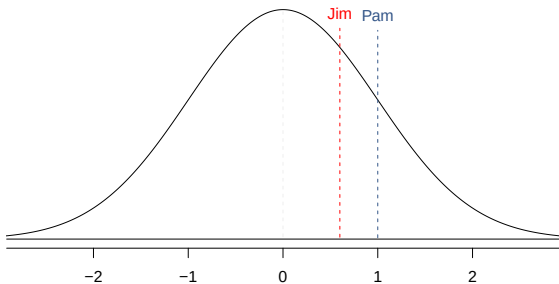
Example

SAT scores are distributed nearly normally with mean 1500 and standard deviation 300. ACT scores are distributed nearly normally with mean 21 and standard deviation 5. A college admissions officer wants to determine which of the two applicants scored better on their standardized test with respect to the other test takers: Pam, who earned an 1800 on her SAT, or Jim, who scored a 24 on his ACT?



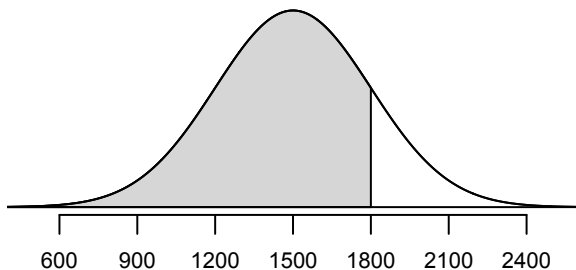
Since we cannot just compare these two raw scores, we instead compare how many standard deviations beyond the mean each observation is.

- ▶ Pam's score is $\frac{1800-1500}{300} = 1$ standard deviation above the mean.
- ▶ Jim's score is $\frac{24-21}{5} = 0.6$ standard deviations above the mean.



Percentiles

- ▶ **Percentile** is the percentage of observations that fall below a given data point.
- ▶ Graphically, percentile is the area below the probability distribution curve to the **left** of that observation.



Z-table

Z		Second decimal place of Z									
		0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	

Understanding Percentiles

- ▶ A percentile is the proportion of values **below** a given value in a distribution.
- ▶ *Example:* A score at the 90th percentile means **90% of values are lower**.
- ▶ **Graphically**, percentiles are the area under the density curve **to the left** of a value.
- ▶ Percentiles are useful for comparing scores across different distributions.

Finding Percentiles Using the Z-Score

- ▶ To find the percentile of a value, first compute its **Z-score**:

$$Z = \frac{x - \mu}{\sigma}$$

- ▶ Use the Z-table (or a statistical tool) to find the probability (percentile) corresponding to Z .
- ▶ *Example:* If $X = 1800$, $\mu = 1500$, and $\sigma = 300$:

$$Z = \frac{1800 - 1500}{300} = 1$$

- ▶ From the Z-table, $P(Z \leq 1) \approx 0.8413$, meaning $X = 1800$ is at the **84.13th percentile**.

Excel Function: NORM.DIST

- ▶ `NORM.DIST(x, mean, standard_dev, cumulative)`
- ▶ Returns the probability of observing $X \leq x$ (cumulative) or the probability density (non-cumulative).
- ▶ If `cumulative` is `TRUE`, it returns the area under the normal curve to the left of x .
- ▶ *Example:*
 - ▶ `=NORM.DIST(1800, 1500, 300, TRUE)` returns 0.8413.

Excel Function: NORM.INV

- ▶ `NORM.INV(probability, mean, standard_dev)`
- ▶ Finds the value of X given a percentile (inverse normal).
- ▶ Example:
 - ▶ `=NORM.INV(0.9, 1500, 300)` returns 1884, meaning the 90th percentile SAT score is 1884.

Practice Problems

- ▶ **Problem 1:** SAT scores are normally distributed with $\mu = 1500$ and $\sigma = 300$.
 - ▶ What percentile corresponds to a score of 2000?
 - ▶ What is the SAT score at the 95th percentile?
- ▶ **Problem 2:** Heights of adult males are normally distributed with $\mu = 70$ inches and $\sigma = 3$ inches.
 - ▶ What proportion of men are taller than 73 inches?
 - ▶ What height corresponds to the 75th percentile?

The 68-95-99.7 rule

- ▶ Approximately 68% of the data falls within one standard deviation from the mean on either side.
- ▶ Approximately 95% falls within two standard deviations.
- ▶ Almost all (about 99.7%) of the data falls within three standard deviations from the mean.

Applications of the Normal Distribution

- ▶ The Normal distribution is widely used in the natural and social sciences to represent real-valued random variables whose distributions are not known.
- ▶ Its importance is largely due to the **Central Limit Theorem**.
- ▶ The Central Limit Theorem (CLT) is a fundamental concept in Statistics. It tells us that if we have a large enough sample size, the distribution of the sample means will approach a Normal distribution, regardless of the shape of the population distribution.

References

- ▶ Section 4.1 of the textbook
- ▶ Video from the (amazing) channel *StatQuest*:

The Normal Distribution, Clearly Explained!

STOR 155 - Lecture 12

Geometric Distribution

Panagiotis Andreou

UNC Chapel Hill

February 5, 2025

Motivation

- ▶ Imagine a series of repeated trials, like flipping a coin. What if we're interested in how many trials it takes until we get our first success?
- ▶ This is exactly what the **geometric distribution** models: the *number of trials needed to get the first success in repeated Bernoulli trials*.

Bernoulli Trials

- ▶ A **Bernoulli trial** is an experiment that results in a success with probability p and failure with probability $1 - p$.
- ▶ Each trial is independent, meaning the outcome of one trial does not affect the outcome of another.
- ▶ *Examples:*
 - ▶ Toss of fair coin: success probability = 0.5, where we have to determine whether success means H or T.
 - ▶ Toss of biased coin: success probability could be any number in $[0, 1]$.

PMF of the Geometric Distribution

- ▶ If X is a geometrically distributed random variable with probability p , we write $X \sim \text{Geom}(p)$.
- ▶ If $X \sim \text{Geom}(p)$, its probability mass function (PMF) is given by

$$P(X = k) = (1 - p)^{k-1}p$$

for $k = 1, 2, 3, \dots$

- ▶ This formula gives the probability that we need k trials to get the first success.

Mean and Variance

If $X \sim \text{Geom}(p)$, then

- ▶ its expectation is given by

$$E[X] = \frac{1}{p}$$

- ▶ its variance is given by

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Derivation

$$\begin{aligned} E[X] &= \sum_{k=1}^{\infty} k \cdot P(X = k) \\ &= \sum_{k=1}^{\infty} k \cdot (1 - p)^{k-1} \cdot p \\ &= p \cdot \sum_{k=1}^{\infty} k \cdot (1 - p)^{k-1} \\ &= p \cdot (1 + 2(1 - p) + 3(1 - p)^2 + 4(1 - p)^3 + \dots) \\ &= p \cdot \frac{1}{p^2} \quad (\text{calculus fact}) \\ &= \frac{1}{p} \end{aligned}$$

Problem 1

- ▶ Consider flipping a biased coin that comes up heads (success) with probability $p = 0.3$.
- ▶ What is the probability that the first head occurs on the third flip?

Solution

- ▶ We can use the PMF of the geometric distribution to solve this.
- ▶ $P(X = 3) = (1 - 0.3)^{3-1} \cdot 0.3 \approx 0.147$
- ▶ So, the probability that the first head occurs on the third flip is approximately 0.147.

Problem 2 - Binge Watching Dilemma

A popular streaming service claims that a user will find a new show they like 80% of the time. Assuming each suggestion is independent of the others:

- ▶ (a) What is the probability that a user will find a show they like on the k^{th} suggestion?
- ▶ (b) On average, how many suggestions does it take for a user to find a show they like?
- ▶ (c) What is the variability in the number of suggestions?

Solution

- ▶ Let Y represent the number of suggestions it takes until the user finds a show they like.
- ▶ $Y \sim \text{Geom}(q)$, where $q = 0.8$ (probability of liking a show from a suggestion).
- ▶ (a) The pmf of Y is $P(Y = k) = (1 - q)^{k-1}q$, for $k = 1, 2, \dots$
- ▶ For this problem, $q = 0.8$, so
$$P(Y = k) = (1 - 0.8)^{k-1} \cdot 0.8 = 0.2^{k-1} \cdot 0.8.$$

- ▶ (b) The expected (mean) number of suggestions from a geometric distribution is $E[Y] = \frac{1}{q}$.
- ▶ For our problem, $E[Y] = \frac{1}{0.8} = 1.25$. So, on average, it takes about 1.25 suggestions for a user to find a show they like.
- ▶ (c) The variance of a geometric distribution is $Var(Y) = \frac{1-q}{q^2}$.
- ▶ In our context, $Var(Y) = \frac{1-0.8}{(0.8)^2} = \frac{0.2}{0.64} \approx 0.3125$. So, there's a variance of approximately 0.3125 in the number of suggestions.

Problem 3 - Malware detection

A software company develops a new antivirus software. They believe that each individual attempt to detect a certain type of malware with their software is independent and has a success probability of 0.7.

- ▶ (a) What is the probability of needing k attempts for a malware detection?
- ▶ (b) What is the expected number of attempts until detection?
- ▶ (c) What is the variance?

Solution

- ▶ Let X represent the number of attempts it takes until the software detects this specific malware.
- ▶ $X \sim \text{Geom}(p)$, with $p = 0.7$
- ▶ (a) The pmf of X is $P(X = k) = (1 - p)^{k-1}p$, for $k = 0, 1, 2, \dots$
- ▶ In this case, $p = 0.7$, so
$$P(X = k) = (1 - 0.7)^{k-1} \cdot 0.7 = 0.3^{k-1} \cdot 0.7.$$

- ▶ (b) The expectation (mean) of a geometric distribution is $E[X] = \frac{1}{p}$.
- ▶ In this case, $E[X] = \frac{1}{0.7} \approx 1.43$. So, on average, it takes about 1.43 attempts to detect the malware.
- ▶ (c) The variance of a geometric distribution is $Var(X) = \frac{1-p}{p^2}$.
- ▶ In this case, $Var(X) = \frac{1-0.7}{(0.7)^2} = \frac{0.3}{0.49} \approx 0.61$. So, the variance in the number of attempts is approximately 0.61.

References

- ▶ Section 4.2 of the textbook
- ▶ Geometric distribution (Khan Academy)

STOR 155 - Lecture 13

Binomial Distribution

Panagiotis Andreou

UNC Chapel Hill

February 7, 2025

Binomial Distribution: Introduction

- ▶ A binomial distribution is a probability distribution that describes the number of successes in a fixed number of Bernoulli trials.
- ▶ Let's denote X as a random variable that counts the number of successes in n Bernoulli trials.
- ▶ Each trial is independent, and the probability of success, denoted by p , remains constant.

The Binomial Coefficient

- ▶ In how many ways can we select k out of n items?
- ▶ This is given by the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

where $n!$ is pronounced “ n **factorial**” and is defined to be

$$n! = 1 \cdot 2 \cdot 3 \cdots n.$$

Binomial Distribution: PMF

- ▶ The probability mass function (PMF) of a binomial distribution can be written as

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

for $k = 0, 1, 2, \dots, n$.

- ▶ The term p^k is the probability of getting k successes.
- ▶ The term $(1 - p)^{n-k}$ is the probability of getting $n - k$ failures.
- ▶ Multiplying these two terms together gives the probability of any specific order of k successes and $n - k$ failures.
- ▶ We then multiply by $\binom{n}{k}$ to account for all possible orders of these successes and failures.

Mean

- ▶ The expected value or mean of a Binomial Distribution is given by $E[X] = np$
- ▶ This is a straightforward result considering that X is the sum of n Bernoulli random variables each with expectation p .
- ▶ Hence, the mean is just the number of trials times the probability of success on each trial.

Mean Derivation

- ▶ Suppose $X_1, X_2, \dots, X_n$ are independent Bernoulli random variables, where each X_i equals 1 with probability p and 0 with probability $1 - p$.
- ▶ $X = X_1 + X_2 + \dots + X_n$ is a Binomial random variable.
- ▶ By the linearity of expectation,

$$E[X] = E[X_1] + E[X_2] + \dots + E[X_n].$$

- ▶ Since the expectation of each X_i is p , it follows that

$$E[X] = np.$$

Variance

- ▶ The variance of a Binomial Distribution is given by

$$\text{Var}(X) = np(1 - p).$$

- ▶ The intuition behind this is that each Bernoulli trial contributes $p(1 - p)$ to the variance and there are n such trials.
- ▶ Thus, the total variance is $np(1 - p)$.

Example 1 - Flipping a Fair Coin

- ▶ You flip a fair coin 5 times. What is the probability of getting exactly 3 tails?
- ▶ What is the expected number of tails?

Solution

- ▶ Let X be the number of tails.
- ▶ $X \sim \text{Binomial}(5, 0.5)$.
- ▶ Using the PMF of the binomial distribution:

$$P(X = 3) = \binom{5}{3} \cdot 0.5^3 \cdot 0.5^2 = 10 \cdot 0.125 \cdot 0.25 = 0.3125$$

- ▶ So, there's a 31.25% chance of getting exactly 3 tails.
- ▶ The expected number of tails is $np = 5 \cdot 0.5 = 2.5$

Example 2 - Flipping a Biased Coin

Consider a biased coin, in which the probability of getting tails in one throw is p . You throw this coin 5 times.

- a. What is the probability that you get T, H, H, T, T?
- b. What is the probability of getting exactly 3 tails?
- c. How many tails do you expect to see?

Solution

- a. The probability of observing T, H, H, T, T is

$$p \cdot (1 - p) \cdot (1 - p) \cdot p \cdot p = p^3(1 - p)^2$$

- b. For the other two, let X denote the number of tails.

- c. Then, $X \sim \text{Bin}(5, p)$.

- d. The probability of getting exactly 3 tails is

$$P(X = 3) = \binom{5}{3} p^3(1 - p)^2 = 10p^3(1 - p)^2$$

(10 times higher than the previous probability, since now the specific order doesn't matter!)

- e. The expected number of tails is $E[X] = 5p$.

Example 3 - Quality Control at a Factory

- ▶ A factory produces light bulbs, 95% of which are perfect. You randomly select 10 light bulbs.
- ▶ What is the probability that at most 2 of them are defective?

Solution

- ▶ Let X be the number of defective bulbs.
- ▶ $X \sim \text{Binomial}(10, 0.05)$.
- ▶ Using the PMF of the binomial distribution:

$$P(X \leq 2) = \sum_{k=0}^2 \binom{10}{k} \cdot 0.05^k \cdot 0.95^{10-k}$$

- ▶ Calculating, $P(X \leq 2) \approx 0.9884$.
- ▶ So, there's a 98.84% chance that at most 2 out of the 10 bulbs are defective.

Example 4 - Clinical Trial

- ▶ A new drug is believed to be effective in 60% of patients. In a clinical trial, it's administered to 7 patients.
- ▶ What is the probability that it is effective in at least 6 patients?
- ▶ What is the expected number of patients for which the drug is effective?

Solution

- ▶ Let X be the number of patients for whom the drug is effective.
- ▶ $X \sim \text{Binomial}(7, 0.6)$.
- ▶ Using the PMF of the binomial distribution:

$$P(X \geq 6) = \sum_{k=6}^7 \binom{7}{k} \cdot 0.6^k \cdot 0.4^{7-k} = 7 \cdot (0.6)^6 \cdot (0.4) + (0.6)^7$$

- ▶ Calculating, $P(X \geq 6) \approx 0.2611$.
- ▶ So, there's a 26.11% chance that the drug is effective in at least 6 out of the 7 patients.
- ▶ The expected number of patients for which the drug is effective, is $np = 7 \cdot 0.6 = 4.2$.

References

- ▶ Section 4.3 of the textbook
- ▶ Great resource for the more curious:

Binomial distributions (3blue1brown)

STOR 155 - Lecture 14

Practice Problems in Probability

Panagiotis Andreou

UNC Chapel Hill

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Example 1 - Expectation of Bernoulli

- ▶ Suppose that X is a random variable that follows Bernoulli distribution with success probability p . We write this as

$$X \sim \text{Bernoulli}(p).$$

- ▶ Write down the distribution of X .
- ▶ Prove that $E[X] = p$.

Solution

- ▶ If we recall the definition of expectation, the first thing we need to determine is the set of values that X can take.
- ▶ Here, X follows a Bernoulli distribution, so by definition it can only take the values 0 and 1.
- ▶ The next step is to determine with what probability X takes each value.
- ▶ $X = 1$ with probability p , and $X = 0$ with probability $1 - p$.
- ▶ Combining the above, we compute the expectation:

$$E[X] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

Example 2 - Hair Color Genetics

A couple both have black hair but carry genes that make it possible for their children to have:

- ▶ Black hair with probability 0.6,
- ▶ Blonde hair with probability 0.2,
- ▶ Red hair with probability 0.2.

(a) What is the probability the first red-haired child they have is their third child?

(b) On average, how many children would such a pair of parents have before having a red-haired child? What is the standard deviation of the number of children they would expect to have until the first red-haired child?

Solution to (a)

- ▶ For the first red-haired child to be the third child, the first two children should not have red hair, while the third should.
- ▶ Probability that a child has red hair is $p = 0.2$.
- ▶ Let X denote the number of children the couple has until they have a red-haired child. Then, $X \sim \text{Geom}(0.2)$.
- ▶ Using the P.M.F. of the Geometric distribution, we get

$$\begin{aligned} P(\text{1st and 2nd not red, 3rd red}) &= P(X = 3) \\ &= (0.8)^2 \cdot 0.2 = 0.128. \end{aligned}$$

Solution to (b)

- ▶ The expectation of a geometric distribution is $E[X] = \frac{1}{p}$.
- ▶ In this case, $E[X] = \frac{1}{0.2} = 5$. So, on average, they would need 5 children to have a red-haired child.
- ▶ The standard deviation of a geometric distribution is $\sqrt{\frac{1-p}{p^2}}$.
- ▶ Here, $p = 0.2$, so we compute

$$\text{st. dev.} = \sqrt{\frac{1 - 0.2}{(0.2)^2}} = \sqrt{\frac{0.8}{0.04}} = \sqrt{20} = 4.4721.$$

Example 3 - UNC Basketball

Suppose we have the following information:

- ▶ 80% of students who attend at least two UNC basketball games during the season also participate in campus-wide events.
- ▶ 20% of students who attend fewer than two UNC basketball games during the season also participate in campus-wide events.
- ▶ Overall, 40% of students attend at least two UNC basketball games during the season.

The question we wish to answer is:

If a randomly selected student participates in campus-wide events, what is the probability that they attended at least two UNC basketball games during the season?

Solution - Defining Events

Define the following events:

- ▶ Let B be the event that a student attended at least two UNC basketball games.
- ▶ Let E be the event that a student participates in campus-wide events.

Bayes' Theorem:

$$P(B | E) = \frac{P(E|B)P(B)}{P(E)}.$$

Using the law of total probability:

$$P(E) = P(E | B)P(B) + P(E | B^c)P(B^c).$$

Solution - Computing Probabilities

Using the given probabilities:

$$\blacktriangleright P(E \mid B) = 0.8, \quad P(B) = 0.4.$$

$$\blacktriangleright P(E \mid B^c) = 0.2, \quad P(B^c) = 1 - 0.4 = 0.6.$$

Compute $P(E)$:

$$\begin{aligned} P(E) &= P(E \mid B)P(B) + P(E \mid B^c)P(B^c) \\ &= (0.8 \cdot 0.4) + (0.2 \cdot 0.6) \\ &= 0.32 + 0.12 = 0.44. \end{aligned}$$

Now compute $P(B \mid E)$:

$$\begin{aligned} P(B \mid E) &= \frac{P(E \mid B)P(B)}{P(E)} \\ &= \frac{0.8 \cdot 0.4}{0.44} \\ &= \frac{0.32}{0.44} \approx 0.7273. \end{aligned}$$

Final Answer

- ▶ By Bayes' theorem, we find that the probability a student attended at least two UNC basketball games given that they participate in campus-wide events is approximately

$$P(B \mid E) = 0.7273.$$

- ▶ This means that about 72.73% of students who participate in campus-wide events also attended at least two UNC basketball games during the season.

Example 4 - Tetrahedral Dice Problem

Consider a tetrahedral dice with four faces numbered 1, 2, 3, and 4. Each face is equally likely to come up in a single roll of the dice. Suppose you roll this dice four times. Calculate the probability of getting the following:

- (a) At least one '1'.
- (b) Exactly 2 rolls showing '2'.
- (c) Exactly 1 roll showing '3'.
- (d) At most 3 rolls showing '4'.

Solution

(a)

- ▶ Let X denote the number of 1's.
- ▶ Then, $X \sim \text{Bin}(4, 0.25)$.
- ▶ Probability of getting at least one '1' is equal to

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{3}{4}\right)^4 \approx 0.6836.$$

(b)

- ▶ Let Y denote the number of 2's.
- ▶ Then, $Y \sim \text{Bin}(4, 0.25)$.
- ▶ The probability that we get exactly 2 rolls showing '2' is

$$P(Y = 2) = \binom{4}{2} \cdot \left(\frac{1}{4}\right)^2 \cdot \left(\frac{3}{4}\right)^2 \approx 0.2109.$$

(c)

- ▶ Let W denote the number of 3's.
- ▶ Then, $W \sim \text{Bin}(4, 0.25)$.
- ▶ The probability that we get exactly 1 roll showing '3' is

$$P(W = 1) = \binom{4}{1} \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^3 \approx 0.4219.$$

(d)

- ▶ Let Z denote the number of 4's.
- ▶ Then, $Z \sim \text{Bin}(4, 0.25)$.
- ▶ The probability that we get at most 3 rolls showing '4' is

$$P(Z \leq 3) = 1 - P(Z = 4) = 1 - \left(\frac{1}{4}\right)^4 \approx 0.9961.$$

Example 5 - Genetic Traits Problem

A couple, both having black hair, carry genes that make it possible for their offspring to have black hair (probability 0.70), blonde hair (0.20), or red hair (0.10).

- (a) What is the probability that their first child will have red hair and the second will not?
- (b) What is the probability that exactly one of their two children will have red hair?
- (c) If they have six children, what is the probability that exactly two will have red hair?
- (d) If they have six children, what is the probability that at least one will have red hair?
- (e) What is the probability that the first red-haired child will be the 5th child?
- (f) Would it be considered unusual if only 2 out of their six children had black hair?

Solution: Parts (a) and (b)

(a) First child with red hair and second without:

► $P = 0.10 \cdot 0.90 = 0.0900.$

(b)

► Let X denote the number of children with red hair.

► Then, $X \sim \text{Bin}(2, 0.1).$

► Thus, we get $P(X = 1) = \binom{2}{1} \cdot 0.10 \cdot 0.90 = 0.1800$

Solution: Parts (c) and (d)

(c) Working as in the previous part, the probability is given by

$$P(Y = 6) = \binom{6}{2} \cdot (0.10)^2 \cdot (0.90)^4 \approx 0.09841,$$

where $Y \sim \text{Bin}(6, 0.1)$.

(d) A common trick with “at least 1” questions is to think of the complementary event. Here, we get

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - (0.9)^6 \approx 0.4686,$$

where we wrote $P(Y = 0) = (0.9)^6$ because we want all children to not have red hair, each children does not have red hair with probability 0.9, and they are independent with each other.

Solution: Parts (e) and (f)

(e)

- ▶ The first four children don't have red hair. The fifth one does.
- ▶ That's the case of a Geometric distribution. Here,
 $X \sim \text{Geom}(0.1)$.
- ▶ Hence, the probability that the first red-haired child is the 5th child is

$$P(X = 4) = (0.90)^4 \cdot 0.10 \approx 0.0656.$$

(f) Only 2 out of their 6 children having black hair:

- ▶ $E[\text{black hair}] = 6 \cdot 0.70 = 4.2$ children on average.
- ▶ $\text{Var}(\text{black hair}) = 6 \cdot 0.70 \cdot 0.30 = 1.26$.
- ▶ $\sigma(\text{black hair}) = \sqrt{1.26} \approx 1.12$.
- ▶ Observing only 2 black-haired children is 2.2 standard deviations below the mean, thus it would be considered unusual.

Example 6 - The Birthday Problem

- ▶ The STOR 155.007 class has 46 students.
- ▶ What is the probability that at least 2 students share the same birthday?

Solution

- ▶ We can think of this as the complement of the probability that there are no matches among the 46 students.
- ▶ We compute

$$\begin{aligned} &P(\text{no matches among 46}) \\ &= 1 \cdot \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \dots \cdot \left(1 - \frac{45}{365}\right) \\ &= \frac{365 \cdot 364 \cdot \dots \cdot 320}{365^{46}} \\ &= \frac{365!}{365^{46} \cdot (365 - 46)!} \approx 0.0833 \end{aligned}$$

- ▶ Thus,

$$P(\text{at least 1 match among 46}) \approx 1 - 0.0833 = 91.67\%.$$

STOR 155 - Lecture 15

Introduction to Inference

Panagiotis Andreou

UNC Chapel Hill

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The World of Uncertainty

- ▶ In our daily lives, we constantly make decisions based on **incomplete information**.
- ▶ This uncertainty arises because we often cannot observe everything we'd like to.
- ▶ Questions: How do we make informed decisions? How do we predict future events?

The World of Uncertainty

- ▶ In our daily lives, we constantly make decisions based on **incomplete information**.
- ▶ This uncertainty arises because we often cannot observe everything we'd like to.
- ▶ Questions: How do we make informed decisions? How do we predict future events?
- ▶ Answer: **Statistical Inference**.

What is Statistical Inference?

- ▶ It's the process of using data from a **sample** to draw conclusions about a **larger population**.
- ▶ It gives us tools to quantify uncertainty, make predictions, and test hypotheses.
- ▶ The core ideas:
 - ▶ Use available data to make the best guess about the unknown.
 - ▶ Understand and quantify the level of uncertainty associated with our guess.
- ▶ Today, we'll delve into the fundamental concepts underlying statistical inference.

Point estimates and error

- ▶ We are often interested in **population parameters**.
- ▶ Complete populations are difficult to collect data on, so we use **sample statistics** as **point estimates** for the unknown population parameters of interest.
- ▶ **Error** in the estimate = difference between population parameter and sample statistic

Bias and Sampling Error

- ▶ **Bias** is systematic tendency to over- or under-estimate the true population parameter.
- ▶ **Sampling error** describes how much an estimate will tend to vary from one sample to the next.
- ▶ Much of statistics is focused on understanding and quantifying sampling error, and **sample size** is helpful for quantifying this error.

Sampling Distribution: An Intuitive Explanation

- ▶ A sampling distribution is a theoretical distribution that reveals how our sample statistic would behave if we drew all possible samples from a population.
- ▶ It shows the possible values a statistic can take and how likely these values are.
- ▶ For example, imagine we are measuring the average height of sunflowers in a field. The sampling distribution tells us what to expect if we repeat this process over and over again with different samples.

Sampling Distribution: A More Rigorous Definition

- ▶ More formally, a sampling distribution of a statistic is the distribution of that statistic for all possible samples from the same population of a given size.
- ▶ It provides a measure of how much our statistic will vary from sample to sample.
- ▶ Continuing our sunflower example, the average height calculated from one sample might differ from that of another sample. The sampling distribution provides a way to quantify this variability.

How does a sample distribution look?

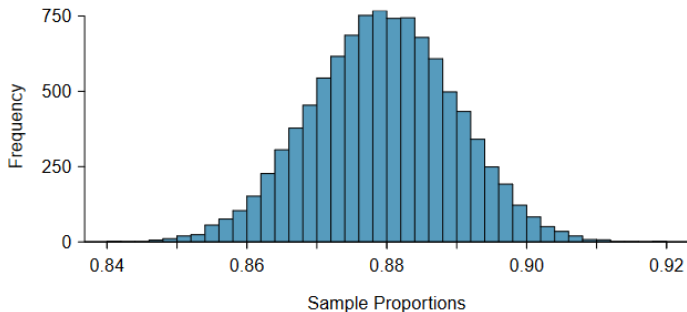


FIGURE: A histogram of 10,000 sample proportions, where each sample is taken from a population where the population proportion is 0.88 and the sample size is $n = 1000$
(Source: Textbook, page 172)

Sampling distributions are never observed

- ▶ In real-world applications, we never actually observe the sampling distribution, yet it is useful to always think of a point estimate as coming from such a hypothetical distribution.
- ▶ Understanding the sampling distribution will help us characterize and make sense of the point estimates that we do observe.

Example 1 - Mean vs. Proportion

For each of the following situations, determine whether the parameter of interest is a mean or a proportion.

- a. A survey reveals that 58% of citizens are concerned about air pollution.
- b. A report states that streaming services saw a 22% increase in subscribers within a three-year period, while cable subscriptions decreased by 5.7%.
- c. In a poll, college students are queried about whether they use digital payment apps on their devices.
- d. In a study, tablet users are asked if they ever use their device for reading e-books.
- e. In a questionnaire, tablet users are asked about the number of e-books they read in the past month.

Solution

- a. This is a proportion. 58% of citizens are concerned.
- b. The parameter of interest is a mean.
- c. This is asking for a yes or no response, so the parameter is a proportion.
- d. This too asks for a binary response. The parameter is a proportion.
- e. This asks for a specific count, indicating the parameter of interest is a mean.

Central Limit Theorem

Theorem (CLT)

Sample proportions will be nearly normally distributed:

$$\hat{p} \sim N \left(\text{mean} = p, SE = \sqrt{\frac{p(1-p)}{n}} \right)$$

- ▶ The sampling distribution is symmetric and centered at the true population proportion.
- ▶ We won't go through a detailed proof of why $SE = \sqrt{\frac{p(1-p)}{n}}$, but note that as n increases SE decreases.
- ▶ As n increases samples will yield more consistent $\hat{p}$ s, i.e., variability among $\hat{p}$ s will be lower.

Galton Board

- ▶ A great illustration of the Central Limit Theorem for the Binomial distribution is given via the **Galton Board**.
- ▶ Sir Francis Galton (1822-1911) said about the Normal curve:
"It is the supreme law of Unreason"
- ▶ Quick illustration: [Galton Board simulation](#)
- ▶ Nice illustration (minute 3.31) as part of a great talk:
[Cédric Villani's TED talk](#)

CLT - conditions

Certain conditions must be met for the CLT to apply:

1. **Independence:** Sampled observations must be independent. This is difficult to verify, but is more likely if
 - ▶ random sampling/assignment is used, and
 - ▶ if sampling without replacement, $n < 10\%$ of the population.
2. **Sample size:** There should be at least 10 expected successes and 10 expected failures in the observed sample, i.e.,

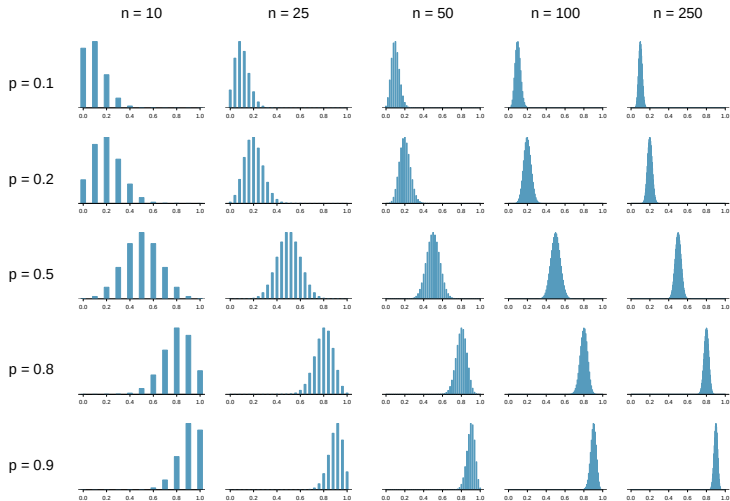
$$np \geq 10 \quad \text{and} \quad n(1 - p) \geq 10.$$

This is difficult to verify if you don't know the population proportion (or can't assume a value for it). In those cases we look for the number of observed successes and failures to be at least 10 (plug-in principle).

When the conditions are not met

- ▶ When either np or $n(1 - p)$ is small, the distribution is more discrete.
- ▶ When np or $n(1 - p) < 10$, the distribution is more skewed. If $n(1 - p) > np$, then it is right-skewed, otherwise it is left-skewed.
- ▶ The larger both np and $n(1 - p)$, the more normal the distribution.
- ▶ When np and $n(1 - p)$ are both very large, the discreteness of the distribution is hardly evident, and the distribution looks much more like a normal distribution.

What happens when np and/or $n(1 - p) < 10$?



Example 2 - Sampling Distribution

In a large university, 15% of the sophomores achieved honors in the current academic year. For a research assignment, students randomly pick 36 sophomores and check if those students achieved honors. They repeat this process 1,200 times and create a distribution of sample proportions.

- a. Name the type of this distribution.
- b. Predict the shape of this distribution: symmetric, right-skewed, or left-skewed. Justify your prediction.
- c. Determine the standard deviation of this distribution.
- d. What's the formal term for the value calculated in (c)?

Solution

- a. Sampling distribution
- b. Since $np = 0.15 \cdot 36 = 5.4 < 10$, the success-failure condition is not satisfied. Since, in addition, $n(1 - p) = 30.6 > np$, the distribution is right-skewed.
- c. The standard deviation is given by

$$\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.15 \cdot 0.85}{36}} \approx 0.06.$$

- d. Standard error

Example 3 - CLT

- ▶ Suppose we are analyzing the proportion of left-handed people in the world.
- ▶ According to various studies, about 10% of people are left-handed.
- ▶ Assume we conduct a study with a sample size of 200 people.
- ▶ What can we say about the sampling distribution of the proportion we'll get?

Solution

- ▶ The sampling distribution of the sample proportion (left-handed people) will be approximately normally distributed due to the Central Limit Theorem.
- ▶ The proportion found by the study is $\hat{p}$. The theoretical quantities are

$$\text{mean} = p = 0.1 \quad \text{and} \quad \text{SE} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.1 \cdot 0.9}{200}} \approx 0.021$$

- ▶ The CLT tells us that $\hat{p} \sim N(0.1, 0.021)$.

Key Observations from Distributions

- ▶ Centers of the distribution always align with the population proportion, p .
- ▶ $\hat{p}$ is unbiased when data are independent and from the given population.
- ▶ As n increases, the variability in the sampling distribution decreases.
- ▶ Variability is highest when $p = 0.5$. Notice the role of p in $SE = \sqrt{\frac{p(1-p)}{n}}$.
- ▶ $\hat{p}$ will always take discrete values (x/n), so perfect normality is not achievable.

Extending the Framework for Other Statistics

- ▶ Using a sample statistic to estimate a parameter is a common strategy across various contexts.
- ▶ Example: Estimating average salary for college graduates.
- ▶ Using the sample mean, $\bar{x}$, to estimate the population mean, μ .
- ▶ Different contexts: Estimating the price difference for products across two websites.
- ▶ While this chapter emphasizes a single proportion, the methods are widely applicable.

References

- ▶ Section 5.1 of the textbook
- ▶ Great resources for the more curious:
 - But what is the Central Limit Theorem? (3blue1brown)
 - The Central Limit Theorem, Clearly Explained (StatQuest)

STOR 155 - Lecture 16

Confidence Intervals

Panagiotis Andreou

UNC Chapel Hill

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Normal Distribution - Reminder

- ▶ Commonly known as the “bell curve”.
- ▶ Symmetric about the mean.
- ▶ Defined by two parameters: mean (μ) and standard deviation (σ). We write $X \sim N(\mu, \sigma)$.
- ▶ The area under the curve represents probabilities.
- ▶ Standard normal distribution: $\mu = 0$ and $\sigma = 1$.

Percentiles in Normal Distribution

- ▶ Percentile: The value below which a percentage of data falls.
- ▶ For example, the 90th percentile is the value below which 90% of the data is found.
- ▶ In a standard normal distribution:
 - ▶ 50th percentile corresponds to $z = 0$.
 - ▶ 84th percentile corresponds to $z = 1$ (since about 34% of the data lies between the mean and $z = 1$).

Z-table

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Using the Z-table

- ▶ A Z-table provides probabilities for different values of Z .
- ▶ To find the probability that Z is less than a certain value a :
 1. Locate the row corresponding to the first two digits of a .
 2. Find the column that corresponds to the second decimal place of a .
 3. The intersection gives the probability $P(Z < a)$.

Using Excel

Instead of a Z-table, we can simply use Excel:

- ▶ `NORM.DIST(x, 0, 1, TRUE)` returns the probability $P(Z \leq x)$.
- ▶ `NORM.INV(y, 0, 1)` finds the value x such that $P(Z \leq x) = y$.

Example 1

- ▶ Compute the probability $P(Z < 1.37)$.
- ▶ Solution:
 1. Go to row 1.3.
 2. Start sliding to the right until you find the intersection with column 0.07.
 3. The intersection yields the value 0.9147.
 4. Hence, $P(Z < 1.37) = 0.9147$.

Example 2

- ▶ Find the 97.5%-percentile, i.e., the value z_α for which $P(Z \leq z_\alpha) = 0.975$.
- ▶ Solution:
 1. Now we have to look inside the table!
 2. Find the value 0.975.
 3. Look at the row and column to which it corresponds.
 4. The row shows the first decimal of the percentile, and the column shows its second decimal.
 5. Here, we find $z_\alpha = 1.96$.

Motivating Confidence Intervals

The Study Hours Dilemma

- ▶ Imagine you're part of a student council survey team. You're trying to figure out the average hours college freshmen spend studying for mid-terms.
- ▶ You can't possibly ask every freshman (that would take forever!), so you decide to take a sample of 100 students.
- ▶ From this sample, you calculate the average study hours to be 15 hours.
- ▶ But can we be confident that this sample average accurately represents the entire freshman class?

Building Our Net

- ▶ Relying solely on that 15-hour average is like betting all your money on a single number in roulette.
- ▶ Instead, let's cast a net and find a range where we're fairly certain the true average lies.
- ▶ This “net” is our Confidence Interval!
- ▶ If we construct a 95% Confidence Interval, we are saying:
“We are 95% confident that the true average study hours of all freshmen lies within this range.”

The Power of Confidence Intervals

- ▶ Suppose our confidence interval gives a range from 13 to 17 hours.
- ▶ This range gives us a much better picture than just saying “15 hours.”
- ▶ It tells us that even though the sample average is 15 hours, we are 95% certain that the true average for all freshmen could be as low as 13 hours or as high as 17 hours.
- ▶ This is the magic and power of Confidence Intervals: providing a realistic and statistically backed range for our estimates.

Confidence Intervals - Intuition

- ▶ A plausible range of values for the population parameter is called a confidence interval.
- ▶ Using only a sample statistic to estimate a parameter is like fishing in a murky lake with a spear, and using a confidence interval is like fishing with a net.
- ▶ We can throw a spear where we saw a fish but we will probably miss. If we toss a net in that area, we have a good chance of catching the fish.
- ▶ If we report a point estimate, we probably won't hit the exact population parameter. If we report a range of plausible values we have a good shot at capturing the parameter.

Mathematical Derivation

- ▶ If $Z \sim N(0, 1)$, define z_α to be the point for which

$$P(Z > z_\alpha) = \alpha.$$

- ▶ Recall CLT:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

- ▶ From the above, we get

$$P\left(-z_{1-\alpha/2} \leq \frac{p - \hat{p}}{\sqrt{\frac{p(1-p)}{n}}} \leq z_{1-\alpha/2}\right) = 1 - \alpha$$

Confidence Interval definition

- Thus, take the interval

$$\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right)$$

to be the $100(1 - \alpha)\%$ confidence interval for p .

- For $\alpha = 0.05$, we have $z_{0.975} = 1.96$, so we get the 95% C.I.

$$(\hat{p} - 1.96 \text{ S.E.}, \hat{p} + 1.96 \text{ S.E.}),$$

where $\text{S.E.} = \sqrt{\frac{p(1-p)}{n}}$.

Frequentist Interpretation of Confidence Interval for p

- ▶ In frequentist statistics, the confidence interval provides a range of plausible values for the unknown population parameter p .
- ▶ A 95% confidence interval means that if we were to collect many samples and build a confidence interval from each of them, approximately 95% of these intervals would contain the true population proportion p .
- ▶ Importantly, the frequentist interpretation doesn't say that there is a 95% chance that a particular interval contains p . Instead, it suggests that in repeated sampling, 95% of such intervals would include p .
- ▶ Note: The interval is random (because it depends on the sample), but the parameter p is fixed but unknown.

Interval Length

- ▶ The length of the confidence interval is

$$2z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}}.$$

- ▶ Hence, if n increases, the standard error decreases and thus we get a shorter confidence interval.
- ▶ If we want *more confidence*, α must be smaller, so $z_{1-\alpha/2}$ is larger.
- ▶ Intuition:
 1. larger sample $\Rightarrow$ more information $\Rightarrow$ equally confident with a smaller interval
 2. If we want to be more certain about our estimate, then we have to “allow more room for error”, i.e., wider confidence interval.

Remarks

1. The confidence interval

$$\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right)$$

is an **approximate** C.I., since it is based on the CLT, which says that the sampling distribution is approximately Normal.

2. The above formula includes p , which is the parameter we want to estimate in the first place! For this reason, we often replace it with our point estimate $\hat{p}$ and get the confidence interval

$$\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right).$$

Example: Confidence Interval Calculation

Problem: A survey reported that 49% of U.S. adults have one or more chronic conditions. However, this value is based on a sample. The standard error is reported as 1.3%, and we assume a normal model is reasonable. We want to construct a 95% confidence interval for the proportion of US adults who live with one or more chronic conditions.

Solution

Gathering the main points from the problem:

- ▶ Sample proportion: $\hat{p} = 0.49$
- ▶ Standard error: $SE = 0.013$
- ▶ Confidence level: 95%
- ▶ Critical value: $z_{0.975} = 1.96$ (from normal distribution)

Use the confidence interval formula:

$$(\hat{p} - z_{0.975}SE, \hat{p} + z_{0.975}SE)$$

Solution

$$\text{Lower Bound} = 0.49 - (1.96 \times 0.013) = 0.49 - 0.02548 = 0.465$$

$$\text{Upper Bound} = 0.49 + (1.96 \times 0.013) = 0.49 + 0.02548 = 0.515$$

Thus, the 95% confidence interval is:

$$(0.465, 0.515)$$

Interpretation: We are 95% confident that the true proportion of U.S. adults with one or more chronic conditions lies within this interval.

References

- ▶ Section 5.2 of the textbook
- ▶ Confidence Interval (Simply explained)

STOR 155 - Lecture 17

Confidence Intervals

Panagiotis Andreou

UNC Chapel Hill

February 26, 2025

Confidence Intervals

- ▶ **General** form of a confidence interval:

point estimate $\pm z^* \cdot$ standard error,

where z^* is the critical value (percentile).

- ▶ The term $z^* \cdot$ S.E. is the **margin of error**.
- ▶ Confidence interval for a **proportion**:

$$\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{n}} \right)$$

- ▶ In practice, we don't know p but only its point estimate $\hat{p}$.
- ▶ Thus, use the confidence interval

$$\left(\hat{p} - z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$$

- ▶ This is an approximation to an already approximate confidence interval.

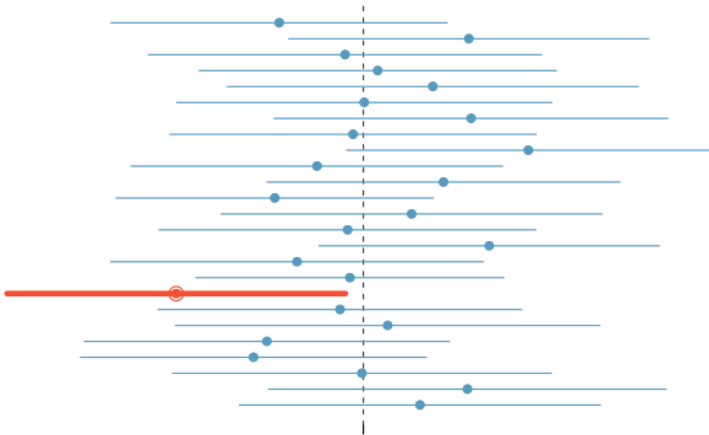


FIGURE: The red line could be one confidence interval that doesn't capture the population proportion.

Confidence Intervals in practice

- ▶ Once we have determined a proportion we are interested in, there are four steps to constructing a confidence interval:
 1. **Prepare:** Identify n and $\hat{p}$, and determine the desired confidence level.
 2. **Check:** Verify the success-failure conditions of the CLT, using $\hat{p}$ instead of (the unknown) p .
 3. **Calculate:** If the CLT conditions are met, compute the standard error, the value z^* , and finally the C.I. itself.
 4. **Conclude:** Interpret the confidence interval in the context of the problem.

Confidence levels

- ▶ For different confidence level α , we get different critical values $z_{1-\alpha/2}$.

- ▶ These values can be found via **z-tables** or Excel commands.

- ▶ If we want a 95% confidence interval, we take

$$z^* = z_{0.975} \approx 1.96.$$

- ▶ If we want a 99% confidence interval, we take

$$z^* = z_{0.995} \approx 2.58.$$

95% vs. 99% Confidence Intervals

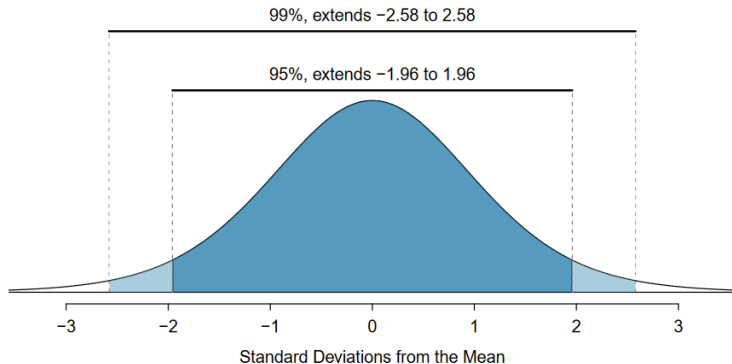


FIGURE: The area between $-z^*$ and z^* increases as z^* becomes larger. If the confidence level is 99%, we choose z^* such that 99% of a normal normal distribution is between z^* and z^* , which corresponds to 0.5% in the lower tail and 0.5% in the upper tail: $z^* = 2.58$.

Example 1 - Twitter

A poll conducted in 2013 found that 52% of U.S. adult Twitter users get at least some news on Twitter, and the standard error for this estimate was 2.4%. Identify each of the following statements as true or false. Provide an explanation to justify each of your answers.

- (a) The data provide statistically significant evidence that more than half of U.S. adult Twitter users get some news through Twitter. Use a significance level of $\alpha = 0.01$.
- (b) Since the standard error is 2.4%, we can conclude that 97.6% of all U.S. adult Twitter users were included in the study.
- (c) If we want to reduce the standard error of the estimate, we should collect less data.
- (d) If we construct a 90% confidence interval for the percentage of U.S. adults Twitter users who get some news through Twitter, this confidence interval will be wider than a corresponding 99% confidence interval.

Solution

Part (a):

- ▶ Construct a 99% confidence interval for the proportion.

- ▶ Given:

$$\hat{p} = 0.52, \quad \text{S.E.} = 0.024$$

- ▶ For 99% CI, $z^* \approx 2.576$.

- ▶ Margin of Error:

$$z^* \cdot \text{S.E.} = 0.0618$$

- ▶ Confidence interval: (0.4582, 0.5818)

- ▶ Conclusion: TRUE.

- ▶ (b) FALSE: S.E. doesn't inform about the population percentage in study.
- ▶ (c) FALSE: Collect more data to reduce S.E.
- ▶ (d) FALSE: 90% CI is narrower than 99% CI.

Example 2 - National Fitness Survey

The National Fitness Survey asked the question:

“On average, for how many days during the past 30 days did you exercise for at least 30 minutes?”

Based on responses from 1,200 US residents, the survey reported a 95% confidence interval of 11.50 to 12.75 days in 2020.

1. Interpret this interval in the context of the data.
2. What does “95% confident” mean? Explain in the context of the application.
3. Suppose the researchers think a 99% confidence level would be more appropriate for this interval. Will this new interval be smaller or wider than the 95% confidence interval?
4. If a new survey were to be done with 500 Americans, do you think the standard error of the estimate would be larger, smaller, or about the same?

Solutions

1. The interval suggests that we are 95% confident that the true average number of days US residents exercised for at least 30 minutes in the past month lies between 11.50 and 12.75 days.
2. Being 95% confident means that if we were to take many samples and compute a 95% confidence interval for each of them, about 95% of these intervals would capture the true average.
3. A 99% confidence level will result in a wider interval because we want to be more certain about our estimate.
4. With fewer respondents (500 Americans instead of 1,200), the standard error will be larger because there is less data to give a precise estimate.

Example 3 - Streaming Service Preferences

- ▶ Let's say you're interested in knowing the proportion of college students who prefer Netflix.
- ▶ A survey of 100 students shows that 60 of them prefer Netflix for streaming.
- ▶ How can we construct a 95% confidence interval for this proportion?

Prepare and Check

- ▶ $n = 100$
- ▶ $\hat{p} = \frac{60}{100} = 0.6$
- ▶ Check CLT conditions:
 - ▶ $n\hat{p} = 60$ (at least 10 successes)
 - ▶ $n(1 - \hat{p}) = 40$ (at least 10 failures)
- ▶ Conditions are met; proceed with the calculations.

Calculate

- ▶ Standard Error: $S.E. = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.6 \cdot 0.4}{100}} \approx 0.049$
- ▶ For a 95% confidence interval, $z^* = 1.96$
- ▶ Confidence Interval: $(\hat{p} - z^* \cdot S.E., \hat{p} + z^* \cdot S.E.)$
- ▶ $(0.6 - 1.96 \cdot 0.049, 0.6 + 1.96 \cdot 0.049)$
- ▶ $(0.504, 0.696)$

Conclude

- ▶ We are 95% confident that the true proportion of college students who prefer Netflix is between 50.4% and 69.6%.
- ▶ In the long run, 95% of such intervals will capture the true proportion, assuming our sampling and calculations are correct.

Example: Interpreting a Confidence Interval

Problem: A survey asked U.S. residents:

"For how many days during the past 30 days was your mental health, which includes stress, depression, and problems with emotions, not good?"

Based on responses from 1,151 U.S. residents, a 95% confidence interval for the mean number of bad mental health days was reported as:

(3.10, 4.24) days per month.

(a) Interpretation:

- ▶ We are **95% confident** that the true mean number of bad mental health days for all U.S. residents falls between 3.10 and 4.24 days per month.

(b) Meaning of "95% confidence"

- ▶ 95% of all possible random samples of size 1,151 will produce confidence intervals that contain the true mean number of bad mental health days.

Effect of Changing Confidence Level and Sample Size

(c) Increasing Confidence Level:

- ▶ A 99% confidence level requires a **larger** interval because we need to be more sure of capturing the true mean.
- ▶ This is achieved by increasing the critical value $z_{0.995}$, which results in a wider interval.

(d) Effect of Sample Size:

- ▶ If we reduce the sample size to 500 participants, we have **less precision**.
- ▶ This means the standard error will be **larger**, leading to a wider confidence interval.

Key Takeaways:

- ▶ Larger sample sizes yield more precise estimates (smaller standard error).
- ▶ Higher confidence levels lead to wider intervals.

References

- ▶ Section 5.2 of the textbook
- ▶ Confidence Intervals video
- ▶ Confidence Interval (Simply explained)

STOR 155 - Lecture 18

Hypothesis Testing

Panagiotis Andreou

UNC Chapel Hill

February 28, 2025

The Question: Can an Energy Drink Increase Student Focus?

- ▶ Imagine you are a college student pulling an all-nighter for an exam.
- ▶ A new energy drink claims it can increase your focus by 20%.
- ▶ Would you buy this drink? How can we test this claim scientifically?

Setting Up the Experiment

- ▶ To test this, we gather 100 students and divide them into two groups.
- ▶ One group gets the energy drink; the other gets a placebo.
- ▶ We then measure their focus through a series of cognitive tests.

Our Hypotheses

- ▶ **Null Hypothesis** (H_0): The energy drink has no effect on student focus.
- ▶ **Alternative Hypothesis** (H_a): The energy drink improves student focus.

Why Should We Care?

- ▶ Consumer Decision: Is it worth buying this energy drink?
- ▶ Scientific Credibility: Is the manufacturer's claim valid?
- ▶ General Understanding: How do different substances affect human focus?

Introduction to Hypothesis Testing

- ▶ Hypothesis testing is a statistical method used to make inferences about a population based on a sample.
- ▶ It allows us to test claims or hypotheses about various characteristics of a population, such as its mean or proportion.
- ▶ Essentially, it's a way to use sample data to answer research questions.

Null and Alternative Hypotheses

- ▶ **Null Hypothesis** (H_0): The hypothesis we want to test against. Often represents the idea of “no effect” or “no difference.” We can think of it as the *status quo*.
- ▶ **Alternative Hypothesis** (H_A or H_1): What we want to prove; the opposite of the null hypothesis.
- ▶ Example: If we're testing a new drug, the null hypothesis might be that the drug has no effect, and the alternative hypothesis would be that it does.

Types of Tests

- ▶ **One-tailed Test:** Tests if a parameter is greater than or less than a certain value, but not both.
- ▶ **Two-tailed Test:** Tests if a parameter is simply not equal to a certain value, irrespective of direction (greater than or less than).
- ▶ Choice between one-tailed and two-tailed tests depends on the research question.

Significance Level and P-Value

- ▶ **Significance Level (α)**: the threshold for determining statistical significance, commonly set at 0.05.
- ▶ **p-value**: the probability of observing data at least as favorable to the alternative hypothesis as our current data set, if the null hypothesis were true.
- ▶ If the p-value is less than α , we reject the null hypothesis.

Intuition for rejection scheme

small p-value



small probability (under the null hypothesis) of observing something
more extreme than what we observed



what we observed is quite extreme under the null hypothesis



the alternative hypothesis should be true



reject the null hypothesis

Decision errors

- ▶ Hypothesis tests are not flawless.
- ▶ In the court system innocent people are sometimes wrongly convicted and the guilty sometimes walk free.
- ▶ Similarly, we can make a wrong decision in statistical hypothesis tests as well.
- ▶ The difference is that we have the tools necessary to quantify how often we make errors in statistics.

Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true		
	H_A true		



Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true		
	H_A true		



Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	
	H_A true		



Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	
	H_A true		Correct



Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true		Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.



Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true	Type 2 Error	Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.
- ▶ A **Type 2 Error** is failing to reject the null hypothesis when H_A is true.
- ▶

Decision errors (cont.)

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true	Type 2 Error	Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.
- ▶ A **Type 2 Error** is failing to reject the null hypothesis when H_A is true.
- ▶ We (almost) never know if H_0 or H_A is true, but we need to consider all possibilities.

Hypothesis Test as a trial

If we again think of a hypothesis test as a criminal trial then it makes sense to frame the verdict in terms of the null and alternative hypotheses:

H_0 : Defendant is innocent

H_A : Defendant is guilty

Which type of error is being committed in the following circumstances?

- ▶ Declaring the defendant innocent when they are actually guilty
- ▶ Declaring the defendant guilty when they are actually innocent

Hypothesis Test as a trial

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Type 2 error
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Type 1 error

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H_A : Defendant is guilty

Which type of error is being committed in the following circumstances?

- ▶ Declaring the defendant innocent when they are actually guilty
Type 2 error
- ▶ Declaring the defendant guilty when they are actually innocent
Type 1 error

Which error do you think is the worse error to make?

Hypothesis Test as a trial

If we again think of a hypothesis test as a criminal trial then it makes sense to frame the verdict in terms of the null and alternative hypotheses:

H_0 : Defendant is innocent

H_A : Defendant is guilty

Which type of error is being committed in the following circumstances?

- ▶ Declaring the defendant innocent when they are actually guilty
Type 2 error
- ▶ Declaring the defendant guilty when they are actually innocent
Type 1 error

Which error do you think is the worse error to make?

“better that ten guilty persons escape than that one innocent suffer”

– William Blackstone

Type-I Error: The False Positive

- ▶ A Type-I error occurs when we **incorrectly reject** a **true** null hypothesis.
- ▶ The probability of making a Type-I error is denoted by α (alpha).
- ▶ α is usually set beforehand, commonly at 0.05 or 5%.

Type-II Error: The False Negative

- ▶ A Type-II error occurs when we **fail to reject** a **false** null hypothesis.
- ▶ The probability of making a Type-II error is denoted by β (beta).
- ▶ Unlike α , β varies based on the sample size, variance, and effect size.

Balancing Type-I and Type-II Errors

- ▶ Reducing α (Type-I error) often increases β (Type-II error), and vice versa.
- ▶ It's essential to balance the risks of both errors based on the context.
- ▶ For example, in medicine, avoiding a Type-I error may be crucial to prevent harmful treatments.

References

- ▶ Section 5.3 of the textbook
- ▶ Hypothesis Testing and The Null Hypothesis, Clearly Explained!

STOR 155 - Lecture 19

Hypothesis Testing

Panagiotis Andreou

UNC Chapel Hill

March 3, 2025

The setup

- ▶ Two competing hypotheses:

H_0 (“the null”) vs. H_A (“the alternative”)

- ▶ H_0 : the status quo, what we think is true before new data arises.
- ▶ H_A : an alternative claim under consideration, a hypothesis we wish to examine.
- ▶ Typically, we suspect that H_A is true, and we ask whether the data provide enough evidence to reject H_0 .

Motivating Example

- ▶ Suppose we flip a coin 100 times and we observe 64 heads and 36 tails. Is the coin fair?
- ▶ Two possible scenarios:
 1. The coin is actually fair and this unlikely result was due to chance (H_0).
 2. The coin is biased (H_A).
- ▶ This is the typical setup for a hypothesis test:

$$\begin{cases} H_0 : p = \frac{1}{2} \\ H_A : p \neq \frac{1}{2} \end{cases}$$

- ▶ To decide between the two, compute the probability of observing this outcome **under the null**.
- ▶ Let $X \sim \text{Bin}(100, 1/2)$ denote the number of heads. Then, $P(X = 64) \approx 0.006$.
- ▶ This suggests that we should probably reject H_0 in favor of H_A , and conclude that the coin is biased.

How to conduct hypothesis testing?

Two main ways:

1. Use duality with confidence intervals
2. Via p-values (usually the preferred way)

Hypothesis Tests and Confidence Intervals

- ▶ We are interested in estimating a proportion p . Test:

$$\begin{cases} H_0 : p = p_0 \\ H_A : p \neq p_0 \end{cases}$$

- ▶ If we have a point-estimate $\hat{p}$, a confidence interval is

$$[\hat{p} - z^* \text{S.E.}, \hat{p} + z^* \text{S.E.}]$$

- ▶ If p_0 is in this C.I., we fail to reject H_0 . If it isn't, then we reject H_0 .
- ▶ In practice, finding confidence intervals can be hard, so we conduct hypothesis testing via **p-values**.

p-values

- ▶ The probability, under H_0 , of observing something at least as extreme as what we observed.
- ▶ For fixed significance level α , reject H_0 if $\text{p-value} < \alpha$.

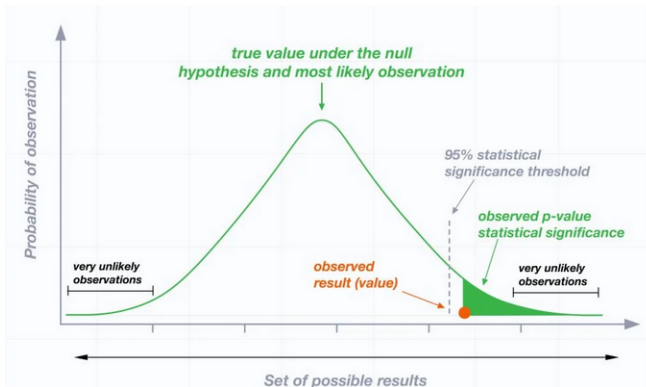


FIGURE: p-value is the green area under the curve
(Source: [Simply Psychology](#))

Case 1: $H_0 : p = p_0$ vs. $H_A : p < p_0$

- ▶ Suppose we observe $\hat{p}$. Then, the Z -score is $\frac{\hat{p} - p_0}{\text{S.E.}}$, where

$$\text{S.E.} = \sqrt{\frac{p_0(1 - p_0)}{n}}.$$

Note: unlike in confidence intervals, here we use p_0 and not $\hat{p}$ in the standard error!

- ▶ The p-value is the probability of observing something more extreme under the null, i.e., something less than $\hat{p}$.
- ▶ Hence,

$$\text{p-value} = P\left(Z \leq \frac{\hat{p} - p_0}{\text{S.E.}}\right),$$

where $Z \sim N(0, 1)$.

Case 2: $H_0 : p = p_0$ vs. $H_A : p > p_0$

- ▶ Suppose we observe $\hat{p}$. Then, the Z-score is $\frac{\hat{p} - p_0}{\text{S.E.}}$, where

$$\text{S.E.} = \sqrt{\frac{p_0(1 - p_0)}{n}}.$$

- ▶ The p-value is the probability of observing something more extreme under the null, i.e., something greater than $\hat{p}$.
- ▶ Hence,

$$\text{p-value} = P\left(Z \geq \frac{\hat{p} - p_0}{\text{S.E.}}\right),$$

where $Z \sim N(0, 1)$.

Case 3: $H_0 : p = p_0$ vs. $H_A : p \neq p_0$, and $\hat{p} < p_0$

- ▶ Suppose we observe $\hat{p}$. The Z-score is $\frac{\hat{p} - p_0}{\text{S.E.}}$.
- ▶ The p-value now needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(Z \leq \frac{\hat{p} - p_0}{\text{S.E.}}\right) + P\left(Z \geq \frac{p_0 - \hat{p}}{\text{S.E.}}\right) \\ &= 2P\left(Z \leq \frac{\hat{p} - p_0}{\text{S.E.}}\right),\end{aligned}$$

where $Z \sim N(0, 1)$.

- ▶ In Excel, we can type

$$= 2 * \text{NORM.DIST}\left(\frac{\hat{p} - p_0}{\text{S.E.}}, 0, 1, \text{TRUE}\right)$$

Case 4: $H_0 : p = p_0$ vs. $H_A : p \neq p_0$, and $\hat{p} > p_0$

- ▶ Suppose we observe $\hat{p}$. The Z-score is $\frac{\hat{p} - p_0}{\text{S.E.}}$.
- ▶ The p-value now needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(Z \geq \frac{\hat{p} - p_0}{\text{S.E.}}\right) + P\left(Z \leq \frac{-\hat{p} + p_0}{\text{S.E.}}\right) \\ &= 2P\left(Z \geq \frac{\hat{p} - p_0}{\text{S.E.}}\right),\end{aligned}$$

where $Z \sim N(0, 1)$.

- ▶ In Excel, we can type

$$= 2 * \left(1 - \text{NORM.DIST}\left(\frac{\hat{p} - p_0}{\text{S.E.}}, 0, 1, \text{TRUE}\right)\right)$$

Example 1 - Facebook interest categories

- ▶ The same survey asked the 850 respondents how comfortable they are with Facebook creating a list of categories for them.
- ▶ 41% of the respondents said they are comfortable.
- ▶ Do these data provide convincing evidence that the proportion of American Facebook users who are comfortable with Facebook creating a list of interest categories for them is different than 50%?

Setting the hypotheses

- ▶ The parameter of interest is the **proportion** of all American Facebook users who are comfortable with Facebook creating categories of interests for them.
- ▶ There may be two explanations why our sample proportion is lower than 0.50 (minority).
 1. The true population proportion is different than 0.50.
 2. The true population mean is 0.50, and the difference between the true population proportion and the sample proportion is simply due to natural sampling variability.

Setting the hypotheses

- ▶ We start with the assumption that 50% of American Facebook users are comfortable with Facebook creating categories of interests for them

$$H_0 : p = 0.50$$

- ▶ We test the claim that the proportion of American Facebook users who are comfortable with Facebook creating categories of interests for them is different than 50%

$$H_A : p \neq 0.50$$

Conducting the test

- ▶ Standard error:

$$\text{S.E.} = \sqrt{\frac{0.50 \cdot 0.50}{850}} \approx 0.0171$$

- ▶ Test statistic (Z-score):

$$Z = \frac{0.41 - 0.50}{0.0171} = -5.26$$

- ▶ The sample proportion is 5.26 standard errors away from the hypothesized value. Is this unusually low? In other words, is the difference **significant**?
- ▶ We check this by computing the **p-value**.

- p-value:

$$P(\hat{p} < 0.41 \text{ or } \hat{p} > 0.59 \mid p = 0.50) = P(|Z| > 5.26) < 0.0001$$

- Otherwise, we can compute it as follows:

1. Observe that we are in Case 3 ($\hat{p} < p_0$).

2. That means

$$\text{p-value} = 2P\left(Z \leq \frac{\hat{p} - p_0}{\text{S.E.}}\right)$$

3. Go to Excel and type

$$= 2 * \text{NORM.DIST}(-5.26, 0, 1, \text{TRUE})$$

Making a decision

- ▶ $p\text{-value} < 0.0001$
 - ▶ If 50% of all American FB users are comfortable with FB creating these interest categories, there is less than a 0.01% chance of observing a random sample of 850 American Facebook users where 41% or fewer or 59% or higher feel comfortable with it.
 - ▶ Pretty low probability to think that observed sample proportion, or something more extreme, is likely to happen by chance.
- ▶ Since $p\text{-value}$ is low (lower than 5%) we reject H_0 .
- ▶ The data provide convincing evidence that the proportion of American FB users who are comfortable with FB creating a list of interest categories for them is different than 50%.
- ▶ The difference between the null value of 0.50 and observed sample proportion of 0.41 is not due to chance or sampling variability.

Example 2

- ▶ You are interested in testing the following hypotheses about the proportion p of a product that is defective:

$$\begin{cases} H_0 : p = 0.15 \\ H_A : p \neq 0.15 \end{cases}$$

- ▶ Given that the sample size is $n = 100$, for what sample proportions $\hat{p}$ would the p-value be equal to 0.05? Assume that all conditions necessary for inference are satisfied.

Setup

- ▶ The null hypothesis H_0 states that the proportion p is 0.15.
- ▶ The sample size $n = 100$.
- ▶ Standard error under the null hypothesis:

$$\text{S.E.}_{H_0} = \sqrt{\frac{p_0(1 - p_0)}{n}} = \sqrt{\frac{0.15 \cdot 0.85}{100}} \approx 0.036$$

Critical Value

- ▶ For a two-tailed test at $\alpha = 0.05$, the critical z-value is approximately ± 1.96 .
- ▶ We need to find $\hat{p}$ such that the z-score is ± 1.96 .

$$z = \frac{\hat{p} - p}{\text{S.E.}_{H_0}}$$

Calculating Sample Proportions

- For smaller sample proportion:

$$-1.96 = \frac{\hat{p} - 0.15}{0.036}$$

Solving for $\hat{p}$:

$$\hat{p} = 0.15 - 1.96 \cdot 0.036$$

$$\hat{p} \approx 0.080$$

- For larger sample proportion:

$$1.96 = \frac{\hat{p} - 0.15}{0.036}$$

Solving for $\hat{p}$:

$$\hat{p} = 0.15 + 1.96 \cdot 0.036$$

$$\hat{p} \approx 0.220$$

Conclusion

- ▶ We would reject the null hypothesis H_0 at a significance level of $\alpha = 0.05$ if the sample proportion $\hat{p}$ is less than 0.080 or greater than 0.220.
- ▶ Thus, the sample proportions for which the p-value is 0.05 are approximately 0.080 and 0.220.

References

- ▶ Section 5.3 of the textbook
- ▶ Hypothesis Testing video

STOR 155 - Lecture 20

Hypothesis Testing for One Mean

Panagiotis Andreou

UNC Chapel Hill

March 5, 2025

Previously

- ▶ Hypothesis tests for **proportions**:

$$\begin{cases} H_0 : p = p_0 \\ H_A : p \neq p_0 \end{cases}$$

- ▶ Estimator:

$$\hat{p} \overset{\text{CLT}}{\sim} N\left(p_0, \text{S.E.}_{H_0} = \sqrt{\frac{p_0(1-p_0)}{n}}\right)$$

- ▶ Test statistic:

$$Z = \frac{\hat{p} - p_0}{\text{S.E.}_{H_0}} \sim N(0, 1)$$

Today

- ▶ Dataset $x_1, \dots, x_n$ with mean μ and standard deviation σ .
(Note: we do not assume any distribution on the sample!)
- ▶ Hypothesis tests for **means**:

$$\begin{cases} H_0 : \mu = \mu_0 \\ H_A : \mu \neq \mu_0 \end{cases}$$

- ▶ Estimator:

$$\bar{X} := \frac{1}{n} \sum_{i=1}^n x_i$$

- ▶ Test statistic:

CLT for the Mean

► Conditions to be met:

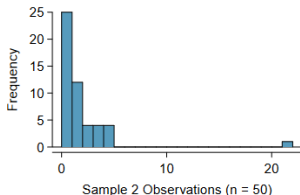
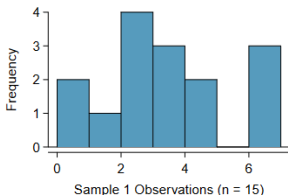
1. **Independence:** typically satisfied when the sample is a simple random sample from the population.
2. **Normality:** trickier to assess. Typically, separate cases:
 - $n < 30$: there should be no outliers.
 - $n \geq 30$: there should be no *particularly extreme* outliers.

► If the above conditions are met, then

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right),$$

approximately.

Example



- ▶ Each samples is from a simple random sample of its respective population, so the **independence** condition is satisfied.
- ▶ Check **normality** conditions:

1. Left figure:

- ▶ Fewer than 30 observations, so we are watching for any clear outliers.
- ▶ None are present, while there is a small gap in the histogram on the right.
- ▶ This gap is small and 20% of the observations in this small sample are represented in that far right bar of the histogram, so we can hardly call these clear outliers.
- ▶ With no clear outliers, the normality condition is reasonably met.

2. Right figure:

- ▶ Sample size greater than 30.
- ▶ Outlier that appears to be roughly 5 times further from the center of the distribution than the next furthest observation.
- ▶ This is an example of a particularly extreme outlier, so the normality condition would not be satisfied.

The Issue

- ▶ The above requires σ to be known in order for CLT to be used.
- ▶ However, in most realistic scenarios σ is unknown.
- ▶ *Idea:* Use the **sample standard deviation** s as an estimate of σ , and consider the statistic

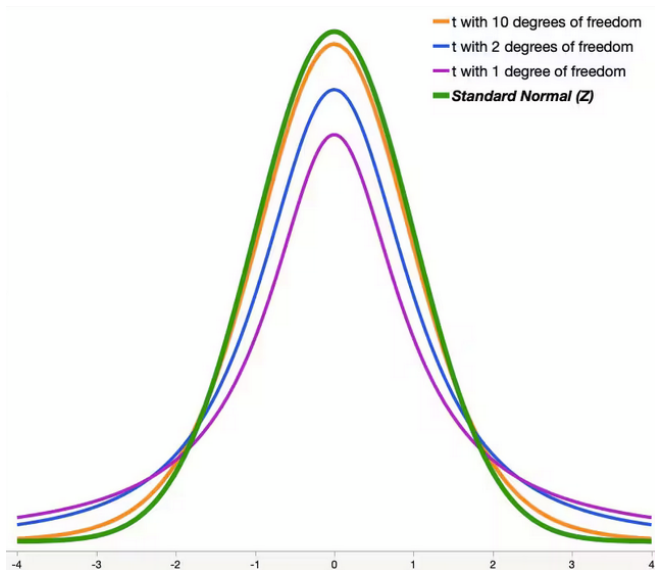
$$T = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}.$$

- ▶ *Issue:* T does not follow $N(0, 1)$, but t_{n-1} distribution.

The t -Distribution

- ▶ The t -distribution, like the standard normal distribution, is bell-shaped and symmetric, but has heavier tails.
- ▶ It is defined by its **degrees of freedom** (df). In the context of hypothesis testing for means, $df = n - 1$, where n is the sample size.
- ▶ As df increases, the t -distribution approaches the standard normal distribution.
- ▶ Critical values from the t -distribution will be further from 0 (larger in absolute value) compared to the standard normal distribution, especially for smaller sample sizes.
- ▶ Always use the t -distribution when estimating a population mean (μ) with a sample standard deviation (s) unless the population standard deviation (σ) is known.

t_d vs. $N(0, 1)$



t -Distribution in Excel

- ▶ Suppose $X \sim t_d$. Then, $P(X \leq x)$ can be calculated by the Excel command:

`=T.DIST(x,d,TRUE)`

- ▶ If we know $P(X \leq x) = p$ where p is known, then x is given by the Excel command:

`=T.INV(p,d)`

Case 1: $H_0 : \mu = \mu_0$ vs. $H_A : \mu < \mu_0$

- Suppose we observe $\bar{x}$. The T -score is $\frac{\bar{x} - \mu_0}{\text{S.E.}}$, where

$$\text{S.E.} = \frac{s}{\sqrt{n}}.$$

- The p-value is the probability of observing something more extreme under the null, i.e., something less than $\bar{x}$.
- Hence,

$$\text{p-value} = P\left(T \leq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right),$$

where $T \sim t_{n-1}$.

Case 2: $H_0 : \mu = \mu_0$ vs. $H_A : \mu > \mu_0$

- ▶ Suppose we observe $\bar{x}$. The T -score is $\frac{\bar{x} - \mu_0}{\text{S.E.}}$.
- ▶ The p-value is the probability of observing something more extreme under the null, i.e., something greater than $\bar{x}$.
- ▶ Hence,

$$\text{p-value} = P\left(T \geq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right),$$

where $T \sim t_{n-1}$.

Case 3: $H_0 : \mu = \mu_0$ vs. $H_A : \mu \neq \mu_0$, and $\bar{x} < \mu_0$

- ▶ Suppose we observe $\bar{x}$. The T -score is $\frac{\bar{x} - \mu_0}{\text{S.E.}}$.
- ▶ The p-value now needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(T \leq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right) + P\left(T \geq \frac{-\bar{x} + \mu_0}{\text{S.E.}}\right) \\ &= 2P\left(T \leq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right),\end{aligned}$$

where $T \sim t_{n-1}$.

- ▶ In Excel, we can type

$$= 2 * \text{T.DIST}\left(\frac{\bar{x} - \mu_0}{\text{S.E.}}, n - 1, \text{TRUE}\right)$$

Case 4: $H_0 : \mu = \mu_0$ vs. $H_A : \mu \neq \mu_0$, and $\bar{x} > \mu_0$

- ▶ Suppose we observe $\bar{x}$. The T -score is $\frac{\bar{x} - \mu_0}{\text{S.E.}}$.
- ▶ The p-value now needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(T \geq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right) + P\left(T \leq \frac{-\bar{x} + \mu_0}{\text{S.E.}}\right) \\ &= 2P\left(T \geq \frac{\bar{x} - \mu_0}{\text{S.E.}}\right),\end{aligned}$$

where $T \sim t_{n-1}$.

- ▶ In Excel, we can type

$$= 2 * \left(1 - \text{T.DIST}\left(\frac{\bar{x} - \mu_0}{\text{S.E.}}, n - 1, \text{TRUE}\right)\right)$$

Example

- ▶ The average time for all runners in 2006: 93.29 minutes.
- ▶ We aim to determine if runners in 2017 have the same finishing time as in 2006.
- ▶ Sample data from 2017 ($n=100$):
 - ▶ Sample mean ($\bar{x}$): 97.32 minutes.
 - ▶ Sample standard deviation (s): 16.98 minutes.

Setting up the Hypotheses

The hypothesis test is:

$$\begin{cases} H_0 : \mu = 93.29 \\ H_A : \mu \neq 93.29 \end{cases}$$

Calculating the t -statistic

- ▶ The formula for T-score is:

$$T = \frac{\bar{x} - \mu_0}{\text{S.E.}}$$

- ▶ Given:

$$\bar{x} = 97.32$$

$$\mu_0 = 93.29$$

$$\text{S.E.} = \frac{s}{\sqrt{n}} = \frac{16.98}{\sqrt{100}} = 1.698$$

- ▶ Calculating the t -statistic:

$$T = \frac{97.32 - 93.29}{1.698} = 2.375$$

Finding the p-value

- ▶ Using the t -distribution with $df = n - 1 = 99$ degrees of freedom.
- ▶ Since our hypothesis test is two-tailed:

$$\text{p-value} = 2 \cdot P(T \geq 2.375)$$

- ▶ This p-value can be determined using Excel:

$$=2*(1-T.DIST(2.375, 99, TRUE))$$

Decision and Conclusion

- ▶ $p\text{-value} = 0.019$
- ▶ $p\text{-value} < 0.05$
- ▶ We reject H_0 in favor of H_A at significance level $\alpha = 5\%$.
- ▶ We conclude that runners in 2017 did not have the same finishing time as in 2006.

References

- ▶ Section 7.1 of the textbook
- ▶ Hypothesis Testing for Means

STOR 155 - Lecture 21

Hypothesis Testing - Paired Data

Panagiotis Andreou

UNC Chapel Hill

March 7, 2025

The general setup

- ▶ Two competing hypotheses:

H_0 (“the null”) vs. H_A (“the alternative”)

- ▶ H_0 : the status quo, what we think is true before new data arises.
- ▶ H_A : an alternative claim under consideration, a hypothesis we wish to examine.
- ▶ Typically, we suspect that H_A is true, and we ask whether the data provide enough evidence to reject H_0 .

Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true		
	H_A true		



Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true		
	H_A true		



Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	
	H_A true		



Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	
	H_A true		Correct



Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true		Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.



Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true	Type 2 Error	Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.
- ▶ A **Type 2 Error** is failing to reject the null hypothesis when H_A is true.
- ▶

Decision errors

There are two competing hypotheses: the null and the alternative. In a hypothesis test, we make a decision about which might be true, but our choice might be incorrect.

		Decision	
		Fail to reject H_0	Reject H_0
Truth	H_0 true	Correct	Type 1 Error
	H_A true	Type 2 Error	Correct

- ▶ A **Type 1 Error** is rejecting the null hypothesis when H_0 is true.
- ▶ A **Type 2 Error** is failing to reject the null hypothesis when H_A is true.
- ▶ We (almost) never know if H_0 or H_A is true, but we need to consider all possibilities.

p-values

- ▶ The probability, under H_0 , of observing something at least as extreme as what we observed.
- ▶ For fixed significance level α , reject H_0 if $\text{p-value} < \alpha$.

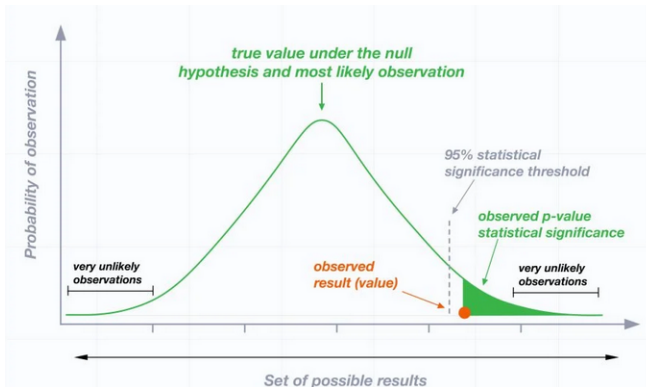


FIGURE: p-value is the green area under the curve
(Source: [Simply Psychology](#))

Hypothesis test for Proportions

- ▶ Hypothesis tests for **proportions**:

$$\begin{cases} H_0 : p = p_0 \\ H_A : p \neq p_0 \end{cases}$$

- ▶ Estimator:

$$\hat{p} \overset{\text{CLT}}{\sim} N\left(p_0, \text{S.E.}_{H_0} = \sqrt{\frac{p_0(1-p_0)}{n}}\right)$$

- ▶ Test statistic:

$$Z = \frac{\hat{p} - p_0}{\text{S.E.}_{H_0}} \sim N(0, 1)$$

Hypothesis test for Means

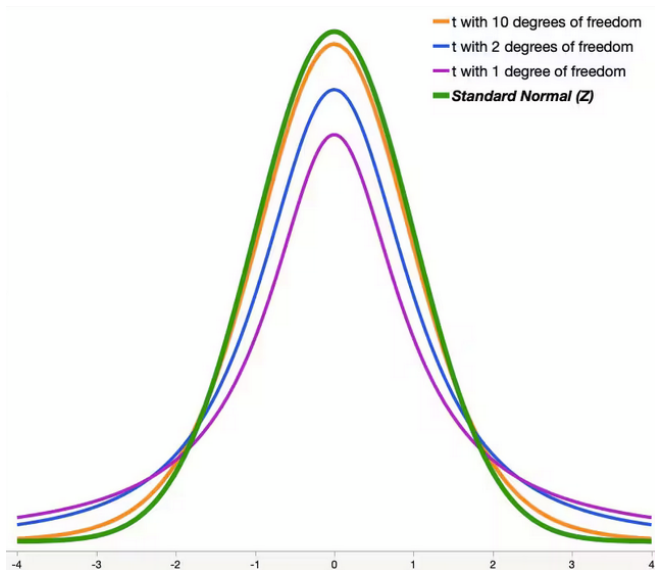
- ▶ Dataset $x_1, \dots, x_n$ with mean μ and standard deviation σ .
- ▶ Hypothesis tests for **means**:

$$\begin{cases} H_0 : \mu = \mu_0 \\ H_A : \mu \neq \mu_0 \end{cases}$$

- ▶ Estimator:

$$T = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}.$$

t_d vs. $N(0, 1)$



t -Distribution in Excel

- ▶ Suppose $X \sim t_d$. Then, $P(X \leq x)$ can be calculated by the Excel command:

`=T.DIST(x,d,TRUE)`

- ▶ If we know $P(X \leq x) = p$ where p is known, then x is given by the Excel command:

`=T.INV(p,d)`

Today: Paired Data

- ▶ Two sets of observations are **paired** if each observation in one set has a special correspondence or connection with exactly one observation in the other data set.
- ▶ Goal: analyze the **difference between the means** of these two sets of observations.

Example 1

Determine if the data in each scenario are paired.

1. Compare the exam scores of students before and after attending a special coaching class.
2. Assess the influence of city living by comparing pollution levels in various urban and rural areas.
3. Compare the blood pressure readings of patients at the beginning of a month and at the end of the month after consistently practicing meditation.
4. Assess the efficiency of a new solar panel model by comparing its performance with older models in various households.

Solutions

1. **Paired.** The same students' scores are being compared before and after coaching.
2. **Not Paired.** Different areas (urban vs rural) are being compared; there's no specific one-to-one correspondence.
3. **Paired.** The same patients' readings are compared at two different times.
4. **Not Paired.** Different solar panel models are being compared in various households, without a specific one-to-one correspondence between old and new models in the same setting.

Inference for Paired Data

- ▶ To analyze a paired data set, we simply analyze the differences.
- ▶ We can use the same t -distribution techniques we applied in the previous lecture, but this time for $\bar{x}_1 - \bar{x}_2$ instead of $\bar{x}$.

Paired Data Hypothesis Testing

- ▶ For paired data, our interest lies in analyzing the **differences** between the paired observations.
- ▶ Given paired data $\{(x_{1i}, x_{2i})\}_{i=1}^n$, define differences:

$$d_i := x_{1i} - x_{2i},$$

$i = 1, \dots, n$, and the average difference

$$\bar{x}_{\text{diff}} := \bar{x}_1 - \bar{x}_2.$$

- ▶ The hypothesis test takes the form

$$\begin{cases} H_0 : \mu_d = 0 \\ H_A : \mu_d \neq 0 \end{cases}$$

where $\mu_d := \mu_1 - \mu_2$ is the (theoretical) mean difference.

Test Statistic for Paired Data

- ▶ We use the sample differences to compute the sample mean difference, $\bar{x}_{\text{diff}}$, and sample standard deviation of differences, s_d .
- ▶ The test statistic for paired data is given by:

$$T = \frac{\bar{x}_{\text{diff}}}{\frac{s_d}{\sqrt{n}}}.$$

- ▶ T follows a t-distribution with $n - 1$ degrees of freedom under the null hypothesis.

Example 2

- ▶ Consider a study assessing the effect of a meditation program on blood pressure.
- ▶ For 15 patients, we measure blood pressure before and after a 6-week program. Check the Excel dataset `BloodPaired` on Canvas for the data.
- ▶ We want to test if the program has an effect on blood pressure.

Setting up the problem

- ▶ First, we formulate the problem as a hypothesis test problem:

$$\begin{cases} H_0 : \mu_d = 0 \\ H_A : \mu_d \neq 0 \end{cases}$$

- ▶ To decide whether we reject or fail to reject the null hypothesis, we need to compute two things:
 1. The **test statistic**
 2. The **p-value**
- ▶ We perform all these computations on Excel.

Test Statistic

- ▶ Suppose Columns A, B contain the original dataset, and in Column C we have stored the differences.
- ▶ Average difference:

$$=AVERAGE(C2:C16)$$

We find $\bar{x}_{\text{diff}} \approx -2.33$.

- ▶ Sample size: $n = 15$
- ▶ Sample standard deviation:

$$=STDEV.S(C2:C16)$$

We find $s \approx 3.04$.

- ▶ Test statistic:

$$T = \frac{\bar{x}_{\text{diff}}}{s/\sqrt{n}} \approx \frac{-2.33}{3.04/\sqrt{15}} \approx -2.97.$$

p-value and Conclusion

- ▶ Two-tailed test, so

$$\text{p-value} = 2P(t_{n-1} \leq T).$$

- ▶ In Excel, we compute this by

$$=2*\text{T.DIST}(T, 14, \text{TRUE})$$

- ▶ We find

$$\text{p-value} \approx 0.01 < 0.05,$$

so we **reject** the null hypothesis and conclude that there is enough evidence to suggest that the meditation program has a significant effect on reducing blood pressure.

Interpretation of the p-value

- ▶ Which of the following is the correct interpretation of the p-value?
 - a) Probability that the average blood pressure before and after the program is equal.
 - b) Probability that the average blood pressure before and after the program is different.
 - c) Probability of obtaining a random sample of 15 patients where the average difference in blood pressure (after-before) is at least 2.33 mm Hg (in either direction), if in fact the true average difference in blood pressure (after-before) is 0 mm Hg.
 - d) Probability of incorrectly rejecting the null hypothesis if in fact the null hypothesis is true.

- ▶ **Correct Answer: c)**

Example 3

Consider the differences between the math and science scores of a random sample of 250 students who took the FutureGenius High School Test. The mean and standard deviation of the differences are given by

$$\bar{x}_{\text{math-science}} = 1.235 \quad \text{and} \quad s = 9.342 \text{ points.}$$

- a) Calculate a 95% confidence interval for the average difference between the math and science scores of all students.
- b) Interpret this interval in context.
- c) Does the confidence interval provide convincing evidence that there is a real difference in the average scores? Explain.

Solution

a) Calculation of 95% Confidence Interval

- ▶ The C.I. is of the form

$$CI = \bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$$

- ▶ We compute t^* by typing =T.INV(0.975, 249) in Excel. We find $t^* \approx 1.97$. (Note how close this is to $z^* = 1.96$ now that the sample size is large!)
- ▶ Plugging in the values, we get

$$CI = 1.235 \pm 1.97 \cdot \frac{9.342}{\sqrt{250}} = [0.076, 2.394].$$

Solution (cont'd)

b) Interpretation of the Interval

If we were to take many random samples of 250 students from the population and calculate the confidence interval for each sample, about 95% of these intervals would contain the true average difference between the math and science scores for the entire student population.

(c) Evidence of Difference

The confidence interval does not contain 0, which suggests a statistically significant difference in the scores.

References

- ▶ Section 7.2 of the textbook

STOR 155 - Lecture 22

Difference of two means

Panagiotis Andreou

UNC Chapel Hill

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Difference of two means

- ▶ We consider the difference in two population means, $\mu_1 - \mu_2$, under the condition that the data are not paired.
- ▶ Key difference between this problem and the paired data problem:
 1. In paired data case, we consider two different aspects for each entity in a single population. Then, we pick up a random sample from that population, and our sample data has paired observations on the sampled entities.
 2. In difference of two means case, we have two entirely different populations from each of which we get a random sample independently.

Point Estimate

- ▶ We want to conduct the hypothesis test

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_A : \mu_1 \neq \mu_2 \end{cases}$$

- ▶ Natural point estimate for difference of means:

$$\bar{x}_1 - \bar{x}_2$$

- ▶ We need a new standard and some kind of CLT to conduct the test.

Standard Error

- ▶ Standard Error:

$$\text{S.E.} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}},$$

where n_1 and n_2 denote the sample sizes of the 1st and 2nd sample, respectively.

- ▶ Key assumption today: σ_1 and σ_2 are **known**!

Test Statistic

- ▶ The general test statistic in hypothesis tests:

$$\frac{\text{point estimate} - \text{theoretical quantity under } H_0}{\text{standard error}}$$

- ▶ In our case, we take

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- ▶ Using CLT, we can assume that Z follows roughly $N(0, 1)$ and the rest follows as in the previous lectures.

Case 1: $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$, and $\bar{x}_1 - \bar{x}_2 < 0$

- The p-value needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(Z \leq \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right) + P\left(Z \geq -\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right) \\ &= 2P\left(Z \leq \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right),\end{aligned}$$

where $Z \sim N(0, 1)$.

- In Excel, we can type

$$= 2 * \text{NORM.DIST}\left(\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}, 0, 1, \text{TRUE}\right)$$

Case 2: $H_0 : \mu_1 = \mu_2$ vs. $H_A : \mu_1 \neq \mu_2$, and $\bar{x}_1 - \bar{x}_2 > 0$

- The p-value needs to take into account **both tails**:

$$\begin{aligned}\text{p-value} &= P\left(Z \geq \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right) + P\left(Z \leq -\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right) \\ &= 2P\left(Z \geq \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}\right),\end{aligned}$$

where $Z \sim N(0, 1)$.

- In Excel, we can type

$$= 2 * \left(1 - \text{NORM.DIST} \left(\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}, 0, 1, \text{TRUE} \right) \right)$$

Example

- ▶ You are an avid wine collector and are particularly interested in the price dynamics of old wines.
- ▶ You have observed that there seems to be a significant price hike between 49-year-old wines and 50-year-old wines, even though the taste might be almost indistinguishable.
- ▶ To investigate this, you collected price data from random samples of both ages.
- ▶ Quantities you're interested in:

$$\mu_{49}, \mu_{50}, \mu_{49} - \mu_{50}$$

- ▶ Suppose we know from our data that:

$$\bar{x}_1 = 60.45, \bar{x}_2 = 70.05, \sigma_1 = 15.24, \sigma_2 = 18.20$$

Hypotheses

- ▶ If we want to test whether the average prices of the 49-year-old and 50-year-old wines are different, what are the null and alternative hypotheses?
- ▶ Hypothesis tests:

$$\begin{cases} H_0 : \mu_{49} = \mu_{50} \\ H_A : \mu_{49} \neq \mu_{50} \end{cases}$$

- ▶ We tend to work with $\mu_{49} - \mu_{50}$ to have a numerical value for comparison.

Point Estimate

- ▶ Using our data, the natural point estimate for $\mu_{49} - \mu_{50}$ is:

$$\bar{x}_{49} - \bar{x}_{50}$$

- ▶ From our data: $\bar{x}_{49} = 60.45, \bar{x}_{50} = 70.05$

- ▶ Hence, the point estimate is:

$$\bar{x}_{49} - \bar{x}_{50} = -9.60$$

Standard Error and Confidence Interval

- ▶ Standard Error (S.E.) is given by:

$$\text{S.E.} = \sqrt{\frac{\sigma_{49}^2}{n_{49}} + \frac{\sigma_{50}^2}{n_{50}}}$$

- ▶ Plugging in our data:

$$\text{S.E.} = \sqrt{\frac{15.24^2}{23} + \frac{18.20^2}{23}} \approx 4.65$$

- ▶ Critical value for a 95% confidence interval:

$$z^* = \text{NORM.INV}(0.975, 0, 1) \approx 1.96$$

- ▶ A 95% confidence interval is:

$$(-9.60) \pm 1.96 \cdot 4.65 \approx [-18.71, -0.49]$$

- ▶ Since 0 is not in this interval, we reject H_0 and conclude that there's a significant difference in the average prices of the wines.

Hypothesis Test

- ▶ For hypothesis tests, the Z-score is:

$$Z = \frac{\bar{x}_{49} - \bar{x}_{50}}{\text{S.E.}}$$

- ▶ Using our data:

$$Z = \frac{-9.60}{4.65} \approx -2.06$$

- ▶ The p-value is:

$$2 \cdot P(Z \leq -2.06)$$

- ▶ This gives a p-value of approximately 0.0394, so we reject H_0 at a 5% significance level.

References

- ▶ Section 7.3 of the textbook

STOR 155 - Lecture 23

Difference of two means with the t -distribution

Panagiotis Andreou

UNC Chapel Hill

March 21, 2025

Short Review of z , p , t

- ▶ Hypothesis test:

$$\begin{cases} H_0 : \mu = 70 \\ H_A : \mu \neq 70 \end{cases}$$

- ▶ Data might be $\bar{x} = 74, \sigma = 10, n = 32$.

Obtain $z = 2.26, p = \dots = 0.024$.

- ▶ Data might be $\bar{x} = 72, s = 11, n = 32$.

Obtain $t = 1.03, p = \dots = 0.312$.

- ▶ Very similar datasets can lead to very different decisions!

Previously

- ▶ We are interested in testing the equality of two means:

$$\begin{cases} H_0 : \mu_1 - \mu_2 = 0 \\ H_A : \mu_1 - \mu_2 \neq 0 \end{cases}$$

- ▶ Assuming that σ is known, we got

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1) \quad \text{and} \quad \text{p-value} = 2P(Z \geq |z|)$$

- ▶ However, in practice usually σ is unknown.

Today

- ▶ Estimate σ from the sample standard deviation s .
- ▶ *Problem:* We no longer have the usual CLT.
- ▶ *Solution:* Use the t -distribution!

Difference of two means

- ▶ We want to conduct the hypothesis test

$$\begin{cases} H_0 : \mu_1 - \mu_2 = 0 \\ H_A : \mu_1 - \mu_2 \neq 0 \end{cases}$$

- ▶ σ is not assumed to be known, so **estimate** it by s .
- ▶ Use the statistic

$$T = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- ▶ T follows a t -distribution with $\min\{n_1 - 1, n_2 - 1\}$ degrees of freedom.

Conditions to check

- ▶ As in the hypothesis test for proportions and the test for single mean, we first need to check two conditions:
 1. **Independence:** the observations must be independent, both within and between samples. Typically, this is satisfied when we have a random sample.
 2. If the datasets have more than 30 observations, we should check that there are no particularly extreme **outliers**.

Example 1

- ▶ You have graduated UNC with an eye-catching 3.6 GPA, have worked a couple of years and have decided to go back to school to obtain an MBA (Masters in Business Administration).
- ▶ You are deciding between Duke (Fuqua) and UNC (Kenan). As part of your decision, you are interested to know if students at the two programs tend to differ in age.
- ▶ Quantities you're interested in:

$$\mu_D, \mu_{NC}, \mu_D - \mu_{NC}$$

Data

- ▶ Find the dataset MBA_DukeUNC in Canvas.
- ▶ If we are interested in testing for whether the mean ages at Duke and UNC are different, what are good null and alternative hypotheses?
- ▶
$$\begin{cases} H_0 : \mu_D = \mu_{NC} \\ H_A : \mu_D \neq \mu_{NC} \end{cases} \quad \text{or} \quad \begin{cases} H_0 : \mu_D - \mu_{NC} = 0 \\ H_A : \mu_D - \mu_{NC} \neq 0 \end{cases}$$
- ▶ We tend to work with $\mu_D - \mu_{NC}$, so that we have a numerical value to work with.

Point Estimate

- ▶ Using our data, how can we get a point-estimate for $\mu_D - \mu_{NC}$?

- ▶ The natural estimate to take is

$$\bar{x}_D - \bar{x}_{NC}.$$

- ▶ From our data, we compute $\bar{x}_D \approx 27.41$, $\bar{x}_{NC} \approx 26.3$, so we get the point estimate

$$\bar{x}_D - \bar{x}_{NC} \approx 1.11$$

- ▶ While we have the spreadsheet open, we might as well find:

$$s_D, s_{NC}, n_D, n_{NC}$$

Standard Error and Confidence Interval

- ▶ In this scenario, the standard error is given by

$$\text{S.E.} = \sqrt{\frac{s_D^2}{n_D} + \frac{s_{NC}^2}{n_{NC}}}.$$

- ▶ In our dataset, $s_D \approx 3.28$, $s_{NC} \approx 2.6$, so

$$\text{S.E.} \approx 0.6998$$

- ▶ Therefore, a 95% confidence interval is given by

$$(\bar{x}_D - \bar{x}_{NC}) \pm 2.03 \cdot \sqrt{\frac{s_D^2}{n_D} + \frac{s_{NC}^2}{n_{NC}}} \approx [-0.29, 2.55]$$

- ▶ 0 is in the interval, so we fail to reject H_0 and conclude that the mean ages at Duke and UNC are not different!

Hypothesis Test

- ▶ We tested this claim using Confidence Intervals. Let's do the same using hypothesis testing.
- ▶ The statistic we'll use is the Z-score. In general,

$$Z = \frac{(\bar{x}_D - \bar{x}_{NC}) - (\mu_D - \mu_{NC})}{S.E.}.$$

- ▶ For hypothesis tests, we are working **under the null**, so $\mu_D - \mu_{NC} = 0$ and we get

$$T = \frac{\bar{x}_D - \bar{x}_{NC}}{S.E.} = \frac{\bar{x}_D - \bar{x}_{NC}}{\sqrt{\frac{s_D^2}{n_D} + \frac{s_{NC}^2}{n_{NC}}}}.$$

- ▶ In our dataset,

$$T \approx 1.61$$

- ▶ The p-value is

$$2 \cdot P(t_{33} \geq 1.61),$$

since

$$\min\{n_D - 1, n_{NC} - 1\} = \min\{33, 38\} = 33.$$

- ▶ We can compute this probability in Excel:

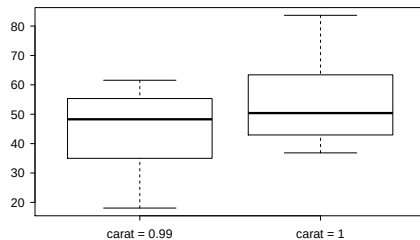
$$=2 * T.DIST(-1.61, 33, TRUE)$$

- ▶ We find a p-value of 0.11598, so we fail to reject at significance level $\alpha = 5\%$.

Example 2

- ▶ Weights of diamonds are measured in carats.
- ▶ $1 \text{ carat} = 100 \text{ points}$, $0.99 \text{ carats} = 99 \text{ points}$, etc.
- ▶ The difference between the size of a 0.99 carat diamond and a 1 carat diamond is undetectable to the naked human eye, but does the price of a 1 carat diamond tend to be higher than the price of a 0.99 diamond?
- ▶ We are going to test to see if there is a difference between the average prices of 0.99 and 1 carat diamonds.
- ▶ In order to be able to compare equivalent units, we divide the prices of 0.99 carat diamonds by 99 and 1 carat diamonds by 100, and compare the average point prices.

Data



	0.99 carat pt99	1 carat pt100
$\bar{x}$	44.50	53.43
s	13.32	12.22
n	23	30

Parameter and point estimate

- ▶ Parameter of interest: Average difference between the point prices of all 0.99 carat and 1 carat diamonds.

$$\mu_{pt99} - \mu_{pt100}$$

- ▶ Point estimate: Average difference between the point prices of sampled 0.99 carat and 1 carat diamonds.

$$\bar{x}_{pt99} - \bar{x}_{pt100}$$

Hypotheses

Which of the following is the correct set of hypotheses for testing if the average point price of 1 carat diamonds (μ_{pt100}) is higher than the average point price of 0.99 carat diamonds (μ_{pt99})?

1. $H_0 : \mu_{pt99} = \mu_{pt100}$
 $H_A : \mu_{pt99} \neq \mu_{pt100}$
2. $H_0 : \mu_{pt99} = \mu_{pt100}$
 $H_A : \mu_{pt99} > \mu_{pt100}$
3. $H_0 : \mu_{pt99} = \mu_{pt100}$
 $H_A : \mu_{pt99} < \mu_{pt100}$
4. $H_0 : \bar{x}_{pt99} = \bar{x}_{pt100}$
 $H_A : \bar{x}_{pt99} < \bar{x}_{pt100}$

Answer: 3.

Conditions

Which of the following does not need to be satisfied in order to conduct this hypothesis test using theoretical methods?

1. Point price of one 0.99 carat diamond in the sample should be independent of another, and the point price of one 1 carat diamond should independent of another as well.
2. Point prices of 0.99 carat and 1 carat diamonds in the sample should be independent.
3. Distributions of point prices of 0.99 and 1 carat diamonds should not be extremely skewed.
4. Both sample sizes should be at least 30.

Answer: 4.

Test statistic

- ▶ General form of test statistic:

$$T = \frac{\text{point estimate} - \text{null value}}{SE}$$

- ▶ Here:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \quad \text{and} \quad df = \min(n_1 - 1, n_2 - 1)$$

- ▶ *Note:* The calculation of the df is actually much more complicated. For simplicity we'll use the above formula to estimate the true df when conducting the analysis by hand.

Test statistic (cont.)

	0.99 carat pt99	1 carat pt100
$\bar{x}$	44.50	53.43
s	13.32	12.22
n	23	30

$$\begin{aligned} T &= \frac{\text{point estimate} - \text{null value}}{SE} \\ &= \frac{(44.50 - 53.43) - 0}{\sqrt{\frac{13.32^2}{23} + \frac{12.22^2}{30}}} \\ &= \frac{-8.93}{3.56} \\ &= -2.508 \end{aligned}$$

Test statistic (cont.)

Which of the following is the correct df for this hypothesis test?

1. 22
2. 23
3. 30
4. 29
5. 52

Answer: 1.

p-value

- ▶ We can find this in Excel by typing

=T.DIST(-2.508, 22, TRUE)

- ▶ We find p-value= 0.010007.
- ▶ Conclusions:
 1. p-value smaller than 5%, so **reject** H_0 at significance level $\alpha = 0.05$.
 2. In the context of the problem, the data provide convincing evidence to suggest that the point price of 0.99 carat diamonds is lower than the point price of 1 carat diamonds.
 3. Maybe buy a 0.99 carat diamond? It looks like a 1 carat, but is significantly cheaper.

References

- ▶ Section 7.3 of the textbook

STOR 155 - Lecture 24

Difference of Proportions

Panagiotis Andreou

UNC Chapel Hill

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Inference for Proportions

- ▶ Hypothesis Test:

$$\begin{cases} H_0 : p = p_0 \\ H_A : p \neq p_0 \end{cases}$$

- ▶ Test statistic:

$$z = \frac{\hat{p} - p_0}{S.E.},$$

where

$$S.E. = \sqrt{\frac{p_0(1 - p_0)}{n}}$$

- ▶ We can assume that $Z \sim N(0, 1)$ only if **CLT conditions** are satisfied.

Central Limit Theorem

Theorem (CLT)

Sample proportions will be nearly normally distributed:

$$\hat{p} \sim N \left(\text{mean} = p, SE = \sqrt{\frac{p(1-p)}{n}} \right)$$

- ▶ The sampling distribution is symmetric and centered at the true population proportion.
- ▶ Note that as n increases SE decreases.
- ▶ As n increases, samples will yield more consistent $\hat{p}$'s, i.e., variability among $\hat{p}$ s will be lower.

CLT - conditions

Certain conditions must be met for the CLT to apply:

1. **Independence:** Sampled observations must be independent. This is difficult to verify, but is more likely if
 - ▶ random sampling/assignment is used, and
 - ▶ if sampling without replacement, $n < 10\%$ of the population.
2. **Success-Failure Condition:** There should be at least 10 expected successes and 10 expected failures in the observed sample, i.e.,

$$np \geq 10 \quad \text{and} \quad n(1 - p) \geq 10.$$

- ▶ Typically we don't know the true proportion p , so we substitute some value to check conditions and estimate the standard error.
- ▶ For confidence intervals, the sample proportion $\hat{p}$ is used.
- ▶ For hypothesis tests, typically the null value p_0 is used.

Two ways of testing

1. **Confidence interval:**
2. **p-value:**

Example

- ▶ A simple random sample of 826 payday loan borrowers was surveyed to better understand their interests around regulation and costs.
- ▶ 70% of the responses supported new regulations on payday lenders.
- ▶ Construct a 95% confidence interval for p , the proportion of payday borrowers who support increased regulation for payday lenders.

First step

► First, we check the conditions:

1. The data are a random sample, so the observations are independent and representative of the population of interest.
2. Success-failure condition:

$$np \approx 826 \cdot 0.7 = 578 > 10 \quad \text{and} \quad n(1-p) \approx 826 \cdot 0.3 = 248 > 10,$$

so the conditions are satisfied.

- Note: we use $\hat{p}$ instead of p to check the success-failure condition, that's why the result is approximate!

The remaining steps

- ▶ Standard error:

$$S.E. = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{0.7 \cdot 0.3826} = 0.016$$

- ▶ Critical value:

$$z^* = z_{0.975} = \text{NORM.INV}(0.975, 0, 1) \approx 1.96$$

- ▶ 95% Confidence Interval:

$$0.7 \pm 1.96 * 0.016 \longrightarrow (0.669, 0.731)$$

Hypothesis Testing with p-values

- ▶ Hypothesis test:

$$\begin{cases} H_0 : \hat{p} = p_0 \\ H_A : \hat{p} \neq p_0 \end{cases}$$

- ▶ Test statistic:

$$z = \frac{\hat{p} - p_0}{S.E.}$$

- ▶ p-value:

$$2 \cdot P(Z \geq |z|)$$

- ▶ If $\text{p-value} < \alpha$, reject H_0 .

Example

- ▶ One possible regulation for payday lenders is that they would be required to do a credit check and evaluate debt payments against the borrower's finances.
- ▶ We would like to know: would borrowers support this form of regulation?
- ▶ From a random sample of 826 borrowers, 51% said they would support such a regulation.
- ▶ Set up hypotheses to evaluate whether borrowers have a majority support or majority opposition for this type of regulation.

First steps

- ▶ Hypothesis test:

$$\begin{cases} H_0 : p = 0.5 \\ H_A : p \neq 0.5 \end{cases}$$

- ▶ CLT conditions:

1. The data are a random sample, so the observations are independent and representative of the population of interest.
2. Success-failure condition:

$$np_0 = 826 \cdot 0.5 = 413 > 10 \quad \text{and} \quad n(1-p_0) = 826 \cdot 0.5 = 413 > 10,$$

so the conditions are satisfied.

- ▶ Note: we use p_0 now, since we assume H_0 is the ground truth!

Conducting the test

- ▶ Standard Error:

$$S.E. = \sqrt{\frac{p_0(1 - p_0)}{n}} = \frac{0.5}{\sqrt{826}} = 0.017$$

- ▶ Test statistic:

$$z = \frac{\hat{p} - p_0}{S.E.} = \frac{0.51 - 0.5}{0.017} = 0.59$$

- ▶ p-value:

$$2 \cdot P(Z \geq 0.59) = 2 * (1 - \text{NORM.DIST}(z, 0, 1, \text{TRUE})) = 0.5552$$

- ▶ Fail to reject H_0 at any reasonable significance level.

Sample Size

- ▶ The **margin of error** for a sample proportion p is defined as

$$z^* \sqrt{\frac{p(1-p)}{n}}$$

- ▶ *Problem:* For a $100(1 - \alpha)\%$ confidence interval and a fixed number $c > 0$, find the **minimum sample size** n such that the margin of error is less than c .

Example

- ▶ Find the minimum sample size so that the margin of error for a 95% confidence interval is less than 0.04.
- ▶ First obstacle: what is p ?
- ▶ When p is not given, we can take $p = 0.5$, since this maximizes the function $p(1 - p)$ in $[0, 1]$ (proof by Calculus).
- ▶ We want the min n for which

$$z^* \sqrt{\frac{0.5(1 - 0.5)}{n}} < 0.04.$$

► 95% confidence interval, so $z^* = 1.96$.

► Solve for n :

$$z^* \sqrt{\frac{p(1-p)}{n}} < 0.04 \Leftrightarrow 1.96 \cdot \frac{0.5}{\sqrt{n}} < 0.04 \Leftrightarrow n > 600.25$$

► Since n is an integer, we take

$$n_{\min} = 601.$$

The goal

- ▶ Extend what we did last time from a single proportion p to a difference of two proportions, $p_1 - p_2$.
- ▶ Similar to the setup of “difference of means”, but this time for categorical data.
- ▶ We'll follow the usual steps:
 1. Identify a reasonable point estimate of $p_1 - p_2$.
 2. Check CLT conditions for Normal approximation.
 3. Find standard error and proceed with finding a confidence interval or conducting a hypothesis test via p-values.

Point Estimate and Conditions

- ▶ The natural point estimate to take is $\hat{p}_1 - \hat{p}_2$.
- ▶ Conditions:
 1. **Independence (extended)**: The data are independent within and between the two groups. Generally this is satisfied if the data come from two independent random samples or if the data come from a randomized experiment.
 2. **Success-Failure**: The success-failure condition must hold for **both** groups, where we check successes and failures in each group separately.

Standard Error

- ▶ When the CLT conditions are met, the standard error of $\hat{p}_1 - \hat{p}_2$ is defined to be

$$S.E. = \sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}}$$

- ▶ Using this, we can construct confidence intervals and perform hypothesis tests.

Confidence Interval

- ▶ General form of confidence interval:

point estimate $\pm z^*$ standard error

- ▶ Here, we get the $100(1 - \alpha)\%$ confidence interval

$$\hat{p}_1 - \hat{p}_2 \pm z_{1-\alpha/2} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

for the difference $p_1 - p_2$.

- ▶ We can also follow the same Prepare, Check, Calculate, Conclude steps for computing a confidence interval or completing a hypothesis test. The details change a little, but the general approach remain the same.

Example

- ▶ We consider an experiment for patients who underwent cardiopulmonary resuscitation (CPR) for a heart attack and were subsequently admitted to a hospital.
- ▶ These patients were randomly divided into a treatment group where they received a blood thinner or the control group where they did not receive a blood thinner.
- ▶ The outcome variable of interest was whether the patients survived for at least 24 hours. The results are:

	Survived	Died	Total
Control	11	39	50
Treatment	14	26	40
Total	25	65	90

TABLE: Results for the CPR study. Patients in the treatment group were given a blood thinner, and patients in the control group were not.

CLT Conditions

1. **Independence:** since this is a randomized experiment, this condition is satisfied.
2. **Success-Failure:** We have at least 10 successes and 10 failures in each experiment arm (11, 14, 39, 26), so this condition is also satisfied

Confidence Interval

- ▶ We want to create and interpret a 90% confidence interval of the difference for the survival rates in the CPR study.
- ▶ We'll use p_t for the survival rate in the treatment group, and p_c for the control group.
- ▶ **Point estimate:**

$$\hat{p}_t - \hat{p}_c = \frac{14}{40} - \frac{11}{50} = 0.35 - 0.22 = 0.13$$

- ▶ **Standard error:**

$$S.E. = \sqrt{\frac{0.35 \cdot 0.65}{40} + \frac{0.22 \cdot 0.78}{50}} = 0.095$$

Confidence Interval (cntd.)

- ▶ Critical value for 90% C.I.:

$$z^* = z_{0.95} = \text{NORM.INV}(0.95, 0, 1) \approx 1.65$$

- ▶ General form of C.I.:

$$\text{point estimate} \pm z^* \cdot \text{standard error}$$

- ▶ Here, we get the 90% confidence interval

$$0.13 \pm 1.65 \cdot 0.095 \longrightarrow (-0.027, 0.287)$$

for the difference $p_t - p_s$.

- ▶ **Conclusion:** Since 0 is contained in the interval, we do not have enough information to say whether blood thinners help or harm heart attack patients who have been admitted after they have undergone CPR.

References

- ▶ Section 6.1 of the textbook
- ▶ Section 6.2 of the textbook

STOR 155 - Lecture 25

Difference of Proportions

Panagiotis Andreou

UNC Chapel Hill

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Hypothesis Testing

- ▶ We want to test:

$$\begin{cases} H_0 : p_1 = p_2 \\ H_A : p_1 \neq p_2 \end{cases}$$

- ▶ Equivalently:

$$\begin{cases} H_0 : p_1 - p_2 = 0 \\ H_A : p_1 - p_2 \neq 0 \end{cases}$$

- ▶ We'll introduce a new type of estimate, the **pooled proportion**.

Pooled Proportion

- ▶ When the null hypothesis is that the proportions are equal, use the pooled proportion ($\hat{p}_{\text{pooled}}$) to verify the success-failure condition and estimate the standard error:

$$\hat{p}_{\text{pooled}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2} = \frac{\text{number of "successes"}}{\text{number of cases}}$$

- ▶ $\hat{p}_i n_i$ represents the number of successes in sample 1, since

$$\hat{p}_i = \frac{\text{number of successes in sample } i}{n_i},$$

for $i = 1, 2$.

Example

- ▶ A mammogram is an X-ray procedure used to check for breast cancer.
- ▶ Whether mammograms should be used is part of a controversial discussion.
- ▶ A 30-year study was conducted with nearly 90,000 female participants.
- ▶ During a 5-year screening period, each woman was randomized to one of two groups: in the first group, women received regular mammograms to screen for breast cancer, and in the second group, women received regular non-mammogram breast cancer exams.

- ▶ No intervention was made during the following 25 years of the study, and we'll consider death resulting from breast cancer over the full 30-year period. Results from the study:

	Death from breast cancer?	
	Yes	No
Mammogram	500	44,425
Control	505	44,405

- ▶ If mammograms are much more effective than non-mammogram breast cancer exams, then we would expect to see additional deaths from breast cancer in the control group.
- ▶ On the other hand, if mammograms are not as effective as regular breast cancer exams, we would expect to see an increase in breast cancer deaths in the mammogram group.

Hypothesis Test setup

- ▶ Formulate the scenario as a hypothesis testing problem.
- ▶ H_0 : the breast cancer death rate for patients screened using mammograms is the same as the breast cancer death rate for patients in the control.

H_A : the breast cancer death rate for patients screened using mammograms is different than the breast cancer death rate for patients in the control

- ▶ Difference of proportions:

$$\begin{cases} H_0 : p_{mgm} - p_{ctrl} = 0 \\ H_A : p_{mgm} - p_{ctrl} \neq 0 \end{cases}$$

Pooled Proportion

- ▶ To check the CLT conditions in the case of $H_0 : p_1 - p_2 = 0$, we use

$$\begin{aligned}\hat{p}_{\text{pooled}} &= \frac{\text{\#patients who died from breast cancer}}{\text{\#of patients}} \\ &= \frac{500 + 505}{500 + 44,425 + 505 + 44,405} = 0.0112\end{aligned}$$

- ▶ This proportion is an estimate of the breast cancer death rate across the entire study, and it's our best estimate of the proportions p_{mgm} and p_{ctrl} if $H_0 : p_{mgm} = p_{ctrl}$ is true.
- ▶ We will also use this pooled proportion when computing the standard error.

Conditions

- ▶ We need to check the CLT conditions before we conduct the test.
- ▶ **Independence:** Because the patients are randomized, they can be treated as independent, both within and between groups.
- ▶ **Success-Failure:** We check it for each group using $\hat{p}_{\text{pooled}}$:

$$\hat{p}_{\text{pooled}} \cdot n_{\text{mgm}} = 0.0112 \cdot 44,925 \approx 503 \geq 10$$

$$(1 - \hat{p}_{\text{pooled}}) \cdot n_{\text{mgm}} = 0.9888 \cdot 44,925 \approx 44,422 \geq 10$$

$$\hat{p}_{\text{pooled}} \cdot n_{\text{ctrl}} = 0.0112 \cdot 44,910 \approx 503 \geq 10$$

$$(1 - \hat{p}_{\text{pooled}}) \cdot n_{\text{ctrl}} = 0.9888 \cdot 44,910 \approx 44,407 \geq 10$$

Point Estimate

- ▶ Subtract the individual point estimates.
- ▶ The point estimate of the difference in breast cancer death rates is

$$\begin{aligned}\hat{p}_{mgm} - \hat{p}_{ctrl} &= \frac{500}{500 + 44,425} - \frac{505}{505 + 44,405} \\ &= 0.01113 - 0.01125 = -0.00012\end{aligned}$$

Standard Error

- ▶ Use $\hat{p}_{\text{pooled}}$ for the point estimate we need for the standard error.
- ▶ The only difference is in the sample sizes:

$$\begin{aligned} S.E. &= \sqrt{\frac{\hat{p}_{\text{pooled}}(1 - \hat{p}_{\text{pooled}})}{n_{\text{mgm}}} + \frac{\hat{p}_{\text{pooled}}(1 - \hat{p}_{\text{pooled}})}{n_{\text{ctrl}}}} \\ &= 0.00070 \end{aligned}$$

Test Statistic and p-value

- ▶ Test statistic:

$$z = \frac{\text{point estimate} - \text{null value}}{\text{standard error}} = \frac{-0.00012 - 0}{0.0007} = -0.17$$

- ▶ p-value:

$$2P(Z \geq |z|) = 2P(Z \geq 0.17) = 0.865$$

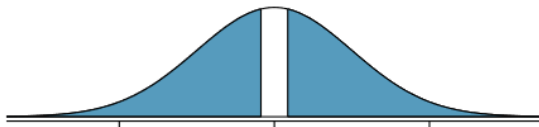


FIGURE: The p-value as area below the $N(0,1)$ density curve.

Conclusion

- ▶ The p-value is larger than any reasonable significance level, so we fail to reject the null hypothesis.
- ▶ In the context of the problem, we don't have sufficient evidence to conclude that mammograms reduce or increase breast cancer deaths.
- ▶ If mammograms are helpful or harmful, the data suggest the effect isn't very large.

Example

A survey asked 850 randomly sampled registered voters:

“Do you support Candidate A in the upcoming election?”

Responses were split based on age group:

	Age < 40	Age $\geq$ 40
Support	180	200
Do not support	220	250

Set up Hypothesis Test

We want to know if there's a significant difference in support between the two age groups.

$$H_0 : p_1 = p_2 \quad (\text{no difference in support})$$

$$H_A : p_1 \neq p_2 \quad (\text{there is a difference})$$

Where:

- ▶ p_1 = proportion of voters under 40 who support Candidate A
- ▶ p_2 = proportion of voters at least 40 who support Candidate A

Sample Proportions

Calculate the sample proportions:

$$\hat{p}_1 = \frac{180}{180 + 220} = \frac{180}{400} = 0.45$$

$$\hat{p}_2 = \frac{200}{200 + 250} = \frac{200}{450} \approx 0.4444$$

$$\hat{p}_1 - \hat{p}_2 = 0.45 - 0.4444 = 0.0056$$

Pooled Proportion

Under $H_0 : p_1 = p_2$, we use the pooled proportion:

$$\hat{p}_{\text{pooled}} = \frac{180 + 200}{400 + 450} = \frac{380}{850} \approx 0.4471$$

Independence satisfied due to random sampling.

Check success-failure condition:

$$\hat{p}_{\text{pooled}} \cdot 400 = 0.4471 \cdot 400 \approx 179 \geq 10$$

$$(1 - \hat{p}_{\text{pooled}}) \cdot 400 \approx 221 \geq 10$$

$$\hat{p}_{\text{pooled}} \cdot 450 \approx 201 \geq 10$$

$$(1 - \hat{p}_{\text{pooled}}) \cdot 450 \approx 249 \geq 10$$

Conditions are satisfied.

Standard Error and Test Statistic

Standard Error (SE):

$$SE = \sqrt{\frac{\hat{p}_{\text{pooled}}(1 - \hat{p}_{\text{pooled}})}{n_1} + \frac{\hat{p}_{\text{pooled}}(1 - \hat{p}_{\text{pooled}})}{n_2}}$$

$$SE = \sqrt{\frac{0.4471(1 - 0.4471)}{400} + \frac{0.4471(1 - 0.4471)}{450}} \\ \approx 0.034166$$

Test Statistic (z):

$$z = \frac{0.0056 - 0}{0.034166} \approx 0.16$$

p-value and Conclusion

p-value:

$$p = 2P(Z > 0.16) = 2(1 - P(Z \leq 0.16)) \approx 0.82$$

Conclusion:

- ▶ Since $p = 0.82 > 0.05$, we fail to reject H_0 .
- ▶ There is not enough evidence to conclude that support for Candidate A differs by age group.

References

- ▶ Section 6.2 of the textbook

STOR 155 - Lecture 26

Correlation

Panagiotis Andreou

UNC Chapel Hill

April 7, 2025

Introduction to Correlation

- ▶ Correlation measures the strength and direction of a linear relationship between two numerical variables.
- ▶ The value of correlation ranges between -1 and 1.
- ▶ A value close to 1 implies a strong positive correlation: as one variable increases, the other also tends to increase.
- ▶ A value close to -1 implies a strong negative correlation: as one variable increases, the other tends to decrease.
- ▶ A value close to 0 implies little to no linear correlation between the variables.

Pearson's r

- ▶ The most common method to determine correlation is Pearson's r .
- ▶ Formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right),$$

where $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$.

- ▶ r is sensitive to outliers.

Historical note

- ▶ Karl Pearson (1857-1936) was a British mathematician who created the field of Mathematical Statistics.
- ▶ He founded the first Statistics department ever (at UCL)!
- ▶ Quote: *“All great scientists have, in a certain sense, been great artists; a person with no imagination may collect facts, but cannot make great discoveries.”*



Interpretation of Correlation

- ▶ $r > 0$: Positive correlation.
- ▶ $r < 0$: Negative correlation.
- ▶ $r = 0$: No correlation.
- ▶ The closer r is to 1 or -1, the stronger the correlation.
- ▶ Correlation is **not** robust to outliers!

Scatterplots and Correlation

- ▶ Scatterplots visually represent the relationship between **two** numerical variables.
- ▶ The correlation coefficient r can often be estimated by the tightness and slope of the points in the scatterplot.
- ▶ Scatterplots also help in identifying outliers or any non-linear patterns in the data.

Examples of Correlations



Correlation $r = 0$



Correlation $r = 0.5$



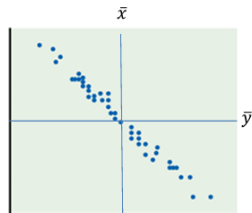
Correlation $r = 0.9$



Correlation $r = -0.3$



Correlation $r = -0.7$



Correlation $r = -0.99$

Correlation and Outliers

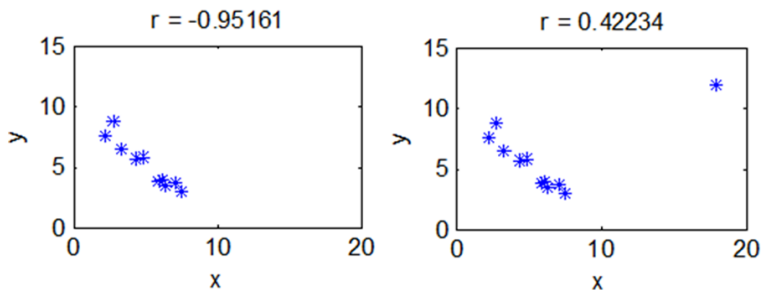


FIGURE: The left figure shows almost perfect linear relationship (correlation close to -1). The right figure has a much smaller correlation in absolute value, which shows how much an outlier is affecting correlation.

Limitations and Cautions

- ▶ Correlation measures linear relationships. Non-linear relationships might be overlooked.
- ▶ Correlation is **not** robust against outliers. A single outlier can significantly change r .
- ▶ Always plot the data to verify the relationship visually.
- ▶ Remember: **Correlation does not imply causation**. A third factor might influence both variables.

Correlation and Non-Linear Data

- ▶ Correlation measures the **linear** relationship between two numerical variables.
- ▶ Non-linear relationships, even strong ones, might have a correlation value close to 0.

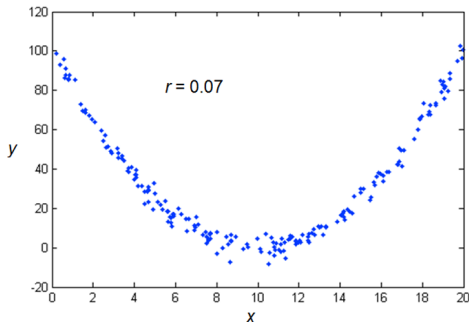


FIGURE: r describes how well the points fit to a straight line, and with the direction of that line.

Non-Linear Data

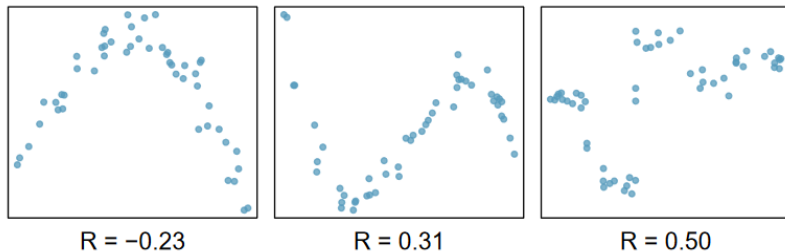
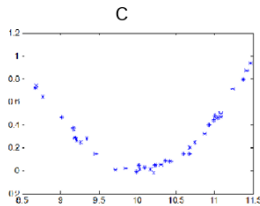
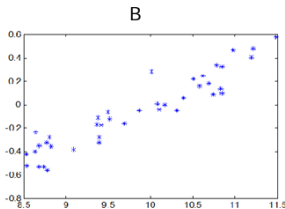
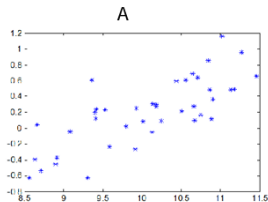


FIGURE: Scatterplots and their correlations. In each case, there is a strong relationship between the variables. However, because the relationship is nonlinear, the correlation is relatively weak.

Practice Question 1 - Pick Scatterplot

► Which scatterplot has the largest correlation?

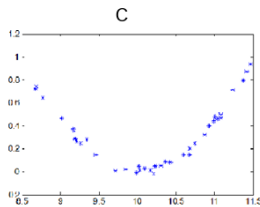
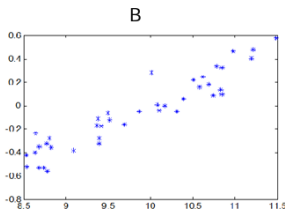
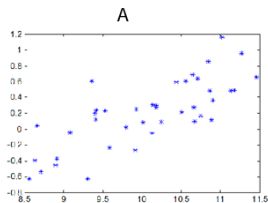
1. Scatterplot A
2. Scatterplot B
3. Scatterplot C
4. Impossible to tell



Practice Question 1 - Pick Scatterplot

► Which scatterplot has the largest correlation?

1. Scatterplot A
2. Scatterplot B
3. Scatterplot C
4. Impossible to tell



Practice Question 2 - Social Rejection and Pain

- ▶ We often describe our emotional reaction to social rejection as “pain”.
- ▶ A study asked whether social rejection causes activity in areas of the brain that are known to be activated by physical pain.
- ▶ If it does, we really do experience social and physical pain in similar ways.
- ▶ Subjects were first included and then deliberately excluded from a social activity while increases in blood flow in their brains were measured.
- ▶ After each activity, the subjects filled out questionnaires that assessed how excluded they felt.

- ▶ You can find the dataset in the Excel spreadsheet SocialDist.xlsx

- ▶ The questions are the following:
 1. Scatterplot
 2. Describe association (direction, strength, form)
 3. Calculate correlation

Recap

- ▶ Correlation is a powerful statistical tool that helps us quantify the strength and direction of a **linear** relationship.
- ▶ Remember, correlation does not imply causation. Further research is often required to establish causality.
- ▶ Always visualize your data with scatterplots to complement the correlation analysis and spot outliers.

References

- ▶ Section 8.1 of the textbook
- ▶ Great resource for the more curious:
Pearson's Correlation, Clearly Explained (StatQuest)

STOR 155 - Lecture 27

Linear Regression

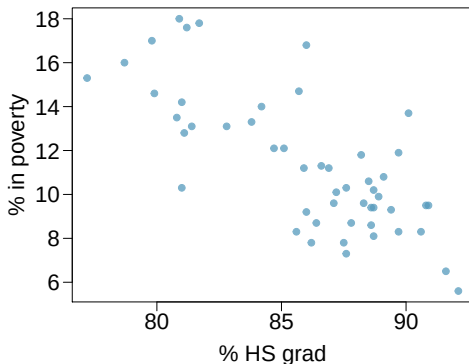
Panagiotis Andreou

UNC Chapel Hill

April 9, 2025

Poverty vs. HS graduate rate

The scatterplot below shows the relationship between HS graduate rate in all 50 US states and DC and the % of residents who live below the poverty line (income below \$23,050 for a family of 4 in 2012).



- ▶ Suppose we want to fit a theoretical model on that data.
- ▶ The simplest possible model is a **linear** one.
- ▶ The linear model for predicting poverty from high school graduation rate in the US is

$$\hat{poverty} = 64.78 - 0.62 \cdot HS_{grad}$$

The “hat” is used to signify that this is an estimate.

- ▶ The high school graduate rate in Georgia is 85.1%. What poverty level does the model predict for this state?

$$64.78 - 0.62 \cdot 85.1 = 12.018$$

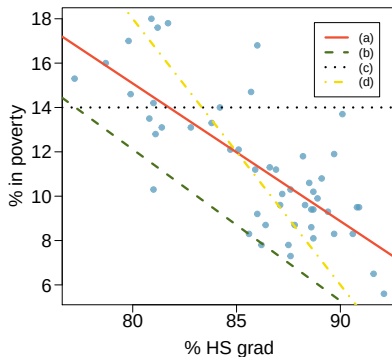
- ▶ **This ability for prediction showcases the power of having a statistical model!**

Technical terms

- ▶ **Correlation:** linear, negative, moderately strong.
- ▶ **Response** variable: “in poverty” (that’s the variable of interest).
- ▶ **Explanatory** variable: “HS grad” (that’s the variable we are using to study the variable of interest).

Eyeballing the line

Which of the following appears to be the line that best fits the linear relationship between % in poverty and % HS grad?

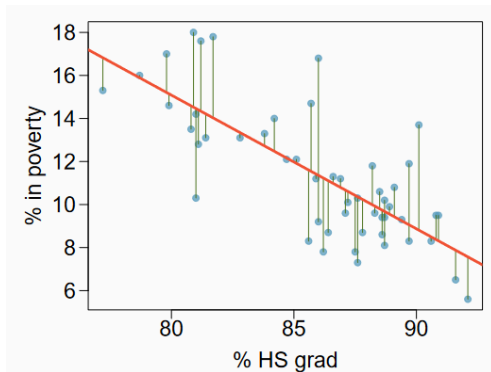


The red line.

Residuals

Residuals are the leftovers from the model fit:

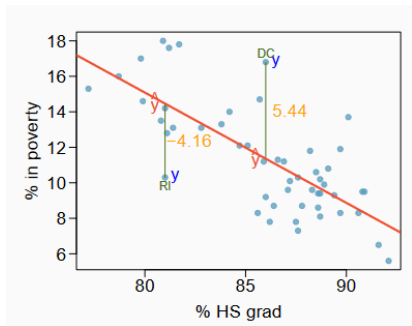
$$\text{Data} \approx \text{Fit} + \text{Residual}$$



Residuals (cont.)

Residual for the i -th observation:

$$e_i = y_i - \hat{y}_i$$



- ▶ % living in poverty in DC is 5.44% more than predicted.
- ▶ % living in poverty in RI is 4.16% less than predicted.

The Problem

- ▶ Pairs of data points:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

- ▶ The observations $x_1, \dots, x_n$ come from a random variable X , while the observations $y_1, \dots, y_n$ come from a random variable Y .
- ▶ We want to construct a model

$$y_i = f(x_i) + \varepsilon_i$$

that connects the random variables X and Y .

- ▶ We focus on linear functions f .

$$y = f(x) = b_0 + b_1x$$

Terminology

- ▶ Such a model allows us to predict Y for any value of X .
- ▶ X is called the **explanatory** or **independent** or **predictor** variable.
- ▶ Y is called the **response** or **dependent** variable.

Different scenarios

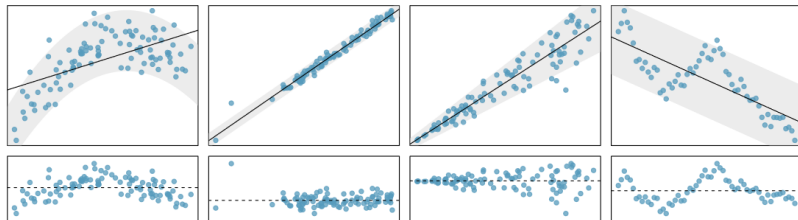


FIGURE: 1st: linearity fails

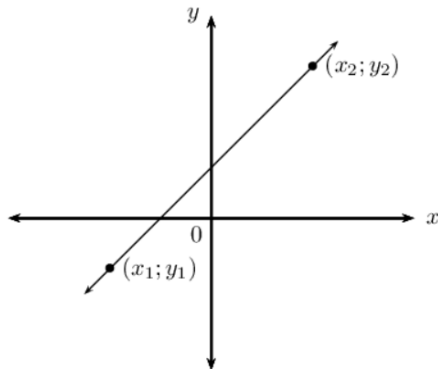
2nd: existence of outliers

3rd: the variability of the errors is related to the value of x

4th: a time series data set is shown, where successive observations are highly correlated

The geometry of straight lines

- ▶ The simplest object of analytic geometry.
- ▶ We need only two parameters to fully describe a straight line.
- ▶ Either slope and one point, or two points.



Equations of a straight line

- ▶ The two-point form of the straight line equation:

$$\frac{y - y_1}{x - x_1} = \frac{y - y_2}{x - x_2},$$

if we know **two points** (x_1, y_1) and (x_2, y_2) of the line.

- ▶ The gradient–point form of the straight line equation:

$$y - y_1 = m(x - x_1),$$

if we know **one point** (x_1, y_1) and the **slope** m of the line.

- ▶ The gradient–intercept form of the straight line equation:

$$y = mx + c,$$

if we know the **intercept** $(0, c)$ and the **slope** m of the line.

Finding the best line

- ▶ What does “best” mean in this case?
- ▶ Collectively small residuals.
- ▶ Potential minimizing criterion:

$$\min(|e_1| + \cdots + |e_n|)$$

- ▶ Other criterion:

$$\min(e_1^2 + \cdots + e_n^2)$$

- ▶ We use the last one, which leads to the **least squares regression**.

Least Squares

- ▶ We need an optimality criterion to find this straight line.
- ▶ Minimize sum of squares of residuals.
- ▶ Residual: Difference between observed value and value predicted by regression line.
- ▶ $SSE = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$
- ▶ The line that minimizes the SSE is our “best” line.

The Regression Equation

- ▶ The equation for the best-fitting (least squares) line is

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x,$$

where

- ▶ $\hat{y}$ is the predicted value of the response variable.
 - ▶ $\hat{b}_0$ is the y-intercept.
 - ▶ $\hat{b}_1$ is the slope.
 - ▶ x is the value of the explanatory variable.
- ▶ $\hat{b}_0$ and $\hat{b}_1$ are calculated based on

$$\arg \min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Finding the Slope and Intercept

Two formulas:

► Slope:

$$\hat{b}_1 = r \frac{s_y}{s_x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

► Intercept:

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

► Once $\hat{b}_0$ and $\hat{b}_1$ are known, any **prediction** for y given a value of x can be made:

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x$$

Coefficient of Determination R^2

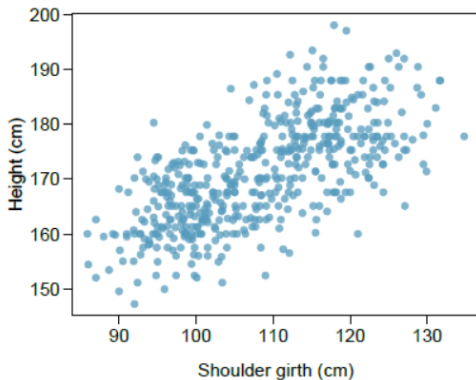
- ▶ Connection with Pearson's correlation.
- ▶ Informally speaking,

$$R^2 \approx \frac{\text{variation explained by the model}}{\text{total variation}}$$

- ▶ Example: If a linear model has a very strong negative relationship with a correlation of -0.97, how much of the variation in the response is explained by the explanatory variable?
- ▶ About $R^2 = (0.97)^2 = 0.94$ or 94% of the variation is explained by the linear model.

Practice Example

Consider the following data on shoulder girth and height of a group of physically active adults. The mean shoulder girth is 107.60 cm with a standard deviation of 10.39 cm. The mean height is 171.15 cm with a standard deviation of 9.42 cm. The correlation between height and shoulder girth is 0.68.



Questions

1. Write the equation of the regression line for predicting height. (Let y represent height in centimeters and let x represent shoulder girth in centimeters. Round your numerical values to three decimal places.)
2. Interpret the slope and intercept in this context.
3. Calculate R^2 of the regression line for predicting height from shoulder girth. Interpret it in the context of the problem.
4. A randomly selected student from your class has a shoulder girth of 100 cm. Predict the height (in cm) of this student using the model.
5. The student from part (d) is 160 cm tall. Calculate the residual (in cm). Did the model underestimate or overestimate the student's height?

Solutions

1. From the given information, we see that

$$\bar{x} = 107.60, \quad s_x = 10.39$$

$$\bar{y} = 171.15, \quad s_y = 9.42$$

$$r = 0.68$$

Hence, we get the regression estimates

$$\hat{b}_1 = r \frac{s_y}{s_x} = 0.68 \cdot \frac{9.42}{10.39} = 0.617$$

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} = 171.15 - 0.617 \cdot 107.60 = 104.813$$

and the equation for the regression line becomes

$$\hat{y} = 0.617x + 104.813$$

2. *Slope:* For each centimeter increase in shoulder girth, we would expect height to increase on average by 0.617 centimeters.

Intercept: People who have a shoulder girth of 0 centimeters are expected to be on average 104.813 centimeters tall.

3. $R^2 = (0.68)^2 = 0.4624$

Interpretation: Approximately 46.24% of the variation in heights is accounted for by the model.

4. The predicted height is

$$\hat{y} = 0.617 \cdot 100 + 104.813 = 166.46 \text{ cm.}$$

5. The residual in general is

$$\text{residual} = \text{observed} - \text{predicted.}$$

Here,

$$e_i = 160 - 166.46 = -6.46 \text{ cm,}$$

meaning that our model **overestimated** the height of that particular student.

Conditions

When fitting a least squares line, we generally require:

- ▶ **Linearity:** The data should show a linear trend.
- ▶ **Nearly normal residuals:** Generally, the residuals must be nearly normal. When this condition is found to be unreasonable, it is usually because of outliers or concerns about influential points.
- ▶ **Constant variability:** The variability of points around the least squares line remains roughly constant.
- ▶ **Independent observations:** Be cautious about applying regression to time series data, which are sequential observations in time such as a stock price each day. Such data may have an underlying structure that should be considered in a model and analysis.

Interpreting the model

- ▶ The slope describes the estimated difference in the y variable if the explanatory variable x for a case happened to be one unit larger.
- ▶ The intercept describes the average outcome of y if $x = 0$ and the linear model is valid all the way to $x = 0$, which in many applications is not the case.
- ▶ The resulting model allows us to give a more refined description of the relationship between variables x and y . However, no causality can be derived. **Correlation does not imply causation!**

Conclusion

- ▶ Linear regression provides a powerful method to **predict** a response variable based on one or more explanatory variables.
- ▶ The **least squares** method ensures the best fit.
- ▶ It's important to remember that **correlation does not imply causation**. Just because two variables are correlated doesn't mean one caused the other.

References

- ▶ Section 8.2 of the textbook
- ▶ Great resource for the more curious:
[Linear Regression, Clearly Explained \(StatQuest\)](#)

STOR 155 - Lecture 28

Linear Regression - Outliers

Panagiotis Andreou

UNC Chapel Hill

April 11, 2025

Regression line

- ▶ Consider a dataset consisting of pairs

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n).$$

- ▶ We would like a straight line that captures the linear relationship between the x -values and the y -values in the “best” possible way.
- ▶ This is called the **regression line**.

The Model

- ▶ The equation for the linear model is

$$y_i = b_0 + b_1 x_i + e_i,$$

where e_i denotes the residual for the i -th observation.

- ▶ The idea is to try and minimize the sum $\sum_{i=1}^n e_i^2$.
- ▶ b_0 and b_1 are calculated based on

$$\arg \min_{b_0, b_1} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Finding the Slope and Intercept

- Slope estimate:

$$\hat{b}_1 = r \frac{s_y}{s_x} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- Intercept estimate:

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

- Once we have the estimates, we get the equation

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x$$

that allows us to predict values of y for new values of x .

- ▶ It is a line that describes how a response variable y changes with an explanatory variables x .
- ▶ The regression line helps to explain or to **predict** y from a given value of x .
- ▶ In regression, unlike correlation, we have to distinguish between an **explanatory** variable x and a **response** variable y .

Why the name regression?

- ▶ The term was coined by Sir Francis Galton (1822 - 1911) in a study on the heights of fathers and sons.
- ▶ He found that fathers taller than average would more likely have shorter sons, and fathers shorter than average, would more likely have taller sons.
- ▶ Galton called this “regression towards the mean height.” (His 1886 paper was titled “Regression towards mediocrity in hereditary stature.”)

ANTHROPOLOGICAL MISCELLANEA.

REGRESSION *towards* MEDIOCRITY *in* HEREDITARY STATURE.

By FRANCIS GALTON, F.R.S., &c.

[WITH PLATES IX AND X.]

THIS memoir contains the data upon which the remarks on the Law of Regression were founded, that I made in my Presidential Address to Section H, at Aberdeen. That address, which will appear in due course in the Journal of the British Association, has already been published in "Nature," September 24th. I reproduce here the portion of it which bears upon regression, together with some amplification where brevity had rendered it obscure, and I have added copies of the diagrams suspended at the meeting, without which the letterpress is necessarily difficult to follow. My object is to place beyond doubt the existence of a simple and far-reaching law that governs the hereditary transmission of, I believe, every one of those simple qualities which all possess, though in unequal degrees. I once before ventured to draw attention to this law on far more slender evidence than I now possess.

It is some years since I made an extensive series of experiments on the produce of seeds of different size but of the same species. They yielded results that seemed very noteworthy, and I used them as the basis of a lecture before the Royal Institution on February 9th, 1877. It appeared from these experiments that the offspring did *not* tend to resemble their parent seeds in size, but to be always more mediocre than they—to be smaller than the parents, if the parents were large; to be larger than the parents, if the parents were very small. The point of convergence was considerably below the average size of the seeds contained in the large bagful I bought at a nursery garden, out of which I selected those that were sown,

FIGURE: The first page of Sir Galton's original publication from 1886.

Types of outliers

- ▶ Points that fall horizontally away from the center of the cloud tend to pull harder on the line, so we call them points with **high leverage**.
- ▶ Points that fall horizontally far from the line are points of high leverage; these points can strongly influence the slope of the least squares line.
- ▶ If one of these high leverage points does appear to actually invoke its influence on the slope of the line, then we call it an **influential point**.
- ▶ Usually we can say a point is influential if, had we fitted the line without it, the influential point would have been unusually far from the least squares line.

Example

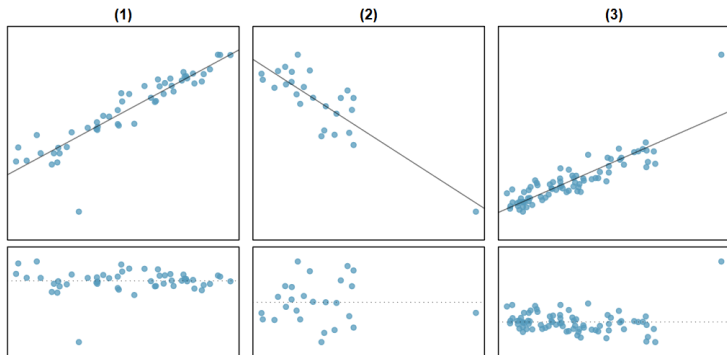


FIGURE: Plot 3 shows an influential point.

Plots description

1. There is one outlier far from the other points, though it only appears to slightly influence the line.
2. There is one outlier on the right, though it is quite close to the least squares line, which suggests it wasn't very influential.
3. There is one point far away from the cloud, and this outlier appears to pull the least squares line up on the right; examine how the line around the primary cloud doesn't appear to fit very well.

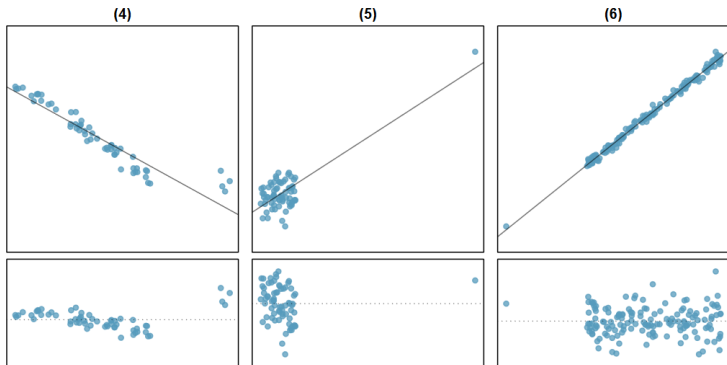


FIGURE: Plots 4 and 5 contain influential points.

Plots description

4. There is a primary cloud and then a small secondary cloud of four outliers. The secondary cloud appears to be influencing the line somewhat strongly, making the least square line fit poorly almost everywhere. There might be an interesting explanation for the dual clouds, which is something that could be investigated.
5. There is no obvious trend in the main cloud of points and the outlier on the right appears to largely control the slope of the least squares line.
6. There is one outlier far from the cloud. However, it falls quite close to the least squares line and does not appear to be very influential.

Conditions

When fitting a least squares line, we generally require:

- ▶ **Linearity:** The data should show a linear trend.
- ▶ **Nearly normal residuals:** Generally, the residuals must be nearly normal. When this condition is found to be unreasonable, it is usually because of outliers or concerns about influential points.
- ▶ **Constant variability:** The variability of points around the least squares line remains roughly constant.
- ▶ **Independent observations:** Be cautious about applying regression to time series data, which are sequential observations in time such as a stock price each day. Such data may have an underlying structure that should be considered in a model and analysis.

Interpreting the model

- ▶ The slope describes the estimated difference in the y variable if the explanatory variable x for a case happened to be one unit larger.
- ▶ The intercept describes the average outcome of y if $x = 0$ and the linear model is valid all the way to $x = 0$, which in many applications is not the case.
- ▶ The resulting model allows us to give a more refined description of the relationship between variables x and y . However, no causality can be derived. **Correlation does not imply causation!**

Example

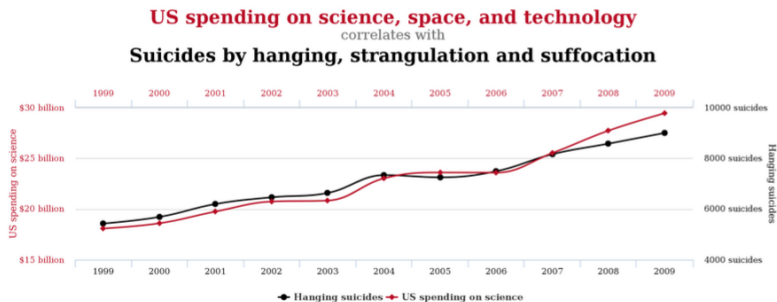


FIGURE: Example of “correlation does not imply causation”.

Extrapolation

- ▶ Linear models can be used to approximate the relationship between two variables.
- ▶ However, these models have real limitations.
- ▶ **Extrapolation:** predicting the response to an input which lies outside of the range of the values of the predictor variable used to fit the model.
- ▶ In general, it is dangerous to extrapolate beyond the scope of the model.

Example 1

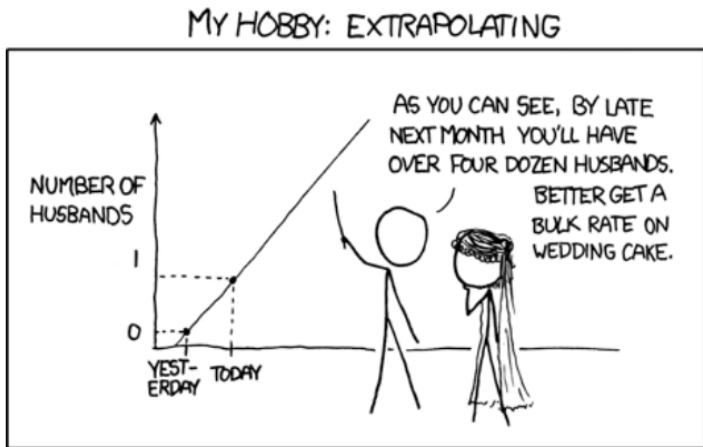


FIGURE: Example of extrapolation.

Example 2

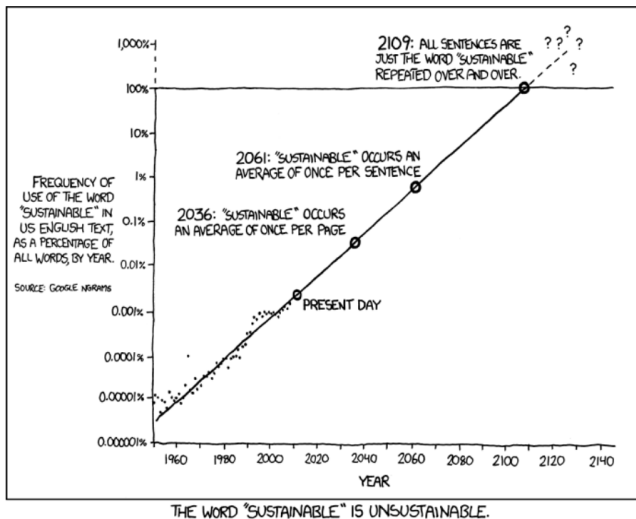


FIGURE: Example of extrapolation.

Extrapolation

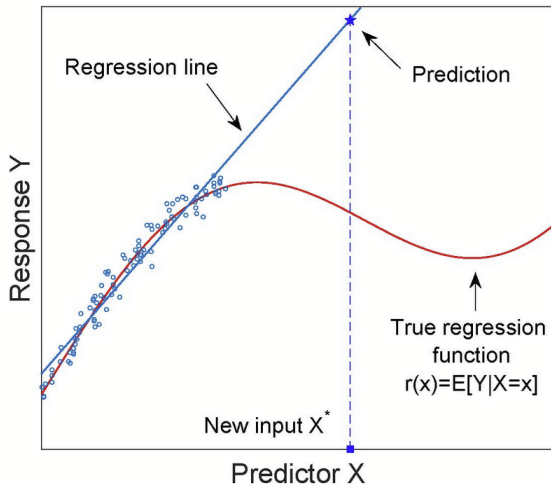


FIGURE: Example of extrapolation.

Inference for Linear Regression

- ▶ The least squares line we fitted contains two estimated quantities, so we want to see how “good” our estimates are.
- ▶ We discuss uncertainty in the estimates of the slope and y -intercept for a regression line.
- ▶ We will show how we can construct confidence intervals and conduct hypothesis tests for these quantities.

References

- ▶ Section 8.3 of the textbook
- ▶ Great resource for the more curious:

Linear Regression, Clearly Explained (StatQuest)

STOR 155 - Lecture 29

Inference for Linear Regression

Panagiotis Andreou

UNC Chapel Hill

April 16, 2025

The setup

- ▶ Consider a dataset consisting of pairs

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n).$$

- ▶ We want to fit a linear model of the form

$$y_i = b_0 + b_1 x_i + e_i.$$

- ▶ Method: **estimate** b_0, b_1 so that $\sum_{i=1}^n e_i^2$ becomes min.
- ▶ The fitted model is

$$\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_i,$$

where $\hat{b}_1 = R \frac{s_y}{s_x}$ and $\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$.

Inference for Linear Regression

- ▶ The least squares line we fitted contains two estimated quantities, so we want to see how “good” our estimates are.
- ▶ We discuss uncertainty in the estimates of the slope and y -intercept for a regression line.
- ▶ We will show how we can construct confidence intervals and conduct hypothesis tests for these quantities.

Motivating Example

- ▶ Elections for members of the United States House of Representatives occur every two years, coinciding every four years with the U.S. Presidential election.
- ▶ The set of House elections occurring during the middle of a Presidential term are called midterm elections.
- ▶ In America's two-party system, one political theory suggests the higher the unemployment rate, the worse the President's party will do in the midterm elections.
- ▶ To assess the validity of this claim, we can compile historical data and look for a connection. We consider every midterm election from 1898 to 2018, with the exception of those elections during the Great Depression.

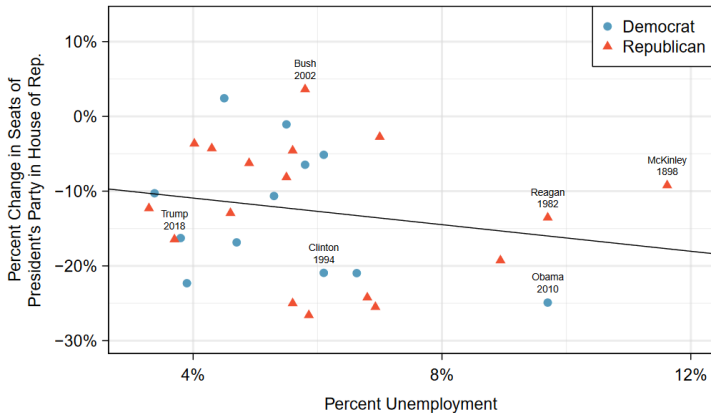


FIGURE: The percent change in House seats for the President's party in each election from 1898 to 2010 plotted against the unemployment rate, along with a fitted least squares regression line.

The fitted model

- ▶ Let y denote the % change in seats of the President's party in the House of Representatives.
- ▶ Let x denote the % of unemployment.
- ▶ We want to see how y is affected by x .
- ▶ Based on the previous data, we find that the estimated linear model is

$$\hat{y} = -7.36 - 0.89x$$

Hypothesis Test

- ▶ There is a negative slope in the fitted line.
- ▶ However, this slope and the intercept are only **estimates** of the parameter values. We might wonder, is this convincing evidence that the “**true**” linear model has a negative slope?
- ▶ That is, do the data provide strong **evidence** that the political theory is accurate, where the unemployment rate is a useful predictor of the midterm election?
- ▶ We can frame this into a statistical **hypothesis test**:

$$\begin{cases} H_0 : b_1 = 0 \\ H_A : b_1 \neq 0 \end{cases}$$

- ▶ The alternative hypothesis H_A corresponds to “The unemployment is predictive of whether the President’s party wins or loses seats in the House of Representatives.”
- ▶ We would reject H_0 in favor of H_A if the data provide strong evidence that the true slope parameter is different than zero.
- ▶ To assess the hypotheses, we identify a standard error for the estimate, compute an appropriate test statistic, and identify the p-value.

Test Statistic

- ▶ Just like other point estimates we have seen before, we can compute a standard error and test statistic for b_1 .
- ▶ We will generally label the test statistic using a T , since it follows the t -distribution. As in all hypothesis tests,

$$T = \frac{\text{point estimate} - \text{null value}}{S.E.}$$

- ▶ We will rely on statistical software to compute the standard error $S.E.$

Regression output from software

- ▶ In practice, we usually rely on statistical software to identify point estimates, standard errors, test statistics, and p-values.
- ▶ However, be aware that software will not generally check whether the method is appropriate, meaning we must still verify conditions are met.

	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	-7.3644	5.1553	-1.43	0.1646
unemp	-0.8897	0.8350	-1.07	0.2961

$df = 27$

FIGURE: Output for the regression line modeling the midterm election losses for the President's party as a response to unemployment.

Reading the output

- ▶ The entries in the first column represent the least squares estimates, b_0 and b_1 .
- ▶ The values in the second column correspond to the standard errors of each estimate.
- ▶ Using the estimates, we could write the equation for the least square regression line as

$$\hat{y} = -7.3644 - 0.8897x,$$

where $\hat{y}$ in this case represents the predicted change in the number of seats for the president's party, and x represents the unemployment rate.

Conducting the test

- ▶ Test statistic:

$$T = \frac{-0.8897 - 0}{0.8350} = -1.07$$

- ▶ p-value:

$$2P(t_{27} \leq -1.07) = 0.2961$$

- ▶ p-value > 0.05 , so we fail to reject the null H_0 at significance level 5%.
- ▶ That is, the data do not provide convincing evidence that a higher unemployment rate has any correspondence with smaller or larger losses for the President's party in the House of Representatives in midterm elections.

Confidence Intervals

- ▶ We can use the above output to construct a 95% confidence interval for the slope b_1 .
- ▶ General formula for a C.I.:

$$\text{point estimate} \pm \text{critical value} \cdot S.E.$$

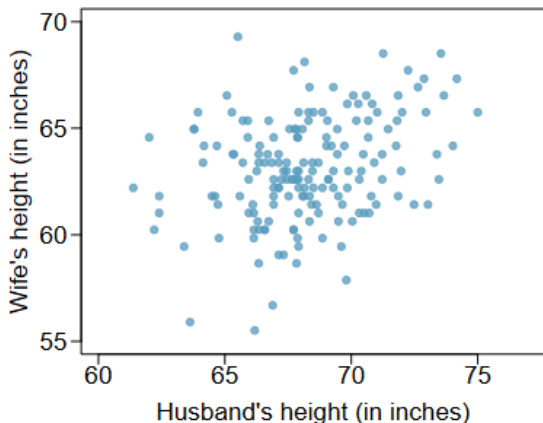
- ▶ Here, $df = 27$, so $t_{27}^* = 2.051831$ and we get the 95% C.I.

$$-0.8897 \pm 2.051831 \cdot 0.835 = (-2.60298, 0.823578)$$

- ▶ **Connection with Hypothesis Test:** 0 is contained in the confidence interval, so we fail to reject the null hypothesis H_0 .

Problem

The scatterplot below summarizes husbands' and wives' heights in a random sample of 170 married couples in Britain, where both partners' ages are below 65 years.



Summary output of the least squares fit for predicting wife's height from husband's height is also provided in the table:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	43.5755	4.6842	9.30	0.0000
height_husband	0.2863	0.0686	4.17	0.0000

1. Is there strong evidence that taller men marry taller women? State the hypotheses and include any information used to conduct the test.
2. Write the equation of the regression line for predicting wife's height from husband's height.
3. Interpret the slope and intercept in the context of the application.

4. Given that $R^2 = 0.09$, what is the correlation of heights in this data set?
5. You meet a married man from Britain who is 5'9" (69 inches). What would you predict his wife's height to be? How reliable is this prediction?
6. You meet another married man from Britain who is 6'7" (79 inches). Would it be wise to use the same linear model to predict his wife's height? Why or why not?

Reading the problem

- ▶ Exploring the relationship between husband's and wife's heights.
- ▶ Dataset: Heights of 170 married couples in Britain.
- ▶ Objective: Determine if there is a correlation between husband's and wife's heights.

Hypothesis Test for Correlation of Heights

- ▶ Null Hypothesis (H_0): There is no linear relationship between husband's and wife's heights ($b_1 = 0$).
- ▶ Alternative Hypothesis (H_A): There is a linear relationship between husband's and wife's heights ($b_1 \neq 0$).
- ▶ Test Statistic: $t = 4.17$ with a p-value of 0.0000.
- ▶ Conclusion: With such a low p-value, we **reject** H_0 . There is strong evidence of a positive linear relationship.

Regression Equation

- ▶ The regression line is given by the equation:

$$\hat{y} = 43.5755 + 0.2863x$$

- ▶ Here, $\hat{y}$ represents the predicted wife's height, and x represents the husband's height.

Interpreting Slope and Intercept

- ▶ Slope (b_1): For every additional inch in husband's height, the wife's height is predicted to increase by 0.2863 inches.
- ▶ Intercept (b_0): If a husband's height were 0 inches, the model predicts the wife's height would be 43.5755 inches. This is a theoretical intercept and not practically interpretable in this context.

Correlation Coefficient (R)

- ▶ Given that $R^2 = 0.09$, the correlation R can be calculated as:

$$R = \pm\sqrt{R^2} = \pm 0.3$$

- ▶ Since we have a positive slope, R is positive. Therefore,

$$R = \sqrt{0.09} = 0.3$$

- ▶ This indicates a moderate positive correlation.

Predicting Wife's Height for a 5'9" Husband

- ▶ Husband's height (x): 69 inches.

- ▶ Predicted wife's height ($\hat{y}$):

$$\hat{y} = 43.5755 + 0.2863 \cdot 69 = 63.3302 \text{ inches}$$

- ▶ Calculation yields a prediction of the wife's height.
- ▶ Reliability: Given $R^2 = 0.09$, only 9% of the variability in the wife's height is explained by the husband's height. Thus, the prediction is not very reliable.

Predicting Wife's Height for a 6'7" Husband

- ▶ Husband's height (x): 79 inches.
- ▶ Use the regression line for the prediction with caution.
- ▶ Given the husband's height is much greater than average, this prediction might not be as reliable due to potential **extrapolation** beyond the range of the data.
- ▶ It is advisable to be cautious when predicting values outside the range of the observed data.

References

- ▶ Section 8.4 of the textbook